Final Year Design Project

College Management System

for Govt. Graduate College, Mandi Bahauddin

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Bachelor of Science in Information Technology (2022-2026)

DEPARTMENT OF INFORMATION TECHNOLOGY
GOVT. GRADUATE COLLEGE, MANDI BAHAUDDIN

College Management System for Govt. Graduate College, Mandi Bahauddin

A project presented to
Govt. Graduate College, Mandi Bahauddin

In partial fulfilment
of the requirement for the degree of

Bachelor of Science in Information Technology (2022-2026)

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CERTIFICATE OF APPROVAL

It is to certify that the final year design project (FYDP) of BSIT, session (2022-2026), titled “College Management System for Govt. Graduate College, Mandi Bahauddin” was developed by Hadia (085668), Sharfa Kiran (085646), and Syeda Laraib Qamar Kazmi (085713) under the supervision of Prof. Ubaid Ullah; in my opinion, it is fully adequate, in scope and quality, for the degree of Bachelor of Science in Information Technology.

Signature: ______________________________________
FYDP Advisor: Prof. Ubaid Ullah

Signatures (Faculty Advisory Committee – FAC)

	FAC1	**FAC2**
Name	Prof. Muhammad Faiyaz	Prof. Ubaid Ullah
Signature		

Signature: ______________________________________
Head of FYDP Coordination Office: Prof. Muhammad Faiyaz

Signature: ______________________________________ Dated: _______________
Head of Department, Information Technology: Prof. Muhammad Faiyaz

Executive Summary

Govt. Graduate College, Mandi Bahauddin runs its academic life on paper. Roughly 3,000 students across Intermediate and BS programmes, and about 50 faculty members, still queue up at the accounts window for a fee challan, wait for a hand-written attendance register to be tallied, and find out their semester result from a notice board rather than a phone. Every one of these steps is a place where a form goes missing, a number gets mistyped, or a student simply doesn't find out in time. That was the starting problem: a college of this size cannot keep running its records by hand without the errors and delays showing up somewhere — a wrong CGPA on a transcript, a fee challan that doesn't match what the accounts office actually has on file, an exam schedule that reaches half the class.

Our answer is a suite of three Android applications — one for admins, one for teachers, one for students — backed by a single shared database and a common design system, so the same piece of data (a student's attendance, a session's fee structure, a published datesheet) is always the same regardless of which app is reading it. The admin app is where the college's actual structure lives: departments, the sessions (intakes) within them, each session's curriculum and timetable, and its fee structure. Teachers mark attendance, enter midterm and sessional marks, submit exam papers, and record semester GPA/CGPA against that same structure. Students see their own slice of it — attendance, marks, fee challan, datesheets, calendar notices — without ever walking into an office.

Midway through the project we made a deliberate architectural change: the backend moved from Firebase (Firestore + Auth) to Supabase (Postgres, with Row-Level Security enforcing who can see and write what, GoTrue for authentication, and Storage for uploaded files like exam papers and documents), and the UI layer moved from XML Activities/Fragments to Jetpack Compose with a proper MVVM/Repository split, wired together with Hilt. The reasoning was straightforward: a relational schema with real foreign keys and row-level policies gives us guarantees a document store couldn't — a mark can't be attributed to a student who was never enrolled in that session, a fee record can't outlive the session it belongs to — and Compose let us build one shared component library (the “Modernist” red/ink design system) that all three apps use identically instead of maintaining three separate sets of XML layouts.

Every major workflow in this report reflects that rebuilt system as it exists in the codebase today, not the original Firebase/Java prototype: department-centric academic structure, per-session fees, a lock-then-request-approval workflow for marks (a teacher enters a score once; changing it after the fact requires the admin's sign-off), semester result recording with GPA/CGPA and supply-subject tracking, fines, a calendar, datesheets, an uploadable-documents library, and a tiered Insights dashboard built directly on Postgres views so that what an admin sees college-wide, a teacher sees scoped to their own classes, purely through the database's own access rules — no extra application logic required. All three apps build and run against the live Supabase project.

Keywords: College Management System, Row-Level Security, Jetpack Compose, Supabase, Android

Acknowledgement

[PLACEHOLDER — write this yourself, in your own words. It's the one section of this report a template can't fill in for you. Typically thanks: Prof. Ubaid Ullah (your FYDP advisor) for supervision and guidance; the FYP Coordination Office / Prof. Muhammad Faiyaz; the Department of Information Technology, Govt. Graduate College, Mandi Bahauddin; and family or anyone else who supported you through the project.]

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Abbreviations

Abbreviations and acronyms used throughout this report, ordered alphabetically.

Abbreviation	Description
API	Application Programming Interface
BS	Bachelor of Science
BSIT	Bachelor of Science in Information Technology
CGPA	Cumulative Grade Point Average
CMS	College Management System
CRUD	Create, Read, Update, Delete
DI	Dependency Injection (Hilt)
DTO	Data Transfer Object
FAC	Faculty Advisory Committee
FR	Functional Requirement
FYDP	Final Year Design Project
GGC MBD	Govt. Graduate College, Mandi Bahauddin
GPA	Grade Point Average
JWT	JSON Web Token
MVVM	Model-View-ViewModel
NFR	Non-Functional Requirement
RLS	Row-Level Security (Postgres)
RPC	Remote Procedure Call (a Postgres function invoked from the app)
SDK	Software Development Kit
SRS	Software Requirements Specification
UI	User Interface
UUID	Universally Unique Identifier

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Chapter 1

Introduction

1. Introduction

This chapter introduces the College Management System developed for Govt. Graduate College, Mandi Bahauddin. It explains the problems that motivated the project, the proposed solution, its objectives and scope, and the six applications that make up the complete system. It also reviews related systems, states the project vision, identifies practical limitations, and presents the main tools, deliverables, and project plan. Together, these sections establish the purpose and boundaries of the project before the later chapters discuss its requirements, design, implementation, testing, and deployment.

1.1 Problem Statement

Many routine activities at Govt. Graduate College, Mandi Bahauddin depend on paper registers, printed notices, and visits to administrative offices. Attendance recorded in class registers has to be counted manually before a student's attendance position can be known. Marks and semester results pass through several paper-based steps, which makes checking, correcting, and sharing them slow. Timetables, examination dates, and college notices can be missed when a student is absent or does not visit the noticeboard. Students may also need to visit the college simply to obtain information about fees, fines, results, or schedules. Teachers repeat clerical work when the same information is required for class records and administrative reports.

The administration faces a related problem: information about departments, academic sessions, teachers, and students is spread across separate records. Preparing a complete report therefore takes time and may involve comparing several registers. A correction made in one place may not be reflected everywhere else. Paper records are also difficult to search and cannot provide timely information to people who are away from campus. These difficulties become more noticeable as the number of students, classes, and academic activities increases. The project addresses the need for one coordinated system through which administrators, teachers, and students can perform their work and view the information relevant to them from either a mobile phone or a desktop computer.

1.2 Problem Solution

The proposed solution is a College Management System made up of six connected applications. Administrators, teachers, and students each receive a mobile application and a desktop application designed around their own responsibilities. All six applications use the same college records, so information entered by an authorised user becomes available to the other affected users without being recorded again. A secure sign-in process identifies the user's role and opens the correct application area. The system keeps administrative, academic, financial, and communication information together while still limiting each person to the information and actions appropriate to their role.

The two admin applications support the management of departments, academic sessions, students, teachers, other administrators, timetables, fees, attendance reports, examinations, documents, and notices. The two teacher applications support everyday academic work such as viewing assigned classes, marking attendance, entering marks, submitting examination papers, checking schedules, and requesting authorised corrections. The two student applications provide a clear view of personal attendance, timetable, marks, results, fee information, datesheets, notices, and shared documents. A student can also request that an account be linked with the correct college record.

Mobile applications make the system convenient during classes and while users are away from a desk. Desktop applications provide the same role-based services on a larger screen for office work, extended data entry, document handling, and printing. The two formats are intended to complement each other rather than create separate records or separate procedures. Important actions, such as changing a recorded mark or approving a student link request, follow a visible approval process. This improves accountability while reducing repeated paperwork. The overall goal is to make college information easier to maintain, quicker to access, and more dependable for all three user groups.

1.3 Objectives of the Proposed System

The project is guided by the following specific and measurable objectives:

- Provide six working applications: mobile and desktop versions for administrators, teachers, and students.
- Maintain one consistent record for each department, academic session, teacher, student, timetable, attendance entry, examination result, and fee structure.
- Allow authorised teachers to record a complete class attendance register digitally and make the updated attendance visible to the administration and affected students after synchronisation.
- Ensure that every change requested for an already-recorded mark is reviewed by an administrator rather than being changed without approval.

- Allow students to view routine academic and fee information without requiring an office visit for information that is already available in the system.
- Publish college notices, calendar events, documents, and examination datesheets to the intended users through both mobile and desktop applications.
- Provide administrators with current summaries of departments, sessions, teachers, students, attendance, and academic performance without manually combining separate registers.
- Apply role-based access throughout the system so that users can view and perform only the actions assigned to their role.
- Keep the mobile and desktop versions of each role consistent in content, navigation, and major workflows.
- Complete functional and user-interface testing for the six applications before final project submission.

1.4 Scope

The project covers the main academic and administrative activities of Govt. Graduate College, Mandi Bahauddin through six applications. The primary stakeholders are college administrators, teaching staff, and enrolled students. Each stakeholder group can use either a mobile application or a desktop application, while both versions work with the same college information.

Included in the project:

- Management of departments, academic sessions, subjects, teachers, administrators, students, and student account-linking requests.
- Preparation and viewing of class timetables, teacher schedules, examination datesheets, and college calendar events.
- Attendance recording by teachers, personal attendance viewing by students, and attendance reporting for administrators.
- Marks entry, approved mark corrections, semester results, examination-paper submission, and academic summaries.
- Session fee structures, student fee challans, fines, notices, notifications, and shared college documents.
- Mobile and desktop access for all three roles, with consistent information and role-appropriate actions.

Outside the current project boundary:

- Online fee payment or direct connection with a bank; the system presents fee information and challans only.
- Automatic grading of examination papers or replacement of academic judgement by software.
- Public website services, online admissions, hostel management, payroll, and library circulation.
- Applications for iPhone, browser-based access, or use by multiple colleges within one installation.

Work was prioritised in stages: first the college structure and user accounts, then attendance, timetables, marks and results, followed by fees, notices, documents, reporting, and desktop availability. This order ensured that later features were built on complete and verified academic records.

1.5 System Components

The College Management System has six user-facing applications. The mobile and desktop versions for each role provide the same essential information and follow the same college procedures, while their layouts are adapted to the available screen size.

1.5.1 Admin Mobile Application

- Dashboard showing key college totals, pending requests, notifications, and shortcuts to common administrative work.
- Department, academic session, subject, timetable, fee, student, teacher, and administrator management.
- Review of student account-linking requests and teacher mark-correction requests.
- Attendance reports, academic summaries, examination datesheets, calendar events, documents, and notices.
- Designed for quick checks, approvals, and updates while the administrator is away from the office desk.

1.5.2 Admin Desktop Application

- Provides the same administrative areas and permissions as the Admin Mobile Application.

- Uses wider card and table layouts for managing larger lists of departments, sessions, teachers, students, and records.
- Supports office-oriented work such as preparing reports, selecting or saving documents, opening files, printing, and exporting information.
- Retains the same approval processes, status information, navigation order, and college data as the mobile version.

1.5.3 Teacher Mobile Application

- Home view showing the teacher's next class, assigned work, notices, and shortcuts.
- Daily class attendance entry with access to previous attendance records.
- Marks entry, mark-correction requests, semester results, and examination-paper submission.
- Personal teaching schedule, assigned students, datesheets, calendar events, documents, notifications, and profile.
- Designed for use during lessons and other day-to-day teaching activities.

1.5.4 Teacher Desktop Application

- Provides the same teaching responsibilities and access limits as the Teacher Mobile Application.
- Offers more screen space for class lists, attendance registers, marks entry, schedules, and result review.
- Supports desktop file selection, examination-paper handling, document opening, and printing where required.
- Keeps the teacher's classes, records, notices, and approval status consistent with the mobile version.

1.5.5 Student Mobile Application

- Student registration and a request process for linking the account with the correct college record.
- Home view showing important academic information, upcoming classes, and notices.
- Personal attendance history, timetable, marks, semester results, datesheets, and academic calendar.
- Fee challan, fines, shared documents, notifications, and student profile.
- Designed to provide routine college information without requiring an office or noticeboard visit.

1.5.6 Student Desktop Application

- Provides the same personal academic and fee information as the Student Mobile Application.
- Uses the larger display for easier reading of timetables, attendance history, results, datesheets, and documents.
- Supports opening, downloading, saving, and printing available college documents and student records.
- Keeps account status, notifications, and displayed college information consistent with the mobile version.

The six applications are supported by one central information service and a common visual design. This allows a record created in one application to appear in the other authorised applications without maintaining separate copies. Shared sign-in, access rules, error messages, and data checks also help users receive a consistent experience across mobile and desktop devices.

1.6 Related System Analysis / Literature Review

Existing learning platforms and college portals show the value of making academic information available online. However, many focus on either classroom teaching or office administration rather than connecting administrators, teachers, and students through matching mobile and desktop applications. Table 1-1 summarises the main gap addressed by the proposed system.

Table 1-1 Related System Analysis with Proposed Project Solution

Related System	Observed Weakness	Proposed Project Response
Google Classroom	Supports course communication and learning material, but it is not intended to manage a college's departments, sessions, fees, attendance reporting, or administrative approvals.	Adds college-wide academic and administrative records while retaining convenient access to notices and documents.
General college	Often concentrate on office use or browser	Provides six coordinated applications,

Related System	Observed Weakness	Proposed Project Response
management portals	access and may not provide matching role-based applications for teachers and students on both mobile and desktop devices.	giving every role an appropriate mobile and desktop experience using the same information.
Paper registers and noticeboards	Require physical access, repeated entry, manual totals, and separate checking when information changes.	Records information once, makes approved updates visible to the affected users, and preserves clear responsibility for important actions.

1.7 Vision Statement

For the administrators, teachers, and students of Govt. Graduate College, Mandi Bahauddin,

Who need dependable college information without relying on separate paper records, office visits, and noticeboards,

The College Management System

Is a coordinated set of six role-based mobile and desktop applications

That supports college administration, everyday teaching work, and student access to academic, fee, examination, and notice information,

Unlike the present paper-based process and systems that serve only one device type or one part of college work,

Our product gives all three user groups timely access to the same approved records through the device that best suits their situation.

1.8 System Limitations and Constraints

Limitations:

- Users need an internet connection to receive the latest information and submit new or changed records.
- The accuracy of reports depends on correct and timely information being entered by authorised college users.
- Service availability is partly dependent on the external online service used to store and exchange college information.
- The project provides Android mobile applications and Windows desktop applications; it does not include iPhone or browser versions.

Constraints:

- The first deployment is designed for one college and follows the academic practices of Govt. Graduate College, Mandi Bahauddin.
- Fee payment remains outside the application. Students can view fee details and challans, but payment is completed through the college's existing process.
- Examination papers are submitted and managed through the system, but grading is performed by teachers rather than automatically.
- Mobile and desktop applications must remain consistent for each role, while operating-system file and print windows may appear differently.
- The project scope and level of testing are limited by the available academic schedule, team size, devices, and college data used during development.

1.9 Tools and Technologies

The project uses a common development approach so that the six applications can share their main behaviour and visual style. The principal tools and their purposes are listed in Table 1-2.

Table 1-2 Tools and Technologies for the Proposed Project

Tool / Technology	Version	Purpose in the Project
Kotlin	2.1.0	Main programming language used for the mobile and desktop applications.
Jetpack Compose	2024.12.01	Creates the user interfaces of the three Android mobile

Tool / Technology	Version	Purpose in the Project
		applications.
Compose Multiplatform	1.7.3	Creates the user interfaces of the three Windows desktop applications.
Android SDK	Target 36	Provides the platform required to build and test the mobile applications.
Java Development Kit	17	Provides the runtime and build environment for the desktop applications and shared project work.
Supabase Kotlin SDK	3.1.4	Connects the six applications with sign-in, shared college records, and uploaded files.
PostgreSQL	Supabase managed	Stores the college's academic, administrative, financial, and communication records.
Gradle	8.13	Builds and packages the mobile and desktop applications in a repeatable way.
Git	2.x	Maintains the project's version history and supports controlled team development.

1.10 Project Deliverables

- Project planning material, requirements specification, design documentation, test records, and the final project report.
- Three installable Android applications: Admin Mobile, Teacher Mobile, and Student Mobile.
- Three installable Windows applications: Admin Desktop, Teacher Desktop, and Student Desktop.
- The configured online data service containing the project structure, access rules, and file-storage arrangements.
- Complete source material for the six applications and their shared supporting parts.
- Application prototypes, screenshots, diagrams, data descriptions, and selected user guidance included with the report.

1.11 Project Planning

The project was organised into connected stages so that each group of features could be reviewed before the next stage depended on it. Requirements and design established the project boundaries, followed by the three mobile applications and then their desktop counterparts. College data setup, testing, refinement, and report preparation continued alongside the later development stages. The principal activities and milestones are shown in Figure 1-1.

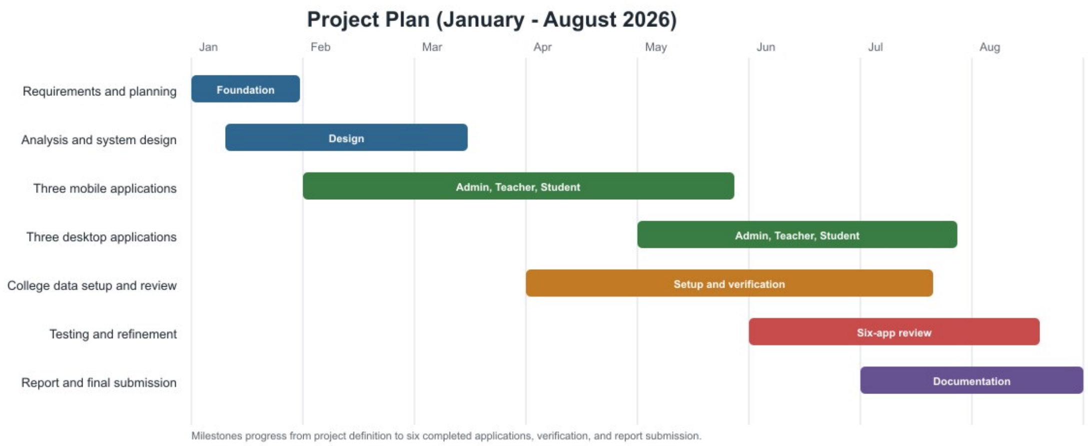

Figure 1-1 High-level project plan and milestones

1.12 Summary

This chapter introduced the need for a coordinated college management system and described the proposed response: six connected applications for administrators, teachers, and students on mobile and desktop devices. It defined the project's objectives, scope, components, limitations, tools, deliverables, and planned stages. The comparison with related systems showed that the project's main contribution is not a single isolated feature, but the connection of routine college work across three roles and two device types. These foundations guide the requirements presented in Chapter 2 and provide the basis against which the completed system can later be evaluated.

Chapter 2

Requirements Analysis

2. Analysis

This chapter defines what the College Management System must provide and the conditions under which it must operate. It identifies the users of all six applications, explains how their needs were translated into use cases and requirements, and records the expected behaviour of the system in a form that can later be tested. Functional requirements describe the services available to each role, while quality and interface requirements define expectations for reliability, usability, performance, security, devices, and communication.

2.1 User Classes and Characteristics

The same administrator, teacher, or student may use both device types. They are listed separately because the working situation, screen size, and available file or printing actions differ between mobile and desktop use.

Table 2-1 User Classes and Characteristics

User Class	Characteristics
Admin Mobile User	An authorised college administrator who needs quick access to summaries, approvals, people, academic records, notices, and reports while moving around the college. The user can perform administrative actions permitted by the assigned account.
Admin Desktop User	An authorised administrator performing longer office tasks on Windows, including management of large lists, timetable and fee work, report review, document handling, exporting, and printing.
Teacher Mobile User	A faculty member using a phone during daily teaching work to check classes, mark attendance, enter marks, request corrections, submit results or papers, and read college information.
Teacher Desktop User	A faculty member completing the same authorised teaching work on Windows, with more space for class registers, marks, schedules, student lists, and document operations.
Student Mobile User	An enrolled or newly registered student using a phone to link an account, check personal attendance, timetable, marks, results, fees, notices, datesheets, documents, and profile information.
Student Desktop User	A student using Windows to access the same personal information, especially when reading, downloading, saving, opening, or printing larger records and documents.

2.2 Requirement Identifying Technique

Use case modelling was selected as the main requirement-identification technique because all six applications are interactive and role based. Each important activity begins with a user goal, follows an expected sequence, and ends with an observable result. Use cases make these responsibilities clear without requiring the reader to understand how the applications are built.

2.2.1 Process and Stakeholder Review

Requirements were derived by reviewing the college activities represented in the original project documentation, examining the existing paper-oriented procedures, and comparing them with the responsibilities of administrators, teachers, and students. Particular attention was given to who creates a record, who approves it, who is allowed to view it, and what should happen when information is missing or invalid.

- Administrative review identified college structure, people management, approvals, fees, notices, reporting, and record-control needs.
- Teaching workflow review identified schedules, attendance, marks, corrections, examination papers, results, student lists, and notices.
- Student workflow review identified registration, account linking, personal academic records, fee information, datesheets, documents, and notifications.

2.2.2 Prototype and Cross-Platform Review

The working screens were reviewed as prototypes to confirm that every required activity had a clear entry point, understandable feedback, and a suitable mobile and desktop presentation. This review also identified desktop-specific needs such as file selection, saving, opening, exporting, and printing. The mobile and desktop versions were treated as two ways of performing the same role, not as separate sources of college information.

2.2.3 Use Case Descriptions

The following fully dressed use cases describe the principal interactions across the six applications. Each table identifies the responsible actor, supported applications, related requirements, normal sequence, exceptions, and the rules that protect college records.

2.2.3.1 UC-01: Sign In and Open the Correct Application Area

Table 2-2 Fully Dressed Description of UC-01

Use Case ID	UC-01
Use Case Name	Sign In and Open the Correct Application Area
Requirement Traceability	FR-1
Purpose	Allow a registered user to enter the system and reach the area assigned to their role.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
Actors	Administrator, Teacher, Student
Description	A registered user signs in and is directed to the correct role-based home area. A valid session may be reused when the application is opened again.
Trigger	The user opens an application or submits the sign-in form.
Preconditions	The user has a registered account. Administrator and teacher accounts have an active college status.
Postconditions	The correct home area is displayed and only authorised actions are available.
Normal Flow	1. The user opens the application. 2. The system checks for an existing valid session. 3. If needed, the user enters the registered email and password. 4. The system verifies the account and role. 5. The role-appropriate home area opens.
Alternative / Exception Flows	Invalid details produce a clear sign-in message. A disabled account is refused access. A role mismatch directs the user to the appropriate application. A connection failure shows a short retry message without displaying service addresses.
Business Rules	A user may access only the role and college records assigned to the account.
Assumptions	The account exists and the user has access to the registered sign-in details.
Includes / Extends	Included by every protected use case.

2.2.3.2 UC-02: Manage Administrator Accounts

Table 2-3 Fully Dressed Description of UC-02

Use Case ID	UC-02
Use Case Name	Manage Administrator Accounts
Requirement Traceability	FR-2
Purpose	Allow an existing administrator to add and maintain other authorised administrators.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator
Description	An administrator views the administrator directory, creates an account, and maintains the account's college identity and status.
Trigger	The administrator opens People and selects Administrators.
Preconditions	The actor is signed in as an active administrator.
Postconditions	The administrator directory reflects the completed addition or update.
Normal Flow	1. The administrator opens the directory. 2. The system displays existing administrators. 3. The administrator selects Add Administrator. 4. Required identity and contact details are entered. 5. The system validates and creates the account. 6. The updated directory is shown.
Alternative / Exception Flows	Missing, invalid, or duplicate details are identified beside the relevant field. The current administrator cannot accidentally remove the only usable administrative access. A service failure leaves the directory unchanged and shows a safe message.
Business Rules	Only an administrator may create or maintain administrator accounts. Email and account identity must be unique.
Assumptions	The college has approved the person who is being granted administrative access.
Includes / Extends	Includes UC-01.

2.2.3.3 UC-03: Manage Academic Structure

Table 2-4 Fully Dressed Description of UC-03

Use Case ID	UC-03
Use Case Name	Manage Academic Structure
Requirement Traceability	FR-3, FR-4, FR-5
Purpose	Maintain departments, academic sessions, semesters, and subjects as the foundation for all academic records.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator
Description	The administrator creates and reviews departments, sessions, semester progress, and the subjects taught in each semester.
Trigger	The administrator opens the Academics area.
Preconditions	The administrator is signed in. A department must exist before a session can be added, and a session must exist before its subjects can be entered.
Postconditions	The approved academic structure is available to timetable, attendance, marks, fee, and reporting activities.
Normal Flow	1. The administrator selects a department or creates one. 2. A session is created or opened. 3. Semester and session details are reviewed. 4. Subjects are added to the appropriate semester. 5. The system confirms the saved structure.
Alternative / Exception Flows	Duplicate identifiers, invalid years, missing names, or incomplete semester details are rejected with field-level guidance. Deletion is blocked or confirmed when dependent records exist.
Business Rules	Every session belongs to one department. Every subject belongs to a defined session and semester.
Assumptions	The administrator has the approved college programme and subject information.
Includes / Extends	Includes UC-01.

2.2.3.4 UC-04: Manage Teachers and Students

Table 2-5 Fully Dressed Description of UC-04

Use Case ID	UC-04
Use Case Name	Manage Teachers and Students
Requirement Traceability	FR-6, FR-7
Purpose	Maintain the people connected with each academic session and the responsibilities granted to teachers.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator
Description	The administrator adds or imports students, reviews student profiles, creates teacher accounts, and manages teacher status and permissions.
Trigger	The administrator opens People, a session roster, or a student profile.
Preconditions	The administrator is signed in. The relevant department and session already exist.
Postconditions	Valid teacher and student records are available to authorised academic workflows.
Normal Flow	1. The administrator selects Teachers or a session's Students area. 2. Existing records are displayed. 3. The administrator adds, imports, or updates the required person. 4. The system validates identity and session information. 5. The directory or roster refreshes with the saved record.
Alternative / Exception Flows	Duplicate roll numbers or emails are rejected. Invalid import rows are reported without hiding successful rows. Disabling a teacher requires confirmation and does not delete historical records.
Business Rules	A roll number must be unique within its session. Teacher permissions are granted only by an administrator.
Assumptions	Approved staff and enrolment information is available to the administrator.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

2.2.3.5 UC-05: Prepare and Maintain Timetables

Table 2-6 Fully Dressed Description of UC-05

Use Case ID	UC-05
Use Case Name	Prepare and Maintain Timetables
Requirement Traceability	FR-8
Purpose	Create an organised weekly schedule without assigning the same teacher or room to conflicting periods.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator, authorised Teacher
Description	An authorised user creates or updates class periods for sessions, subjects, teachers, rooms, days, and times.
Trigger	The user opens Master Timetable or a session timetable.
Preconditions	Relevant sessions, subjects, and teachers exist. The user has timetable permission.
Postconditions	The approved timetable becomes available to affected administrators, teachers, and students.
Normal Flow	1. The user opens the timetable. 2. A day and period are selected. 3. Session, subject, teacher, room, and time are entered. 4. The system checks for conflicts. 5. The period is saved and the timetable refreshes.
Alternative / Exception Flows	A teacher, room, or session conflict prevents saving and identifies the conflict. Incomplete times or missing assignments are rejected. Cancelling leaves the timetable unchanged.
Business Rules	A teacher, room, or session cannot be assigned to overlapping periods.
Assumptions	The college has agreed its teaching periods and room information.
Includes / Extends	Includes UC-01.

2.2.3.6 UC-06: Review Approval Requests

Table 2-7 Fully Dressed Description of UC-06

Use Case ID	UC-06
Use Case Name	Review Approval Requests
Requirement Traceability	FR-14, FR-28
Purpose	Give authorised staff a controlled way to approve student account links and corrections to locked marks.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop; Teacher Mobile and Teacher Desktop for link requests when permission is granted
Actors	Administrator, authorised Teacher
Description	The reviewer examines a pending request, compares its supporting information with college records, and approves or rejects it with a recorded outcome.
Trigger	The reviewer opens a pending request from the People or Requests area.
Preconditions	A pending request exists and the reviewer has the required permission.
Postconditions	The request is marked approved or rejected. Approved changes are applied to the relevant student link or mark record.
Normal Flow	1. The reviewer opens the request queue. 2. A pending request is selected. 3. Submitted and existing information is compared. 4. The reviewer chooses Approve or Reject. 5. The system asks for confirmation and records the decision.
Alternative / Exception Flows	The reviewer may return without deciding. An already-decided request cannot be decided again. A rejection reason is required where the screen requests one. Failure leaves the request pending.
Business Rules	The person who submitted a mark-correction request cannot approve it. Each pending request may receive only one final decision.
Assumptions	The reviewer can verify the request against college information.
Includes / Extends	Includes UC-01; may extend UC-11 or UC-14.

2.2.3.7 UC-07: Publish College Information

Table 2-8 Fully Dressed Description of UC-07

Use Case ID	UC-07
Use Case Name	Publish College Information
Requirement Traceability	FR-18, FR-20, FR-22, FR-24, FR-25, FR-29
Purpose	Create and publish datesheets, fees, fines, calendar events, documents, and notifications for the intended users.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop; permitted Teacher Mobile and Teacher Desktop for selected notices and datesheets
Actors	Administrator, authorised Teacher
Description	An authorised user prepares college information, selects its audience or academic scope, reviews it, and publishes it for viewing in the relevant applications.
Trigger	The user selects an Add, Create, Upload, Issue, or Publish action.
Preconditions	The user is signed in with the required permission. Any referenced session or student exists.
Postconditions	The new information is stored and becomes visible to its intended users according to publication status and audience.
Normal Flow	1. The user opens the relevant records area. 2. The content and dates are entered or a document is selected. 3. Audience and academic scope are chosen. 4. The system validates the information. 5. The user confirms publication. 6. The published item appears in the list.
Alternative / Exception Flows	Missing dates, invalid amounts, unsupported files, or incomplete audience information prevent publication. The user may save or cancel where the feature allows. Upload failure keeps the item unpublished.
Business Rules	Only authorised users may publish. Students see only information addressed to them or their academic group. Fines must identify a student and amount.
Assumptions	The publisher has approved and accurate content.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

2.2.3.8 UC-08: Review Reports and Insights

Table 2-9 Fully Dressed Description of UC-08

Use Case ID	UC-08
Use Case Name	Review Reports and Insights
Requirement Traceability	FR-10, FR-11, FR-12, FR-15, FR-16, FR-31
Purpose	Help authorised staff review attendance, academic progress, and college summaries without combining separate registers.
Priority	Medium
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop
Actors	Administrator, Teacher
Description	The user opens a report or insight, applies available academic filters, and reviews totals, trends, or students needing attention within the user's authority.
Trigger	The user opens Attendance Records, Insights, or an academic summary.
Preconditions	The user is signed in and relevant academic records exist.
Postconditions	The requested report is displayed. Viewing does not change academic records.
Normal Flow	1. The user opens a reporting area. 2. Available scope and filters are shown. 3. The user selects a department, session, semester, subject, or period where applicable. 4. The system prepares the authorised summary. 5. The user reviews or exports the result.
Alternative / Exception Flows	No matching records produces an informative empty state. Invalid filters are ignored or corrected. A loading failure offers retry without exposing service details.
Business Rules	Administrators may review college-wide information. Teachers may review only their assigned academic records.
Assumptions	Reports reflect the latest successfully stored attendance and assessment records.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

2.2.3.9 UC-09: Mark Class Attendance

Table 2-10 Fully Dressed Description of UC-09

Use Case ID	UC-09
Use Case Name	Mark Class Attendance
Requirement Traceability	FR-10
Purpose	Record one class meeting's attendance accurately for every enrolled student.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects an assigned class and date, marks each student, reviews totals, and submits the register.
Trigger	The teacher selects Mark Attendance.
Preconditions	The teacher is signed in, assigned to the class, and the class roster is available.
Postconditions	One attendance record is stored for each student and becomes available to authorised users.
Normal Flow	1. The teacher selects an assigned class. 2. The date and roster are displayed. 3. Each student is marked present, absent, late, or on leave. 4. The teacher reviews the totals. 5. The register is submitted and confirmation is shown.
Alternative / Exception Flows	An incomplete register is identified before submission. A duplicate class register is opened for review rather than silently duplicated. A connection failure keeps the screen state and offers retry.
Business Rules	A teacher may mark only assigned classes. One final status is recorded per student for a class meeting.
Assumptions	The roster and teacher assignment are correct.
Includes / Extends	Includes UC-01.

2.2.3.10 UC-10: Enter Assessment Marks

Table 2-11 Fully Dressed Description of UC-10

Use Case ID	UC-10
Use Case Name	Enter Assessment Marks
Requirement Traceability	FR-12
Purpose	Record valid marks for students in an assigned subject and examination type.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects a class and assessment, enters each student's marks, resolves validation messages, and submits the completed award list.
Trigger	The teacher selects Marks Entry.
Preconditions	The teacher is assigned to the subject. The class roster and allowed maximum marks exist.
Postconditions	Valid marks are stored and become read-only to the teacher unless an approved correction is granted.
Normal Flow	1. The teacher selects session, subject, semester, and assessment type. 2. The roster is displayed. 3. Marks are entered for each student. 4. The system checks values and completeness. 5. The teacher submits the list. 6. Confirmation and locked status are shown.
Alternative / Exception Flows	Negative marks, values above the maximum, and missing required entries are rejected. An already-submitted list opens as locked. A failed submission leaves the entries available for correction and retry.
Business Rules	Marks must be within the permitted range. A submitted mark cannot be changed directly.
Assumptions	The teacher has completed the assessment and has the approved award information.
Includes / Extends	Includes UC-01; may be extended by UC-11.

2.2.3.11 UC-11: Request a Mark Correction

Table 2-12 Fully Dressed Description of UC-11

Use Case ID	UC-11
Use Case Name	Request a Mark Correction
Requirement Traceability	FR-13
Purpose	Allow a teacher to request a justified correction without directly changing an approved mark.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects a locked mark, enters the proposed value and reason, and submits a request for administrative review.
Trigger	The teacher selects the correction action beside a locked mark.
Preconditions	The teacher entered or is responsible for the mark, and the mark is locked.
Postconditions	A pending request is available to an administrator while the original mark remains unchanged.
Normal Flow	1. The teacher opens the locked award list. 2. A student mark is selected. 3. The proposed mark and reason are entered. 4. The system validates the range and checks for another pending request. 5. The request is submitted and its pending status is shown.
Alternative / Exception Flows	An invalid proposed mark or missing reason prevents submission. A second pending request for the same mark is refused. Cancelling leaves the mark unchanged.
Business Rules	Only an authorised reviewer may apply the requested change. The original value remains active until approval.
Assumptions	The teacher can explain why the correction is required.
Includes / Extends	Extends UC-10 and may lead to UC-06.

2.2.3.12 UC-12: Submit Examination Paper and Semester Results

Table 2-13 Fully Dressed Description of UC-12

Use Case ID	UC-12
Use Case Name	Submit Examination Paper and Semester Results
Requirement Traceability	FR-15, FR-17
Purpose	Allow a teacher to submit required examination material and record final semester outcomes for assigned students.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects the relevant academic context, uploads an examination paper or records semester results, and submits the completed information.
Trigger	The teacher opens Submit Exam Paper or Semester Results.
Preconditions	The teacher is assigned to the relevant session or subject. Required students and semester information exist.
Postconditions	The paper or semester result is stored and available to authorised users according to its status.
Normal Flow	1. The teacher selects the academic session and semester. 2. For a paper, the subject and file are selected; for results, each student's result information is entered. 3. The system validates completeness. 4. The teacher reviews and submits. 5. Confirmation is displayed.
Alternative / Exception Flows	Unsupported or missing files, incomplete student results, and invalid academic values prevent submission. Cancelling leaves existing records unchanged. A transfer failure offers retry.
Business Rules	Teachers may submit only for assigned academic work. Result values must follow the college's allowed grading information.
Assumptions	The paper and result information have been approved for submission.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

2.2.3.13 UC-13: View Teaching Schedule and Assigned Students

Table 2-14 Fully Dressed Description of UC-13

Use Case ID	UC-13
Use Case Name	View Teaching Schedule and Assigned Students
Requirement Traceability	FR-6, FR-9
Purpose	Give a teacher a reliable view of assigned classes, periods, and students.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher reviews the weekly schedule, upcoming class, assigned sessions, and student rosters permitted by the timetable.
Trigger	The teacher opens Home, Schedule, or My Students.
Preconditions	The teacher is signed in and has at least one assignment.
Postconditions	The requested schedule or roster is displayed without changing it.
Normal Flow	1. The teacher opens the relevant area. 2. The system identifies the teacher's assignments. 3. Classes or students are grouped by schedule and session. 4. The teacher selects an item for details. 5. The requested information is displayed.
Alternative / Exception Flows	A teacher with no assignment receives a clear empty state. Missing or unavailable information offers refresh. The teacher cannot open an unrelated roster by changing a selection.
Business Rules	Teachers may view only students and schedules connected with their assigned teaching work.
Assumptions	The administration has prepared the current timetable and roster.
Includes / Extends	Includes UC-01.

2.2.3.14 UC-14: Register and Link a Student Account

Table 2-15 Fully Dressed Description of UC-14

Use Case ID	UC-14
Use Case Name	Register and Link a Student Account
Requirement Traceability	FR-27, FR-28
Purpose	Connect a student's sign-in account with the correct college roll-number record after review.
Priority	High
Supported Applications	Student Mobile, Student Desktop; reviewed in Admin Mobile, Admin Desktop, and permitted Teacher applications
Actors	Student, Administrator, authorised Teacher
Description	A student registers, enters identifying college information, submits a link request, and waits for an authorised decision before personal records are opened.
Trigger	A new or unlinked student selects Register or Link College Record.
Preconditions	The student has a valid email and an existing college enrolment record.
Postconditions	A pending request is created. After approval, the account is linked to one student record; after rejection, the reason or status is shown.
Normal Flow	1. The student registers or signs in. 2. The system identifies the account as unlinked. 3. The student enters roll number and required identity details. 4. The system validates the form and submits a request. 5. An authorised reviewer verifies it through UC-06. 6. The student sees the final status and gains access after approval.
Alternative / Exception Flows	Invalid identity details, an unknown roll number, or an already-linked record prevents submission. A duplicate pending request is not created. Rejected requests may be corrected and resubmitted where allowed.
Business Rules	One account may link to one student record, and one student record may have one approved account.
Assumptions	The college roster contains accurate identity information for comparison.
Includes / Extends	Includes UC-01 and UC-06.

2.2.3.15 UC-15: View Personal Attendance and Timetable

Table 2-16 Fully Dressed Description of UC-15

Use Case ID	UC-15
Use Case Name	View Personal Attendance and Timetable
Requirement Traceability	FR-9, FR-11
Purpose	Give a linked student a current view of class schedule and personal attendance history.
Priority	High
Supported Applications	Student Mobile, Student Desktop
Actors	Student
Description	The student reviews the weekly timetable, next class, attendance totals, subject percentages, and available attendance history for the linked record.
Trigger	The student opens Attendance or Timetable.
Preconditions	The student is signed in and linked to a college record.
Postconditions	The authorised personal schedule or attendance information is displayed without modification.
Normal Flow	1. The student opens the required area. 2. The system identifies the linked session and roll number. 3. Timetable or attendance information is loaded. 4. The student selects a day, subject, or summary where available. 5. Details are displayed.
Alternative / Exception Flows	No published timetable or attendance produces an informative empty state. A connection failure offers retry. The student cannot select another student's record.
Business Rules	Students may view only their own attendance and the timetable of their linked session.
Assumptions	Teachers and administrators have entered the relevant source records.
Includes / Extends	Includes UC-01.

2.2.3.16 UC-16: View Marks, Results, and Datesheets

Table 2-17 Fully Dressed Description of UC-16

Use Case ID	UC-16
Use Case Name	View Marks, Results, and Datesheets
Requirement Traceability	FR-16, FR-19
Purpose	Allow a linked student to review personal assessment information and published examination schedules.
Priority	High
Supported Applications	Student Mobile, Student Desktop
Actors	Student
Description	The student views subject marks, semester results, progression information, and datesheets published for the linked academic session.
Trigger	The student opens Exams, Marks, Results, or Datesheets.
Preconditions	The student is signed in and linked. Relevant records have been entered or published.
Postconditions	The authorised academic information is displayed without allowing student changes.
Normal Flow	1. The student opens the Exams area. 2. Marks, Results, or Datesheets is selected. 3. The system loads information for the linked record and session. 4. Available semester or examination details are shown. 5. The student reviews the selected item.
Alternative / Exception Flows	Unpublished or unavailable information produces a clear message. Incomplete historical data is shown only where valid. A loading failure offers retry.
Business Rules	A student may view only personal marks and results, and only datesheets published for the relevant audience.
Assumptions	Teachers and administrators have completed the relevant academic records.
Includes / Extends	Includes UC-01.

2.2.3.17 UC-17: View Fee Challan and Fines

Table 2-18 Fully Dressed Description of UC-17

Use Case ID	UC-17
Use Case Name	View Fee Challan and Fines
Requirement Traceability	FR-21, FR-23
Purpose	Provide a student with current personal fee information without a routine information visit to the accounts office.
Priority	High
Supported Applications	Student Mobile, Student Desktop
Actors	Student
Description	The student opens the fee area to review the session fee structure, personal challan information, due dates, payment note, and recorded fines.
Trigger	The student selects Fee Challan or Fines.
Preconditions	The student is signed in and linked. A fee structure or fine record exists where applicable.
Postconditions	The student's current fee and fine information is displayed without recording payment.
Normal Flow	1. The student opens the fee area. 2. The system identifies the linked session and roll number. 3. Fee heads, totals, due information, and fines are prepared. 4. The student reviews the details. 5. On desktop, the available record may be opened or printed.
Alternative / Exception Flows	If no fee structure or fines exist, the system shows an informative empty state. Missing required information is not replaced with an invented amount. Loading failure offers retry.
Business Rules	Students see only their own fee and fine records. Displaying a challan does not mark it paid.
Assumptions	The accounts information has been entered by authorised college staff.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

2.2.3.18 UC-18: View Notices, Events, and Documents

Table 2-19 Fully Dressed Description of UC-18

Use Case ID	UC-18
Use Case Name	View Notices, Events, and Documents
Requirement Traceability	FR-24, FR-26, FR-30
Purpose	Give users one place to read college announcements, calendar events, and published documents intended for them.
Priority	Medium
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
Actors	Administrator, Teacher, Student
Description	The user opens notifications, calendar, datesheets, or documents and views items addressed to the user's role or academic group.
Trigger	The user selects a notice, event, or document from the relevant area or notification entry point.
Preconditions	The user is signed in. The item has been published for the user's audience.
Postconditions	The item is displayed and may be marked read where supported. Published content remains unchanged.
Normal Flow	1. The user opens the relevant information area. 2. The system lists authorised published items. 3. The user selects an item. 4. Details or the document are displayed. 5. The user returns to the list or performs an allowed file action.
Alternative / Exception Flows	An expired, removed, or unavailable item produces a safe message. Unsupported opening is handled through download or an explanatory message. Empty lists explain that nothing has been published.
Business Rules	Only items addressed to the user's role, session, or general college audience may be displayed.
Assumptions	Publishers have selected the correct audience and content.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

2.2.3.19 UC-19: Use Desktop File, Export, and Print Actions

Table 2-20 Fully Dressed Description of UC-19

Use Case ID	UC-19
Use Case Name	Use Desktop File, Export, and Print Actions
Requirement Traceability	FR-33
Purpose	Allow Windows users to complete role-appropriate document and reporting actions through familiar desktop services.
Priority	Medium
Supported Applications	Admin Desktop, Teacher Desktop, Student Desktop
Actors	Administrator, Teacher, Student
Description	A desktop user selects a file, chooses a save location, opens a document, prints available information, or invokes a supported sharing action from an existing workflow.
Trigger	The user selects Upload, Open, Save, Export, Print, or Share where that action is available.
Preconditions	The user is signed in, authorised for the underlying record, and using a Windows desktop application.
Postconditions	The selected operating-system action completes or returns a clear cancellation or failure result to the application.
Normal Flow	1. The user opens an authorised record or form. 2. A desktop action is selected. 3. The application opens the appropriate Windows selection, save, open, or print service. 4. The user completes the operating-system step. 5. The application confirms completion or returns to the original screen.
Alternative / Exception Flows	The user may cancel without changing data. Missing file access, unsupported file type, or unavailable printer produces a clear message. The application does not expose internal service information.
Business Rules	Desktop actions do not bypass the user's role or record permissions. Native Windows windows may differ visually from the application.
Assumptions	Windows has an appropriate file association or printer where the selected action requires one.
Includes / Extends	Extends UC-04, UC-07, UC-08, UC-12, UC-17, or UC-18.

2.2.3.20 UC-20: View and Maintain Personal Profile

Table 2-21 Fully Dressed Description of UC-20

Use Case ID	UC-20
Use Case Name	View and Maintain Personal Profile
Requirement Traceability	FR-32
Purpose	Allow each signed-in user to review account identity, update permitted profile details, and sign out safely.
Priority	Medium
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
Actors	Administrator, Teacher, Student
Description	The user opens the profile area, reviews account and college identity, changes allowed contact or profile information, and may sign out.
Trigger	The user selects Profile or Sign Out.
Preconditions	The user is signed in.
Postconditions	Valid permitted changes are stored, or the session is ended and protected information is no longer accessible after sign-out.
Normal Flow	1. The user opens Profile. 2. Account and college identity information is displayed. 3. The user edits an allowed field if needed. 4. The system validates and saves the change. 5. The user may return or sign out. 6. Sign-out returns to the authentication screen.
Alternative / Exception Flows	Invalid details are identified without replacing valid information. Protected identity and role fields remain read-only. If saving fails, the previous profile remains. Cancelling sign-out keeps the session active.
Business Rules	Users may change only fields permitted for their role. Role, status, and protected college identity require an authorised administrative process.
Assumptions	The account and related college profile are available.
Includes / Extends	Includes UC-01.

2.2.4 Use Case Diagrams

Figures 2-1 to 2-6 show the main actions available in each application. The paired mobile and desktop diagrams deliberately overlap because both versions use the same approved college records; the desktop diagrams additionally show file, export, and print actions where appropriate.

2.2.4.1 Admin Mobile Application

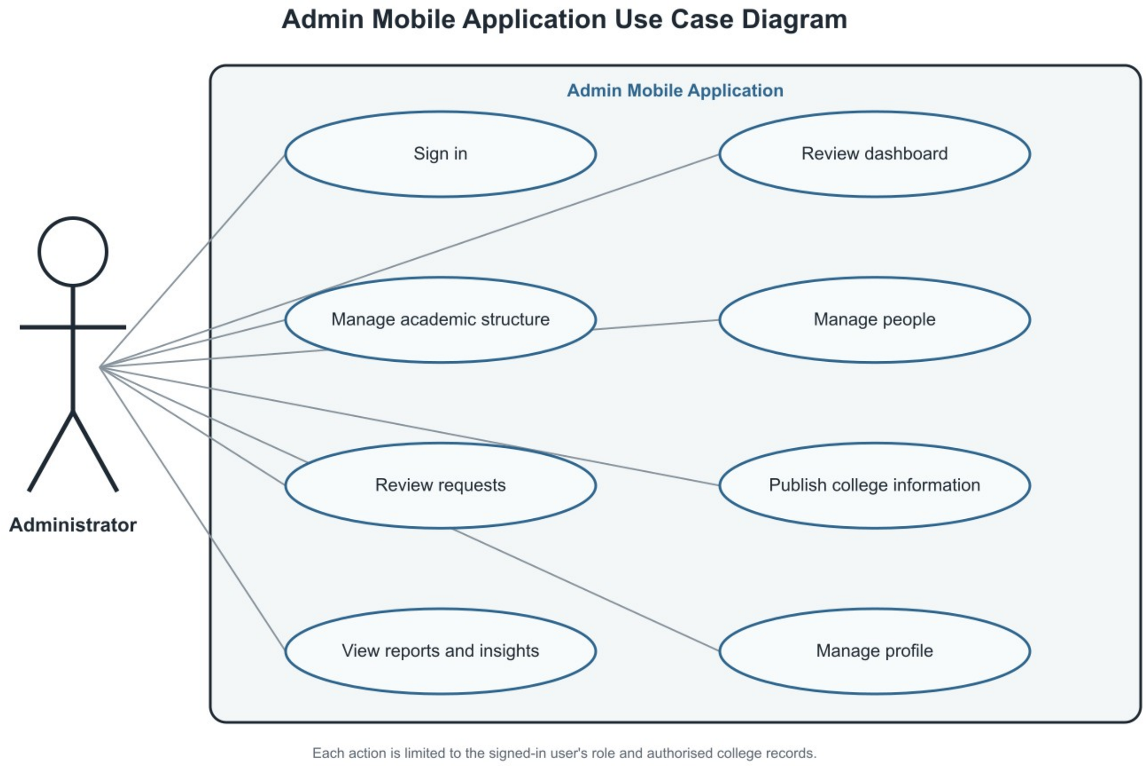

Figure 2-1 Admin Mobile Application use case diagram

2.2.4.2 Admin Desktop Application

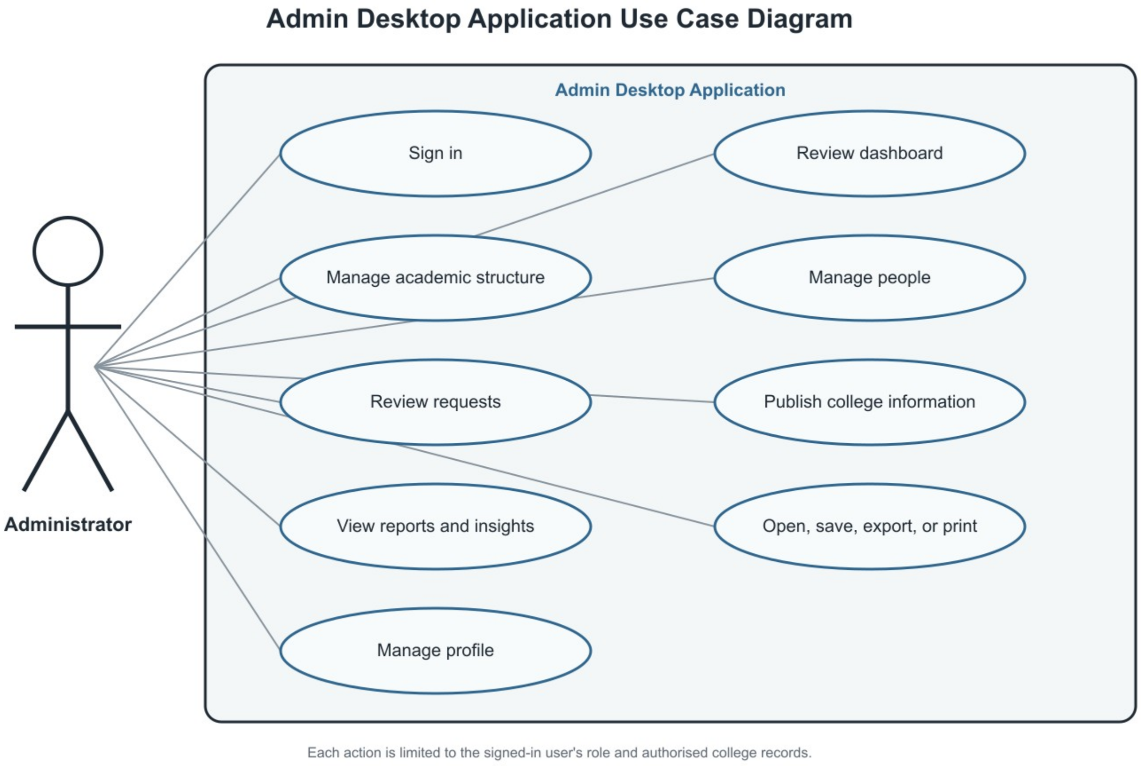

Figure 2-2 Admin Desktop Application use case diagram

2.2.4.3 Teacher Mobile Application

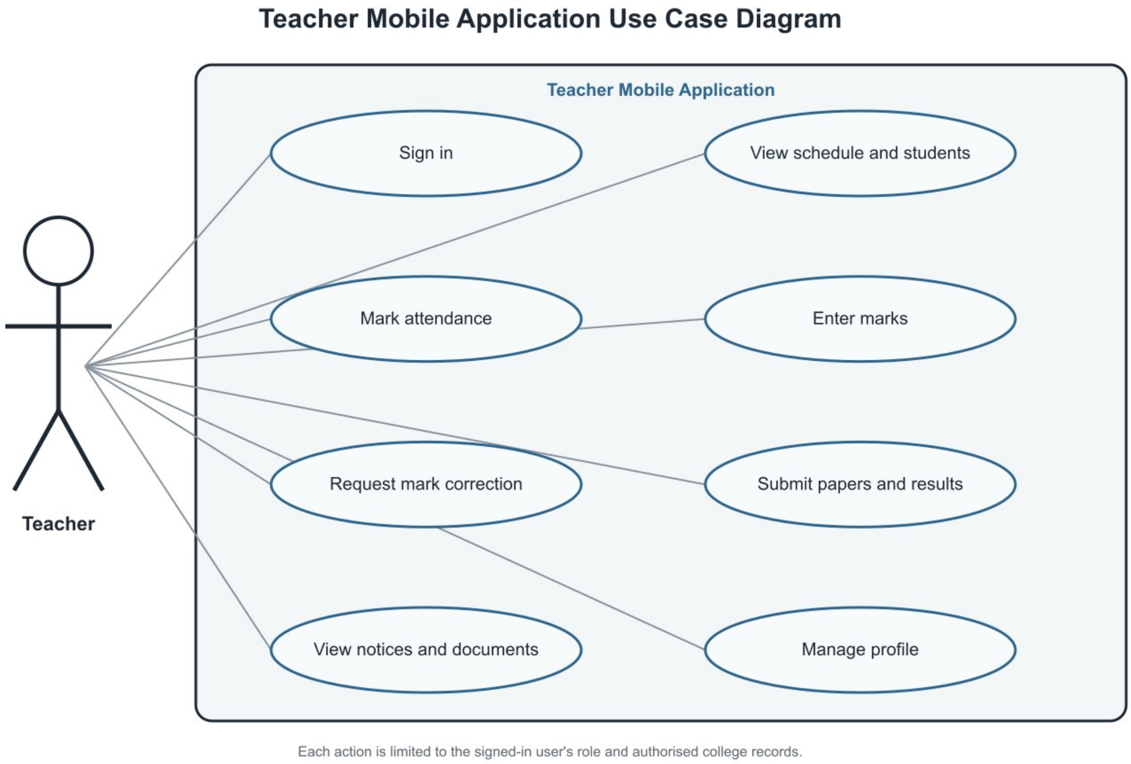

Figure 2-3 Teacher Mobile Application use case diagram

2.2.4.4 Teacher Desktop Application

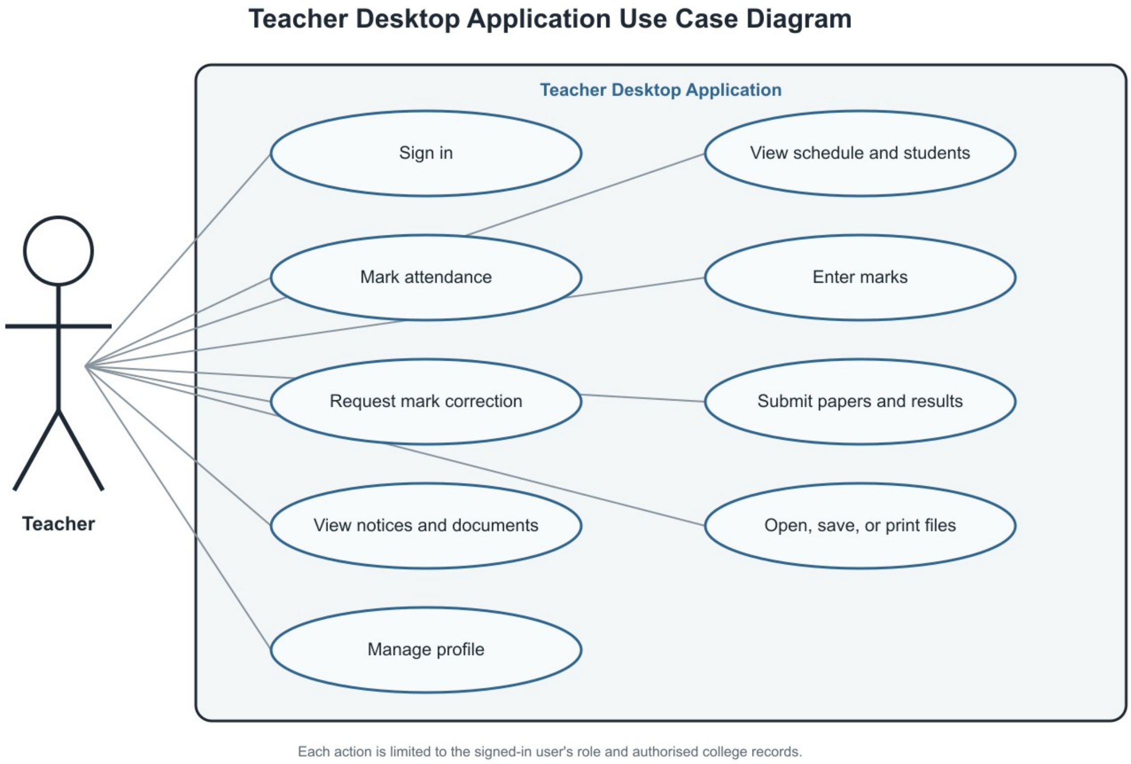

Figure 2-4 Teacher Desktop Application use case diagram

2.2.4.5 Student Mobile Application

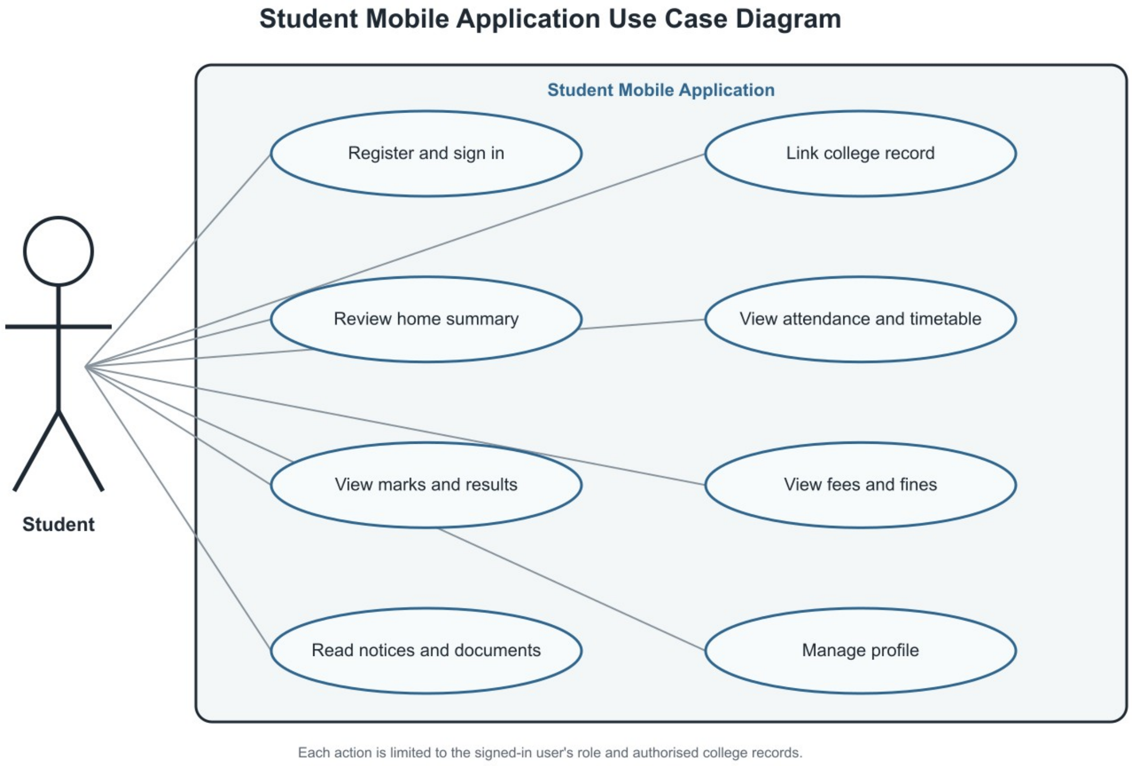

Figure 2-5 Student Mobile Application use case diagram

2.2.4.6 Student Desktop Application

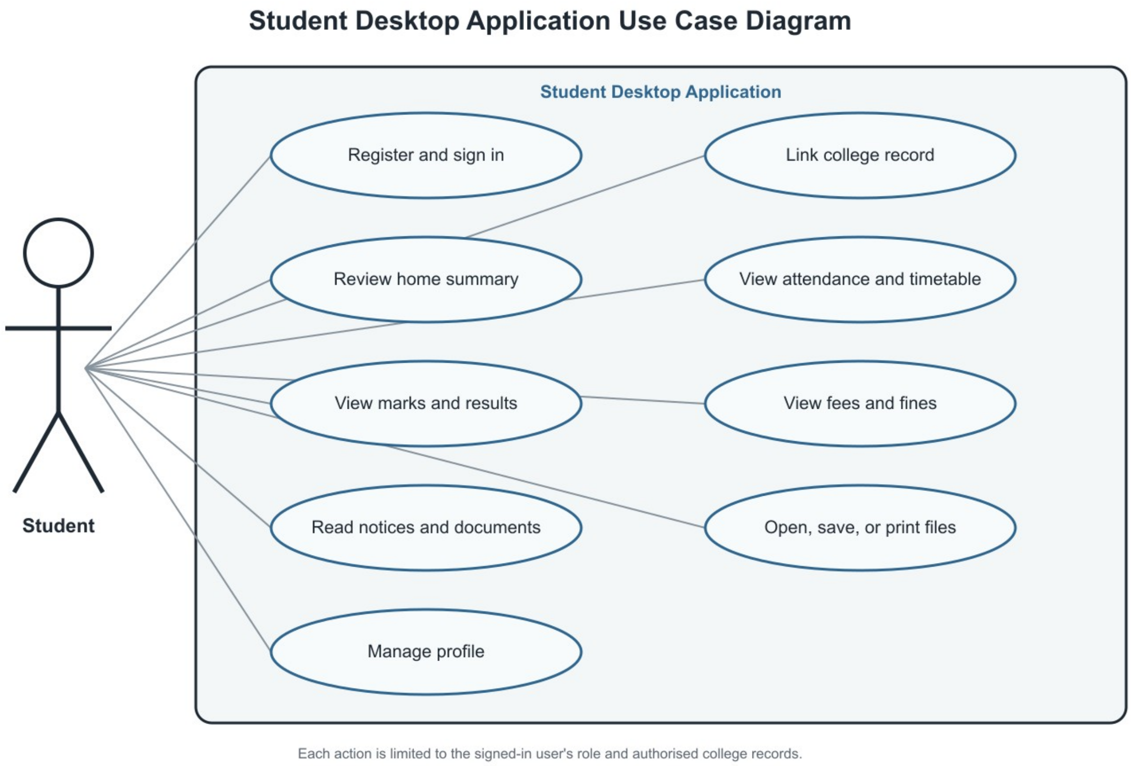

Figure 2-6 Student Desktop Application use case diagram

2.3 Functional Requirements

The following requirements are organised in the same order that information moves through the college: access and people, academic structure, daily teaching records, examinations, fees, communication, reporting, and desktop document actions. Each requirement states an observable service, its source and purpose, the main rule that controls it, its dependencies, and its priority.

2.3.1 FR-1: Sign In and Role Routing

Table 2-22 Description of FR-1

Identifier	FR-1
Title	Sign In and Role Routing
Requirement	A registered administrator, teacher, or student shall be able to sign in through the appropriate mobile or desktop application. The system shall reopen a valid saved session, direct the user to the correct role area, and show a safe message for invalid details, inactive accounts, role mismatch, or connection failure.
Source	All user classes
Rationale	Every protected activity requires a verified identity and role.
Business Rule	A user may enter only the application area assigned to the account's active role.
Dependencies	None
Priority	High

2.3.2 FR-2: Manage Administrator Accounts

Table 2-23 Description of FR-2

Identifier	FR-2
Title	Manage Administrator Accounts
Requirement	An administrator shall be able to view existing administrators and create or maintain another administrator account from both admin applications. Required identity and contact details shall be validated, duplicate accounts shall be refused, and failed creation shall not add a partial account.
Source	College administration
Rationale	The college must be able to share authorised administrative work without external account setup.
Business Rule	Only an active administrator may grant administrative access. Account email and identity must be unique.
Dependencies	FR-1
Priority	High

2.3.3 FR-3: Manage Departments

Table 2-24 Description of FR-3

Identifier	FR-3
Title	Manage Departments
Requirement	An administrator shall be able to create, view, and update departments. A department with missing or duplicate identifying information shall not be saved, and removal shall require confirmation when related academic records may be affected.
Source	College administration
Rationale	Departments are the top-level organisation for sessions, people, and academic records.
Business Rule	Each department must have a unique identifier and a clear display name.
Dependencies	FR-1
Priority	High

2.3.4 FR-4: Manage Academic Sessions

Table 2-25 Description of FR-4

Identifier	FR-4
Title	Manage Academic Sessions
Requirement	An administrator shall be able to create, view, update, and progress academic sessions within a department. Session year, shift, programme details, and semester progression shall be checked before saving, and duplicate or illogical session information shall be rejected.
Source	College administration
Rationale	Students, curriculum, timetables, fees, and results are organised by academic session.
Business Rule	A session belongs to one department and may progress only within its permitted semester range.
Dependencies	FR-3
Priority	High

2.3.5 FR-5: Manage Curriculum

Table 2-26 Description of FR-5

Identifier	FR-5
Title	Manage Curriculum
Requirement	An administrator shall be able to add, view, update, and remove the subjects assigned to each session and semester. Subject code, title, credit information, and semester placement shall be validated, and duplicate subjects within the same curriculum shall be refused.
Source	College administration and teaching staff
Rationale	Attendance, marks, timetables, and results require an agreed subject list.
Business Rule	Every curriculum subject must belong to one session and semester.
Dependencies	FR-4
Priority	High

2.3.6 FR-6: Manage Students and Session Rosters

Table 2-27 Description of FR-6

Identifier	FR-6
Title	Manage Students and Session Rosters
Requirement	An administrator shall be able to add individual students, import a roster, view student cards and profiles, and update valid student details. Duplicate roll numbers, incomplete required identity information, and invalid import rows shall be reported without silently replacing existing students.
Source	College administration and teaching staff
Rationale	Attendance, marks, fees, and account linking depend on an accurate enrolled-student list.
Business Rule	A roll number must identify one student within an academic session. Historical academic records shall not be lost when profile details change.
Dependencies	FR-4
Priority	High

2.3.7 FR-7: Manage Teachers and Permissions

Table 2-28 Description of FR-7

Identifier	FR-7
Title	Manage Teachers and Permissions
Requirement	An administrator shall be able to create teacher accounts, view teacher details, update permitted profile information, grant or remove defined college permissions, and disable, ban, or reactivate an account with confirmation. Invalid or duplicate account details shall not be saved.
Source	College administration
Rationale	Teaching access and selected delegated duties must remain under administrative control.
Business Rule	Only an administrator may change teacher status or permissions. Status changes shall preserve historical teaching records.
Dependencies	FR-1
Priority	High

2.3.8 FR-8: Manage Timetables

Table 2-29 Description of FR-8

Identifier	FR-8
Title	Manage Timetables
Requirement	An administrator or specifically authorised teacher shall be able to create, view, update, and remove class periods in the master and session timetables. Missing assignments and overlapping teacher, room, session, or time selections shall be identified before saving.
Source	College administration and teaching staff
Rationale	A shared timetable coordinates rooms, teachers, subjects, and sessions.
Business Rule	A teacher, room, or session cannot be assigned to overlapping periods.
Dependencies	FR-4, FR-5, FR-7
Priority	High

2.3.9 FR-9: View Personal Schedule

Table 2-30 Description of FR-9

Identifier	FR-9
Title	View Personal Schedule
Requirement	A teacher shall be able to view assigned weekly classes and upcoming work, while a student shall be able to view the timetable of the linked session. If no schedule is published, the application shall show an informative empty state rather than unrelated data.
Source	Teaching staff and students
Rationale	Users need timely access to the schedule relevant to them.
Business Rule	Teachers see assigned periods; students see only the linked session timetable.
Dependencies	FR-8, FR-27
Priority	High

2.3.10 FR-10: Mark Attendance

Table 2-31 Description of FR-10

Identifier	FR-10
Title	Mark Attendance
Requirement	An assigned teacher shall be able to select a class and date, mark every enrolled student present, absent, late, or on leave, review totals, and submit the register. Incomplete entries, duplicate class meetings, unauthorised classes, and failed submission shall be clearly identified.
Source	Teaching staff and college administration
Rationale	Digital attendance replaces manual counting and makes the same record available for review.
Business Rule	One final attendance status is stored per student for each class meeting. Only an assigned teacher may submit it.
Dependencies	FR-6, FR-8
Priority	High

2.3.11 FR-11: View Attendance History

Table 2-32 Description of FR-11

Identifier	FR-11
Title	View Attendance History
Requirement	Teachers shall be able to review attendance history for assigned classes, administrators shall be able to review authorised attendance reports, and students shall be able to view only their personal attendance summary and history. Missing records shall be presented as an empty state, not as zero attendance.
Source	All user classes
Rationale	Attendance information is useful only when the appropriate user can review it in time.
Business Rule	Attendance visibility follows role, teaching assignment, session, and linked student identity.
Dependencies	FR-10, FR-27
Priority	High

2.3.12 FR-12: Enter Assessment Marks

Table 2-33 Description of FR-12

Identifier	FR-12
Title	Enter Assessment Marks
Requirement	An assigned teacher shall be able to enter marks for an approved assessment and submit a complete award list. Negative values, values above the maximum, incomplete required entries, or unauthorised classes shall be rejected with clear guidance. Submitted marks shall become locked against direct editing.
Source	Teaching staff and college administration
Rationale	Marks require accurate entry and protection from unrecorded changes.
Business Rule	A mark must remain within the approved range and becomes locked after first successful submission.
Dependencies	FR-5, FR-6, FR-8
Priority	High

2.3.13 FR-13: Request Mark Correction

Table 2-34 Description of FR-13

Identifier	FR-13
Title	Request Mark Correction
Requirement	A teacher responsible for a locked mark shall be able to propose a corrected value and enter a reason for review. Invalid values, missing reasons, and duplicate pending requests shall not be accepted, and the original mark shall remain unchanged while the request is pending.
Source	Teaching staff and college administration
Rationale	Errors must be correctable without allowing silent changes to submitted marks.
Business Rule	A locked mark changes only after approval by an authorised reviewer.
Dependencies	FR-12
Priority	High

2.3.14 FR-14: Review Mark Correction

Table 2-35 Description of FR-14

Identifier	FR-14
Title	Review Mark Correction
Requirement	An administrator shall be able to review a pending mark-correction request, compare old and proposed values, read the reason, and approve or reject the request. A decided request shall not be decided again, and failure shall leave the original mark and request status unchanged.
Source	College administration
Rationale	Independent review protects the integrity of assessment records.
Business Rule	The requester cannot approve the request. Approval records the decision and applies only the proposed change.
Dependencies	FR-13
Priority	High

2.3.15 FR-15: Record Semester Results

Table 2-36 Description of FR-15

Identifier	FR-15
Title	Record Semester Results
Requirement	An assigned teacher shall be able to record semester results for relevant students, including permitted grade, grade-point, and supplementary-subject information. Missing students, invalid values, and incomplete result entries shall be identified before submission.
Source	Teaching staff and college administration
Rationale	Students and administrators need an agreed semester-level academic outcome.
Business Rule	Results may be recorded only for the assigned session and semester using allowed academic values.
Dependencies	FR-5, FR-6, FR-8
Priority	High

2.3.16 FR-16: View Marks and Semester Results

Table 2-37 Description of FR-16

Identifier	FR-16
Title	View Marks and Semester Results
Requirement	A linked student shall be able to view personal subject marks and semester results, including available progression information. Administrators and authorised teachers shall be able to review results within their permitted scope. Unpublished or missing records shall not be shown as completed results.
Source	Students, teaching staff, and college administration
Rationale	Timely access reduces dependence on manually prepared copies and office inquiries.
Business Rule	Students see only their own results; teachers see assigned records; administrators see authorised college records.
Dependencies	FR-12, FR-15, FR-27
Priority	High

2.3.17 FR-17: Submit Examination Paper

Table 2-38 Description of FR-17

Identifier	FR-17
Title	Submit Examination Paper
Requirement	An assigned teacher shall be able to select the academic context, choose an examination-paper file, and submit it for the relevant subject. Missing files, unsupported file types, unauthorised subjects, and transfer failure shall prevent completion and produce a clear message.
Source	Teaching staff and college administration
Rationale	Paper submission needs a controlled record tied to the correct examination and subject.
Business Rule	A teacher may submit only for assigned academic work and accepted document types.
Dependencies	FR-5, FR-8
Priority	High

2.3.18 FR-18: Build and Publish Datesheet

Table 2-39 Description of FR-18

Identifier	FR-18
Title	Build and Publish Datesheet
Requirement	An administrator or authorised teacher shall be able to create an examination datesheet, enter paper dates and times, review it, and publish it to the intended academic audience. Invalid dates, overlapping papers within the same scope, and incomplete entries shall block publication.
Source	College administration and teaching staff
Rationale	A published datesheet gives all affected users one approved examination schedule.
Business Rule	Only authorised users may publish, and students see only datesheets relevant to their session or audience.
Dependencies	FR-4, FR-5
Priority	High

2.3.19 FR-19: View Datesheet

Table 2-40 Description of FR-19

Identifier	FR-19
Title	View Datesheet
Requirement	Administrators and teachers shall be able to review datesheets within their authority, and students shall be able to view datesheets published for their linked academic session. Draft, unrelated, removed, or unavailable datesheets shall not appear as active schedules.
Source	All user classes
Rationale	Examination schedules must be available beyond the physical noticeboard.
Business Rule	Only published datesheets are visible to students.
Dependencies	FR-18, FR-27
Priority	High

2.3.20 FR-20: Manage Session Fee Structure

Table 2-41 Description of FR-20

Identifier	FR-20
Title	Manage Session Fee Structure
Requirement	An administrator shall be able to define and update fee heads, amounts, due information, frequency, and payment notes for an academic session. Negative amounts, missing required fee information, and invalid dates shall be rejected before saving.
Source	College administration and accounts office
Rationale	Students require an agreed session fee structure before a challan can be presented.
Business Rule	A fee structure belongs to one session and is changed only by an administrator.
Dependencies	FR-4
Priority	High

2.3.21 FR-21: View Fee Challan

Table 2-42 Description of FR-21

Identifier	FR-21
Title	View Fee Challan
Requirement	A linked student shall be able to view personal fee-challan information prepared from the linked session's current fee structure. Missing fee data shall produce an informative message, and opening or printing a challan shall not record payment.
Source	Students and accounts office
Rationale	Routine fee information should be available without an information-only office visit.
Business Rule	Students see only the fee information for their linked record and session.
Dependencies	FR-20, FR-27
Priority	High

2.3.22 FR-22: Issue Student Fine

Table 2-43 Description of FR-22

Identifier	FR-22
Title	Issue Student Fine
Requirement	An authorised administrator or teacher shall be able to issue a fine to an identified student with category, amount, date, and reason. Missing student information, non-positive amounts, and incomplete reasons shall be rejected.
Source	College administration and teaching staff
Rationale	A fine requires a clear, reviewable record rather than an informal note.
Business Rule	Every fine must identify the issuer, student, amount, category, and reason.
Dependencies	FR-6
Priority	Medium

2.3.23 FR-23: View Student Fines

Table 2-44 Description of FR-23

Identifier	FR-23
Title	View Student Fines
Requirement	A linked student shall be able to view personal fines, while administrators and authorised teachers may view fines within their permitted scope. If no fine exists, the system shall state this clearly rather than displaying an unexplained zero.
Source	Students, teaching staff, and college administration
Rationale	Users need a shared understanding of recorded fines and their reasons.
Business Rule	Students see only fines attached to their linked record.
Dependencies	FR-22, FR-27
Priority	Medium

2.3.24 FR-24: Manage Calendar Events

Table 2-45 Description of FR-24

Identifier	FR-24
Title	Manage Calendar Events
Requirement	An administrator shall be able to create, update, publish, and remove college calendar events with title, dates, category, audience, and details. Invalid date ranges, missing titles, or incomplete audience information shall prevent publication.
Source	College administration
Rationale	Holidays, examinations, deadlines, and events need one current college calendar.
Business Rule	Users see only published events addressed to their role or general college audience.
Dependencies	FR-1
Priority	Medium

2.3.25 FR-25: Publish College Documents

Table 2-46 Description of FR-25

Identifier	FR-25
Title	Publish College Documents
Requirement	An administrator shall be able to create or upload a college document, assign its type and audience, review it, and publish it. Missing content, unsupported files, invalid audience selection, or failed upload shall leave the document unpublished.
Source	College administration
Rationale	Rules, reports, prospectuses, and other approved material need controlled distribution.
Business Rule	Only authorised publishers may publish, and a document must have content, type, audience, and publication status.
Dependencies	FR-1
Priority	Medium

2.3.26 FR-26: View and Download Documents

Table 2-47 Description of FR-26

Identifier	FR-26
Title	View and Download Documents
Requirement	A signed-in user shall be able to list, open, and where permitted download published documents addressed to the user's role or academic group. Removed, unsupported, or unavailable files shall produce a clear message and shall not expose internal storage details.
Source	All user classes
Rationale	College documents should be accessible without depending on physical copies.
Business Rule	Users may access only published documents for their authorised audience.
Dependencies	FR-25
Priority	Medium

2.3.27 FR-27: Submit Student Link Request

Table 2-48 Description of FR-27

Identifier	FR-27
Title	Submit Student Link Request
Requirement	An unlinked student shall be able to submit a request containing roll number and required identity information to claim an existing college record. Unknown records, mismatched information, duplicate pending requests, and already-linked records shall be identified before submission.
Source	Students and college administration
Rationale	Self-registration must not grant access to another student's academic information.
Business Rule	One account may link to one student record, and each student record may have one approved account.
Dependencies	FR-1, FR-6
Priority	High

2.3.28 FR-28: Review Student Link Request

Table 2-49 Description of FR-28

Identifier	FR-28
Title	Review Student Link Request
Requirement	An administrator or specifically authorised teacher shall be able to compare a pending student link request with the college roster and approve or reject it. A decided request shall not be processed again, and a failed decision shall leave it pending.
Source	College administration and teaching staff
Rationale	Independent review protects student records from incorrect account linking.
Business Rule	Approval links one account to one matching student record. Rejection records the decision without linking.
Dependencies	FR-27
Priority	High

2.3.29 FR-29: Publish Notifications

Table 2-50 Description of FR-29

Identifier	FR-29
Title	Publish Notifications
Requirement	An administrator or authorised teacher shall be able to create a notification, select its target audience, review the message, and publish it. Empty messages, missing titles, and invalid audience selections shall be rejected.
Source	College administration and teaching staff
Rationale	Important updates require timely delivery to the intended users.
Business Rule	Only authorised users may publish, and every notification must identify an audience.
Dependencies	FR-1
Priority	Medium

2.3.30 FR-30: View Notifications

Table 2-51 Description of FR-30

Identifier	FR-30
Title	View Notifications
Requirement	A signed-in user shall be able to view notifications addressed to the user's role or academic group, open their details, and see an unread count where available. Unrelated notifications shall not be displayed, and loading failure shall offer retry.
Source	All user classes
Rationale	Users need one consistent place for current college alerts.
Business Rule	Notification visibility follows the published target audience.
Dependencies	FR-29
Priority	Medium

2.3.31 FR-31: View Insights and Reports

Table 2-52 Description of FR-31

Identifier	FR-31
Title	View Insights and Reports
Requirement	Administrators shall be able to view college-wide summaries, while teachers shall be able to view summaries limited to assigned academic work. Available filters shall support departments, sessions, semesters, subjects, or periods where appropriate. No-data and loading states shall be explained clearly.
Source	College administration and teaching staff
Rationale	Current summaries help users identify attendance and academic issues without manual compilation.
Business Rule	Report scope must follow the signed-in user's role and assignments.
Dependencies	FR-4, FR-6, FR-10, FR-12, FR-15
Priority	Medium

2.3.32 FR-32: View and Maintain Personal Profile

Table 2-53 Description of FR-32

Identifier	FR-32
Title	View and Maintain Personal Profile
Requirement	A signed-in user shall be able to view the personal account and college profile information available to the role, update permitted fields, and sign out. Invalid updates shall not replace valid information, and protected identity or role fields shall remain read-only unless changed by an authorised administrator.
Source	All user classes
Rationale	Users need to confirm account identity and maintain allowed contact information.
Business Rule	Users may change only profile fields permitted for their role.
Dependencies	FR-1
Priority	Medium

2.3.33 FR-33: Use Desktop File and Print Services

Table 2-54 Description of FR-33

Identifier	FR-33
Title	Use Desktop File and Print Services
Requirement	The Admin Desktop, Teacher Desktop, and Student Desktop applications shall provide role-appropriate file selection, save-location selection, document opening, exporting, printing, and supported sharing actions. Cancellation, unavailable applications or printers, missing permissions, and unsupported files shall return a clear result without changing college data.
Source	Desktop user classes
Rationale	Windows users expect longer office and document tasks to use familiar operating-system services.
Business Rule	Desktop services shall not bypass application access rules. Native Windows windows are outside the application's visual design.
Dependencies	FR-1 and the requirement owning the selected record or document
Priority	Medium

2.4 Non-Functional Requirements

The following quality requirements apply across the six applications. They are stated as observable targets so they can be checked during testing rather than described only as general intentions.

2.4.1 Reliability

Table 2-55 Reliability Requirements

Identifier	Requirement Description	Applicable Applications	Priority
NFR-R1	After the system confirms that a record was saved, the same record shall remain available after the application is closed and reopened.	All six	High
NFR-R2	A temporary loading or communication failure shall not close the application or discard already displayed information; the user shall be offered a clear retry action.	All six	High
NFR-R3	When an administrator enters the main application for the first time in a session, the principal departments, teachers, students, sessions, requests, and summary information shall begin loading automatically without requiring a page-by-page refresh.	Admin Mobile and Admin Desktop	High
NFR-R4	A valid signed-in session shall be restored when the application is reopened, while an expired or signed-out session shall return the user to sign-in without showing protected data.	All six	High
NFR-R5	Every failed create, update, upload, approval, or delete action shall leave the previous valid record unchanged unless the system confirms successful completion.	All six	High

2.4.2 Usability

Table 2-56 Usability Requirements

Identifier	Requirement Description	Applicable Applications	Priority
NFR-U1	A user shall be able to reach each primary role destination within three navigation selections from the role's home area.	All six	High
NFR-U2	Required fields, invalid values, and unavailable actions shall be explained beside the form or action in plain language before the user is asked to repeat the whole task.	All six	High
NFR-U3	Destructive or high-impact actions shall name the affected record and require confirmation before completion.	Admin and Teacher applications	High
NFR-U4	Mobile controls used for primary actions shall provide touch areas of at least 48 by 48 density-independent pixels wherever the screen permits.	Three mobile applications	Medium
NFR-U5	The mobile and desktop versions of the same role shall use the same labels, order of major destinations, status meanings, and form rules.	All six	High
NFR-U6	Every main screen shall provide a clear title, back behaviour, refresh access, loading feedback, and an understandable empty or error state where relevant.	All six	High

2.4.3 Performance

Table 2-57 Performance Requirements

Identifier	Requirement Description	Applicable Applications	Priority
NFR-P1	Under a stable college internet connection, the first load of a normal list or summary shall complete within five seconds for the expected single-college data volume.	All six	High
NFR-P2	A screen that already has valid local or recently loaded information shall display that information within two seconds while checking for updates in the background.	All six	Medium
NFR-P3	An attendance roster for a normal class shall become ready for marking within three seconds after the teacher selects the class under stable conditions.	Teacher Mobile and Teacher Desktop	High
NFR-P4	The admin start-up data load shall begin immediately after role resolution and shall not prevent the dashboard from showing progress or available summary information.	Admin Mobile and Admin Desktop	High
NFR-P5	Operations that take longer than one second shall show visible progress and shall prevent accidental duplicate submission while the operation is active.	All six	High

2.4.4 Security

Table 2-58 Security Requirements

Identifier	**Requirement Description**	**Applicable Applications**	**Priority**
NFR-S1	A user shall be able to view and perform only the actions assigned to the authenticated role and current account status.	All six	High
NFR-S2	A student shall be able to view only the personal records linked to that student's approved college identity and the general information published to the relevant audience.	Student Mobile and Student Desktop	High
NFR-S3	A teacher shall be able to view or change only the students, classes, marks, attendance, and delegated duties connected with assigned teaching work and permissions.	Teacher Mobile and Teacher Desktop	High
NFR-S4	Error messages, logs visible to users, and support text shall not reveal passwords, access tokens, personal secrets, database addresses, or complete service request links.	All six	High
NFR-S5	After sign-out, protected screens and personal records shall not remain accessible through back navigation or reopening the application.	All six	High
NFR-S6	Approval decisions and other sensitive record changes shall preserve the responsible account, decision or action, and time needed for later review.	Admin and authorised Teacher applications	High

2.5 External Interface Requirements

External interface requirements define how the six applications present information to users and interact with devices, operating systems, online services, email, and files. Detailed screen design is presented separately in Chapter 3.

2.5.1 User Interface Requirements

Table 2-59 User Interface Requirements

Application	Required Interface Characteristics
Admin Mobile	Portrait-first dashboard and five primary navigation areas; compact cards for college totals and records; short forms and confirmation windows suitable for touch use.
Admin Desktop	Responsive administrative workspace using wider card, list, and table arrangements; clear support for file, export, and print actions without changing role rules.
Teacher Mobile	Quick access to today's schedule, attendance, examinations, marks, and menu actions; class and student controls suitable for use during a lesson.
Teacher Desktop	Wider attendance, marks, schedule, and student views that preserve the same labels, status meanings, validation, and navigation sequence as the mobile version.
Student Mobile	Simple personal home view and five main areas for attendance, examinations, timetable, and additional information; forms limited to registration, linking, and permitted profile updates.
Student Desktop	Readable personal academic, fee, notice, and document views with appropriate open, save, and print actions; no additional access beyond the student mobile application.
Common interface	Every main screen shall use the GGC-MBD title treatment, clear back behaviour, refresh access, notification entry where relevant, consistent status colours, readable typography, and plain loading, empty, validation, and error messages.

2.5.2 Software Interfaces

Table 2-60 Software Interfaces

Software Interface	Version / Platform	Required Interaction
Supabase services	Kotlin client 3.1.4	Provide account sign-in, shared college records, uploaded documents, notifications, and authorised administrative operations to all six applications.
PostgreSQL database	Supabase managed	Store consistent administrative, academic, fee, approval, communication, and reporting information for the college.
Android operating system	Android 7.0 or later	Run the three mobile applications and provide device storage selection, application lifecycle, and secure internet access.
Windows operating system	Windows 10 or later	Run the three desktop applications and provide native file selection, save location, document opening, printing, and sharing services.
Email delivery service	Configured through the online account service	Deliver account verification and password-recovery messages without requiring a separate college email application.

2.5.3 Hardware Interfaces

Table 2-61 Hardware Interfaces

Hardware	Applications	Interface Requirement
Android phone or tablet	Admin Mobile, Teacher Mobile, Student Mobile	Provide a touch display, sufficient storage for the application and selected documents, and network access. No camera, biometric reader, or specialist sensor is required.
Windows desktop or laptop	Admin Desktop, Teacher Desktop, Student Desktop	Provide keyboard and pointing-device input, a display suitable for responsive application windows, storage access, and network access.
Printer	Three desktop applications where Print is available	Receive a user-approved print job through the Windows print window. The application shall remain usable when no printer is installed.
Local or removable storage	Applications that open, upload, save, or download files	Provide a user-selected file or destination. The application shall not assume access to a fixed personal folder.

2.5.4 Communications Interfaces

Table 2-62 Communications Interfaces

Communication	Requirement
Application data exchange	All six applications shall exchange sign-in, college record, and file information through an encrypted internet connection. Interrupted communication shall produce a retryable state rather than an incomplete confirmed action.
Email	Registration verification and password recovery shall use the registered email address. Invalid or unavailable addresses shall be reported without revealing whether another user's private account exists beyond the information needed for the current request.
File transfer	Uploads and downloads shall preserve the selected file's content and report progress, cancellation, unsupported type, or failure clearly.
Notifications	Published notices shall be delivered or displayed only to the selected role or academic audience and shall remain available in the appropriate application's notification list.

2.6 Summary

This chapter translated the project overview into a complete and reviewable set of requirements. Six user classes represent administrators, teachers, and students on mobile and desktop devices. Twenty fully dressed use cases and six diagrams describe how those users carry out their main work, while thirty-three functional requirements define the services and failure behaviour expected from the system. Reliability, usability, performance, security, and external interface requirements set measurable conditions for all six applications. These requirements provide the reference point for the system design in Chapter 3 and for the testing and evaluation presented later in the report.

Chapter 3

Design and Architecture

3. System Design

The College Management System is a family of six applications rather than one large interface. Administrators, teachers, and students each have a mobile application and a desktop application suited to their work. These applications share the same college records and business rules, so an action completed on one device is reflected wherever the same authorised information is shown. The system also depends on managed account, database, document-storage, and email services.

This chapter explains how those parts fit together. It covers the design constraints, the main information objects, system architecture, data organisation, user interfaces, important interactions, workflow states, and the decisions that kept mobile and desktop behaviour consistent.

3.1 Design Considerations

The design was shaped by the college environment: users may move between office computers and personal phones, internet quality is not always consistent, and access to student information must follow each person's role. The main considerations and the response built into the design are listed below.

Table 3-1 Design Considerations, Constraints, and Risks

Category	Consideration	Design Response
Assumption	Each user has one authorised account and uses the application intended for the assigned role.	The sign-in process verifies the account, role, and status before opening a workspace.
Assumption	Academic sessions, subjects, teachers, and student rosters are maintained by authorised college staff.	Dependent features use these approved records rather than asking users to enter the same information again.
Dependency	All six applications rely on the same online college information service.	Shared validation and access rules keep records consistent regardless of the device used.
Dependency	Documents, password recovery, and account verification rely on managed file and email services.	A failed external operation is reported clearly and does not create a false success state.
Constraint	Mobile screens have limited space, while desktop users expect wider working areas and native file or print actions.	The information and actions remain the same, but each presentation is arranged for its device.
Limitation	Most current records require an internet connection to be confirmed and shared with other users.	Previously loaded information may remain visible where safe, while changes wait for a confirmed connection.
Risk	An incorrect role or permission rule could expose confidential academic information.	Access is checked centrally for every protected record, not only hidden in the visible interface.
Risk	Two users may update related academic information close together.	Important updates are validated against the latest record and rejected when the context is no longer valid.
Risk	A document upload or download may be interrupted.	The system reports progress, cancellation, and failure, and keeps the college record unchanged when publication is incomplete.
Risk	Large initial data loads could delay the administrative home screen.	Essential administrative information is prepared at startup and later screens reuse it instead of requiring manual refreshes.

3.2 Design Models

Three complementary model types are used in this chapter. The domain relationship model explains the main information objects. Interaction diagrams show the order in which a user, an application, and the shared services cooperate. State-transition diagrams show how approval and publication records move from one condition to another. Together they describe structure, behaviour, and change without depending on a particular programming language.

3.2.1 Domain and Class Relationship Model

The main domain objects represent people, academic structure, teaching activity, published information, and financial notices. A department contains academic sessions; sessions contain subjects and enrolled students; teachers are assigned through timetable periods; and attendance, marks, and results connect a student to that teaching context. User accounts provide authorised access to the role-specific view of these objects.

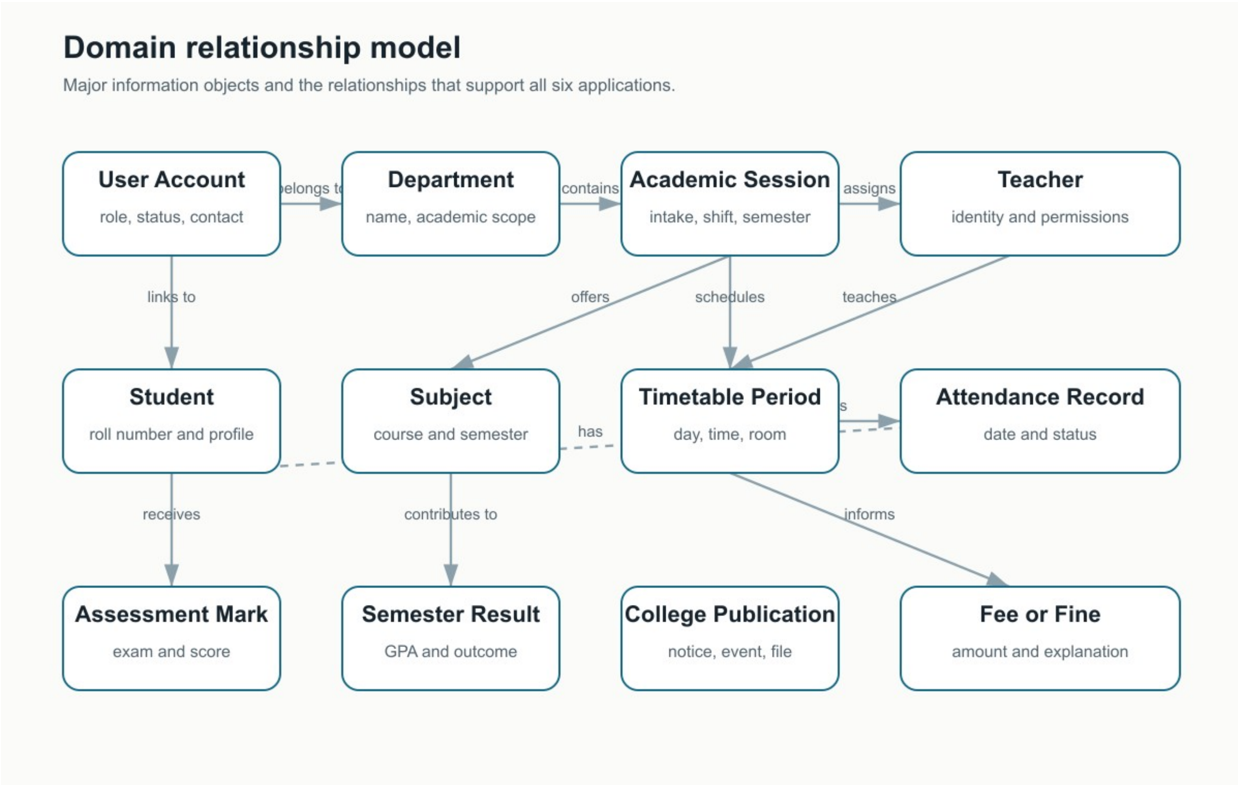

Figure 3-1 Domain and class relationship model

3.2.1.1 Core Object Responsibilities

Table 3-2 Core Object Responsibilities

Object Group	Key Information	Main Operations and Inputs
Identity	Account role, status, contact details, and the approved link to a college person record.	Verify an account using sign-in details; open a workspace using role and status; link a student using account and roster identity; sign out the current account.
Academic structure	Departments, sessions, semesters, subjects, rosters, and timetable periods.	Create a session using department, intake, and shift; assign a subject using session and semester; schedule a period using class, teacher, time, and room.
Teaching records	Attendance, assessment marks, correction requests, and semester outcomes.	Record attendance using class, date, roster, and statuses; enter marks using subject, assessment, and scores; review a correction using request and decision.
College information	Events, datesheets, documents, examination papers, and notifications.	Prepare information using content and audience; publish after validation; open, save, print, or archive using the selected record and device action.
Financial information	Session fee structures, fee headings, exceptions, and student fines.	Prepare a challan view using session and student; issue a fine using student, amount, category, and reason; display financial notices without processing payment.
Reporting	Approved attendance, marks, results, session totals, and progress summaries.	Build a summary using role, academic scope, and selected filters; return only information permitted for that role.

3.2.2 Interaction Models

Interaction models are provided in Section 3.6 for sign-in, administrative publication, teacher attendance and assessment, student registration and linking, and document publication and opening. Each model names the participant, the selected application, and the shared service involved, so the same logic can be understood for both mobile and desktop.

3.2.3 State-Transition Models

State models are also provided in Section 3.6 for student account linking, mark-correction review, and publishable college information. These are the workflows where the current state controls which action is allowed next and what another user can see.

3.3 Architectural Design

The system uses a layered client-server design. The six applications present role-specific workspaces, while shared services apply the same validation, access rules, and record operations. Managed online services hold accounts, college data, and documents. This separation allows each application to remain focused without creating six different versions of the college's rules.

Each application also keeps a role-specific local copy of the records it is authorised to see. On a first use, the application prepares that local copy from the online service. Later refreshes request only records that changed since the last successful refresh, then update the local copy as one operation. This gives the user a useful recent view during weak connectivity without creating a second source of truth: the online college database remains authoritative.

3.3.1 System Context

Administrators and teachers may create or update information within their authority. Students mainly view personal and published information, with limited actions for registration, linking, and profile maintenance. Mobile and desktop applications exchange information with the same central service, so switching devices does not create a separate record.

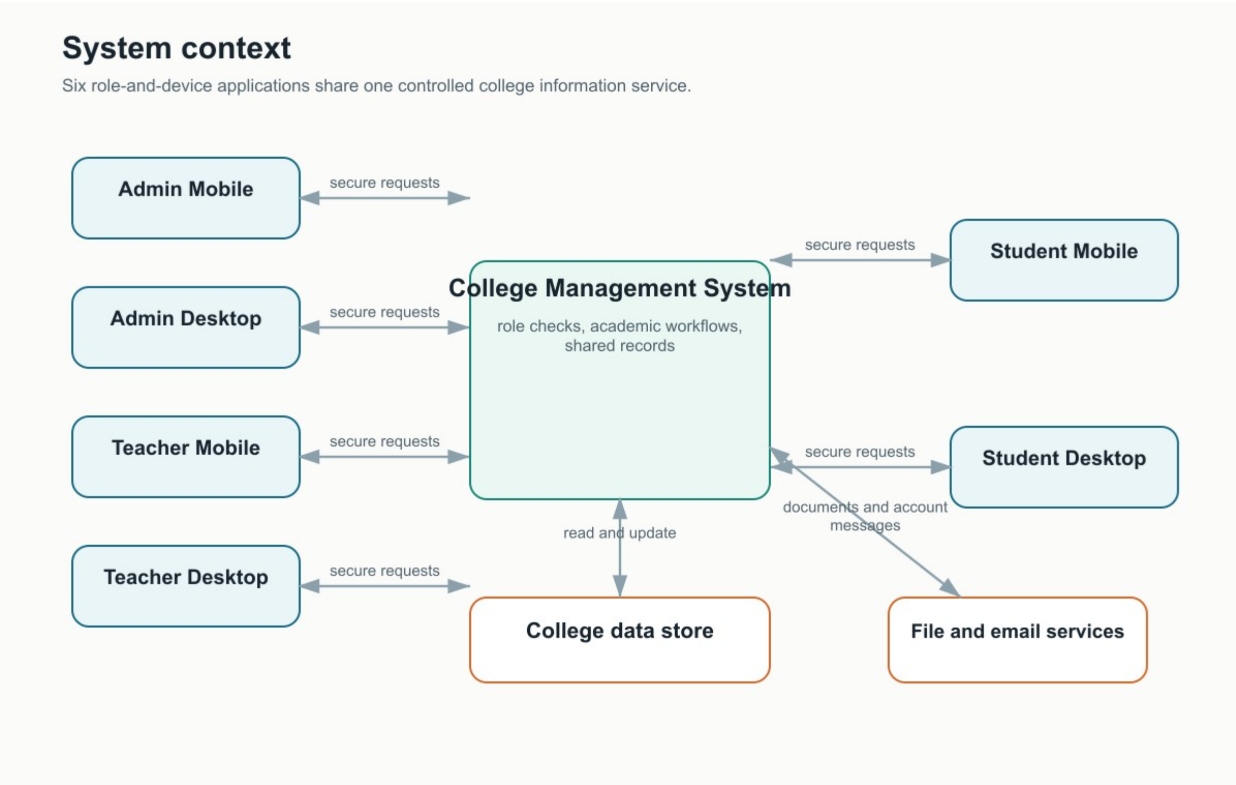

Figure 3-2 System context for all six applications

3.3.2 Component Structure

The component structure separates application shells, role workspaces, shared system services, and managed online services. Application shells handle navigation and device presentation. Role workspaces organise the tasks for administrators, teachers, and students. Shared services apply identity, academic, publishing, and reporting rules before communicating with the online services.

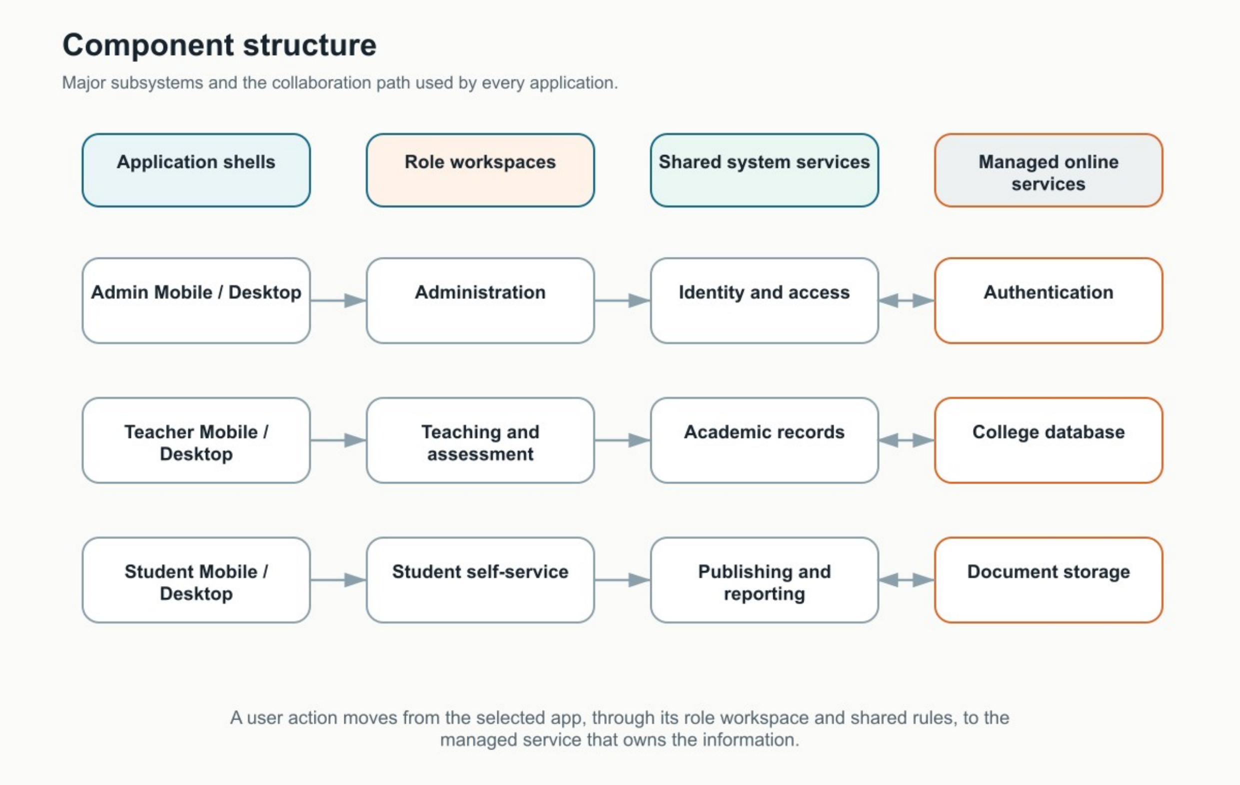

Figure 3-3 UML component structure and subsystem collaboration

3.3.3 Architecture Style and Module Mapping

A layered pattern was selected because responsibilities can be explained and changed independently: a screen presents information, a role workflow decides what the user is trying to do, shared rules validate the action, and the information service stores or retrieves the result. The client-server boundary is between the applications and the managed online services.

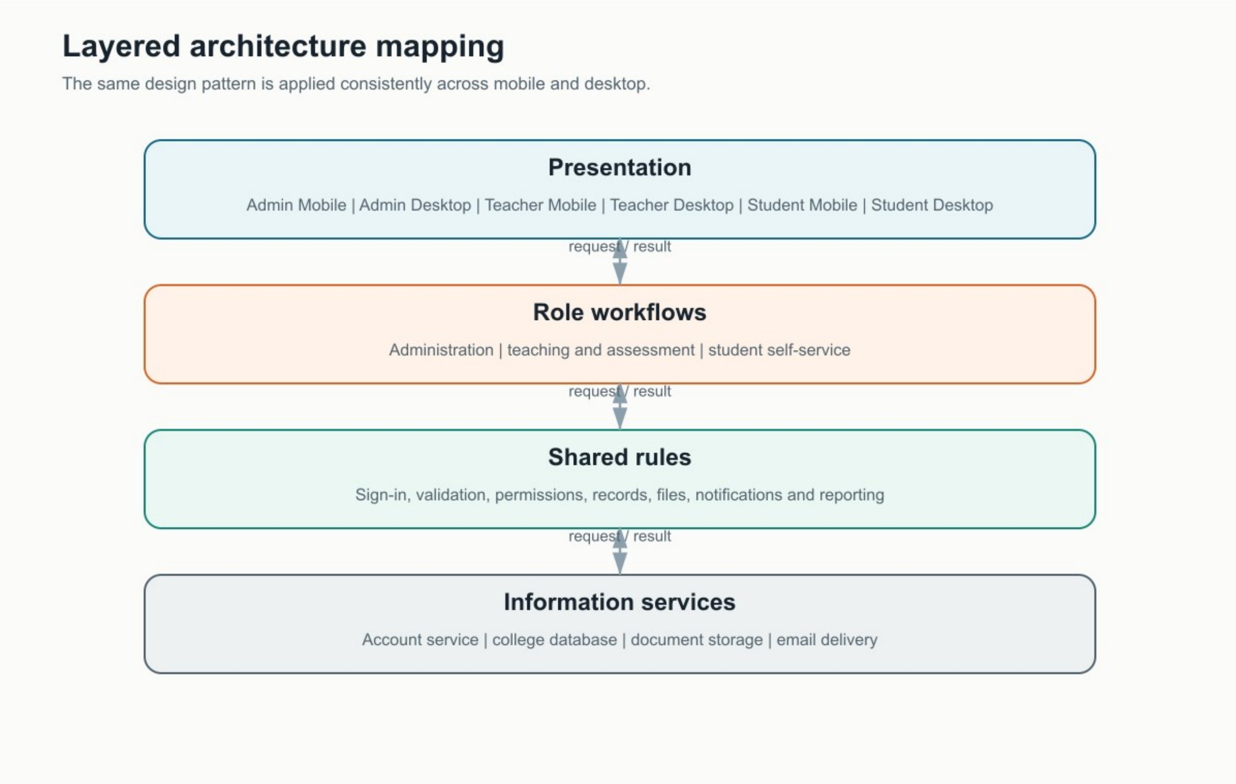

Figure 3-4 Layered architecture and module mapping

Table 3-3 Architecture Layer Mapping

Layer	Included Applications or Components	Responsibility
Presentation	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop	Displays role-appropriate screens, accepts input, shows progress and feedback, and adapts the arrangement to phone or Windows.
Role workflows	Administration, teaching and assessment, student self-service	Coordinates the steps and decisions for the selected role without changing shared business rules.
Shared rules	Identity, validation, permissions, academic records, publishing, files, notifications, and reporting	Checks information and access consistently before any confirmed operation.
Information services	Account service, college database, document storage, and email delivery	Stores authoritative information and returns controlled results to authorised applications.

3.4 Data Design

Information is organised around stable college records rather than around individual screens. Departments and academic sessions provide the structure; people records identify administrators, teachers, and students; teaching records capture timetables, attendance, marks, and results; publication records distribute datesheets, events, documents, and notifications; and financial records explain fees and fines. All six applications read or update the same authorised record, which prevents a mobile and desktop version from becoming separate sources of truth.

Relationships are kept explicit. A student belongs to an academic session, a subject is offered in a semester, and every attendance or assessment entry carries enough academic context to identify the class. Published information records its intended audience, while approval records retain both the requested change and the decision.

Operational records also carry creation and update information so that the system can identify the newest authorised version of a row during refresh. Removed records are retained by the service as controlled deletion markers long enough for other devices to remove their local copy. Normal lists do not show removed records. In the Admin applications, an information control on each persisted record can show its creation and update details without exposing deletion details to routine users.

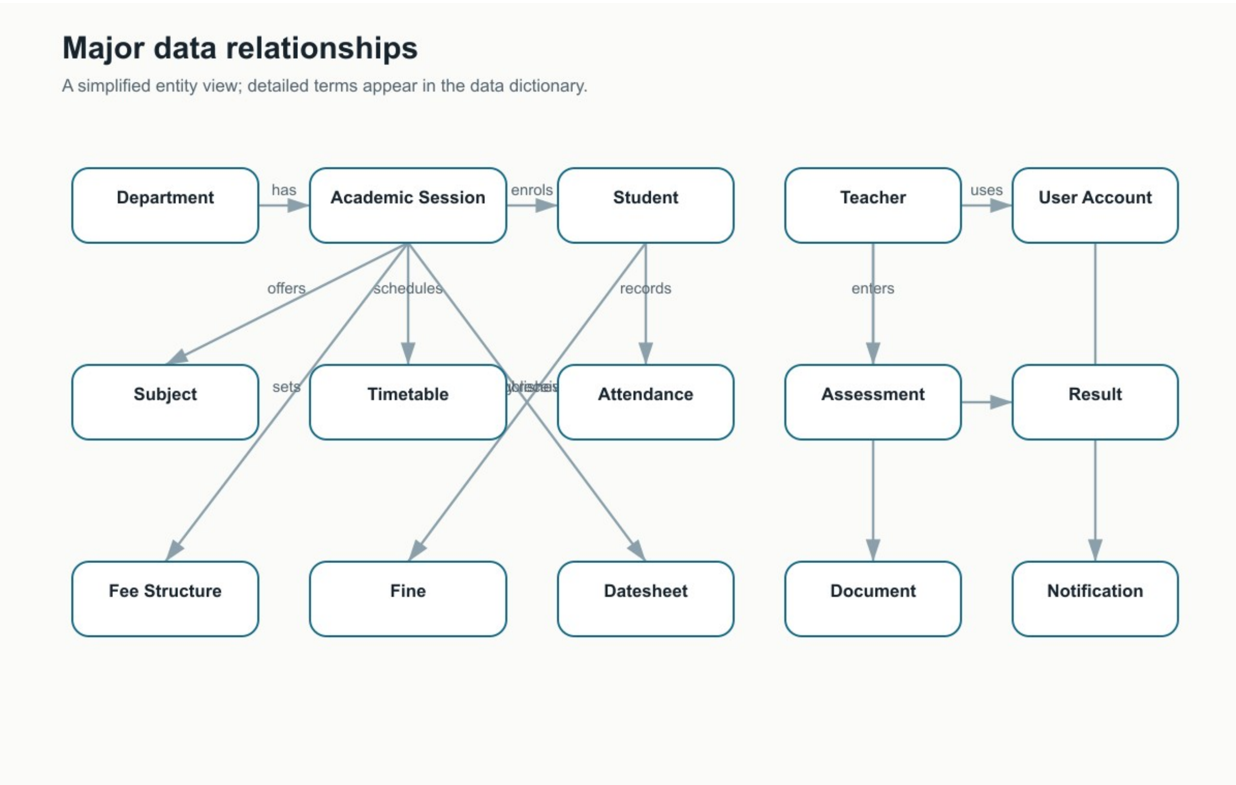

Figure 3-5 Major data relationships

3.4.1 Data Dictionary

The dictionary lists the major information entities and calculated summaries in alphabetical order. Each description identifies the information type and its purpose. Field-level implementation details are intentionally omitted because this chapter explains the system design rather than implementation.

Table 3-4 Alphabetical Data Dictionary

Terminology	Description, Type, and Role
Academic Session	Record. Represents one department intake and shift. It keeps the current semester, session status, and the structure used to connect students, subjects, fees, and timetables.
Administrator Account	Identity record. Represents an authorised college administrator and supports administrative access, profile details, status control, and responsibility for approved actions.
Assessment Mark	Academic record. Stores a student's score for one subject and assessment type, together with the maximum score and the academic context in which it was recorded.
Attendance Record	Academic record. Stores one student's present, absent, or leave status for a dated class meeting. It may also indicate late arrival and an authorised note.
Attendance Summary	Calculated summary. Combines class attendance records into present, absent, leave, total-marked, and percentage values for reporting.
At-Risk Student Summary	Calculated summary. Identifies students whose attendance or academic standing falls below the college's review threshold so authorised staff can follow up.
Calendar Event	Publication record. Describes a holiday, college event, examination period, or deadline, including dates and the audience allowed to view it.
Datesheet	Publication record. Represents an examination schedule for a department, session, or semester and controls whether the schedule is still a draft or visible to users.
Datesheet Slot	Child record. Describes one examination within a datesheet, including subject, date, time, room, and invigilation information.
Department	Master record. Represents an academic department and provides the top-level grouping for sessions, programmes, teachers, and reports.
Document	Publication record. Represents a college file or written notice, its category, description, audience, publication status, and stored-file reference where applicable.
Examination Paper Submission	Academic document record. Connects a teacher's submitted examination paper with its session, semester, subject, file, submission time, and review status.
Examination Statistics	Calculated summary. Provides entered-mark counts, average and range, pass rate, and other authorised comparisons for an assessment.
Fee Exception	Financial record. Stores an authorised adjustment or exception that applies to an individual student's standard session fee information.
Fee Heading	Financial child record. Describes one line of a session fee structure, such as tuition or laboratory charges, with its label and amount.
Fee Structure	Financial record. Defines the expected charges, due information, payment note, and annual or semester schedule for one academic session.
Fine	Financial notice. Records an informational fine issued to a student, including category, amount, reason, date, and issuing authority. It does not represent online payment.

Terminology	Description, Type, and Role
Mark Correction Request	Approval record. Keeps the recorded score, requested score, teacher's reason, review state, reviewer decision, and decision time.
Notification	Communication record. Stores a title, message, priority, publication time, and role, department, or session audience.
Period-Session Assignment	Relationship record. Connects a timetable period to one or more academic sessions when a class is shared.
Semester Result	Academic outcome. Stores a student's semester grade point average, cumulative average, outcome, class position, and any subjects that must be repeated.
Semester Term	Academic date record. Defines the opening and closing dates for a semester within a particular academic session.
Session Overview	Calculated summary. Combines department, shift, semester, enrolment, attendance, and academic standing into one authorised session-level view.
Student	Person and enrolment record. Keeps roll number, name, contact and guardian information, enrolment status, academic standing, and the approved account link.
Student Account Link Request	Approval record. Connects a registered account request with claimed student identity, current review state, reviewer information, and the final decision.
Student Progression Summary	Calculated summary. Presents semester-by-semester academic outcomes and changes in performance for one student.
Subject	Academic record. Defines a course code, title, credit value, theory or laboratory type, and the semester in which it is taught.
Teacher	Person and employment record. Keeps identity, department, account status, contact details, and the additional duties the administrator has authorised.
Timetable Period	Scheduling record. Describes day, start and end time, period type, subject, teacher, room, and the session or sessions attending.
User Account	Identity record. Stores the account's email, assigned role, status, and the approved person record used to open the correct application workspace.

3.5 User Interface Design

The interface is organised around the work each role performs. Administrators see college-wide totals, academic structure, people, approvals, records, and reports. Teachers see assigned classes, attendance, assessment, schedules, students, and permitted college information. Students see only their own attendance, timetable, marks, results, fee information, notices, and profile. Mobile applications use compact navigation and vertically arranged cards; desktop applications use the additional width for side navigation, wider cards, tables, and file or print actions.

The wording and business meaning of an action do not change between devices. A teacher selecting Mark Attendance follows the same class, subject, roster, validation, and confirmation steps on mobile and desktop. The arrangement changes, not the rule.

3.5.1 Screen Images

The mobile images below are captured from working application builds. The desktop images are representative design views generated from the implemented desktop navigation and screen structure. Together they show the role-specific perspective of all six applications.

3.5.1.1 Admin Mobile Application

The mobile dashboard gives the administrator an immediate college summary and short routes to academic, people, record, and approval work.

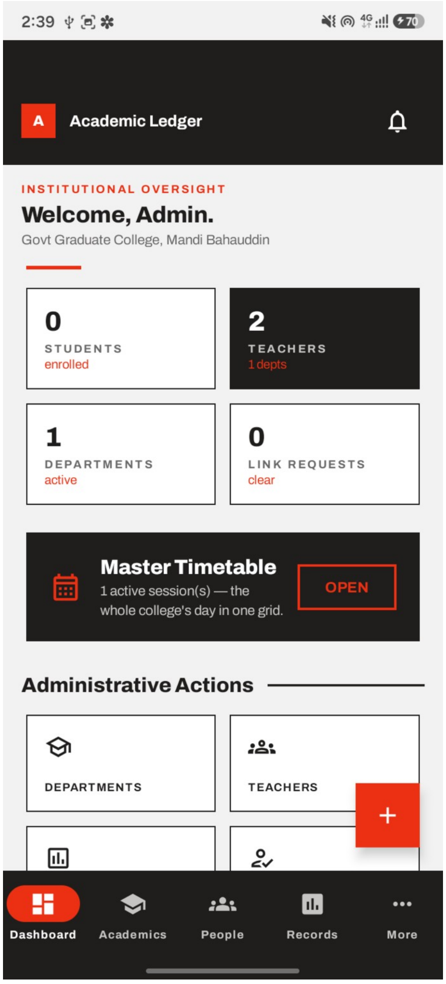

Figure 3-6 Admin Mobile dashboard

3.5.1.2 Admin Desktop Application

The desktop workspace preserves the same administrative areas while using a permanent navigation rail and wider information cards for office work.

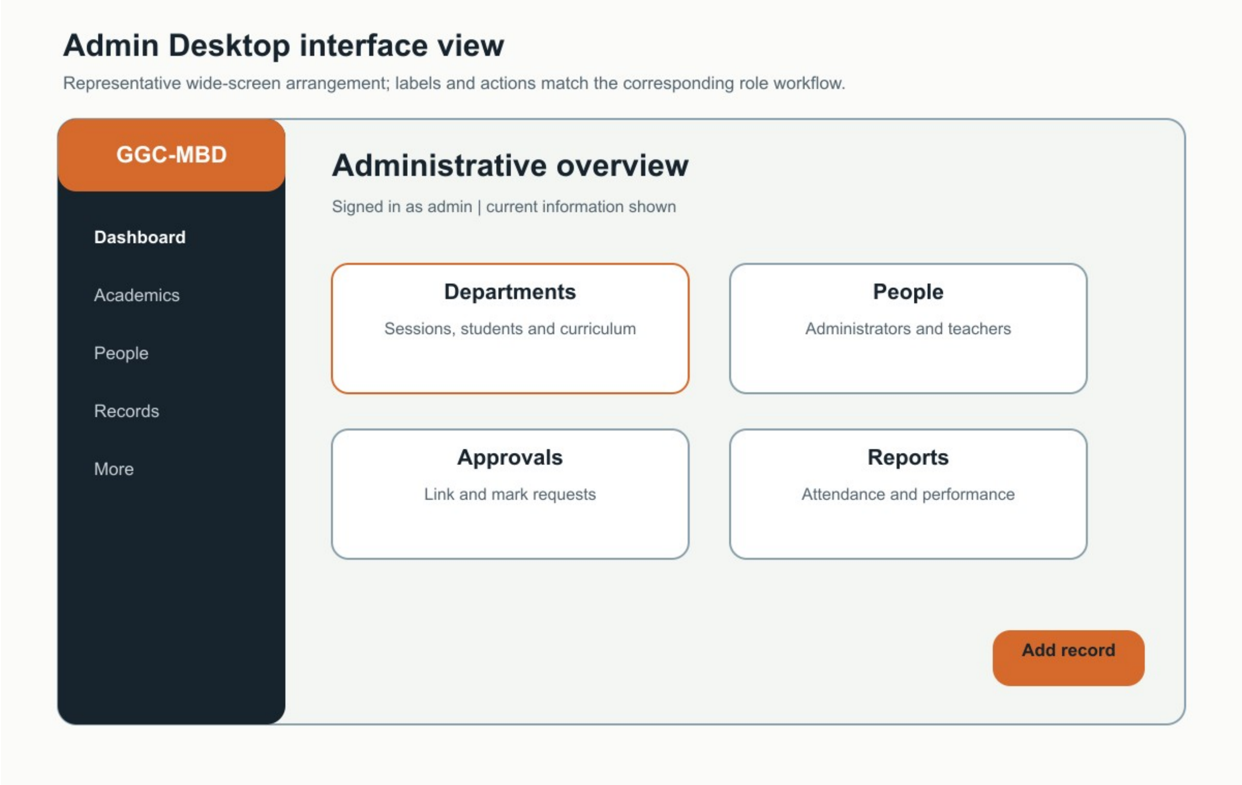

Figure 3-7 Admin Desktop representative interface

3.5.1.3 Teacher Mobile Application

The teacher home screen prioritises today's classes and the actions normally needed during a lesson, including attendance and assessment.

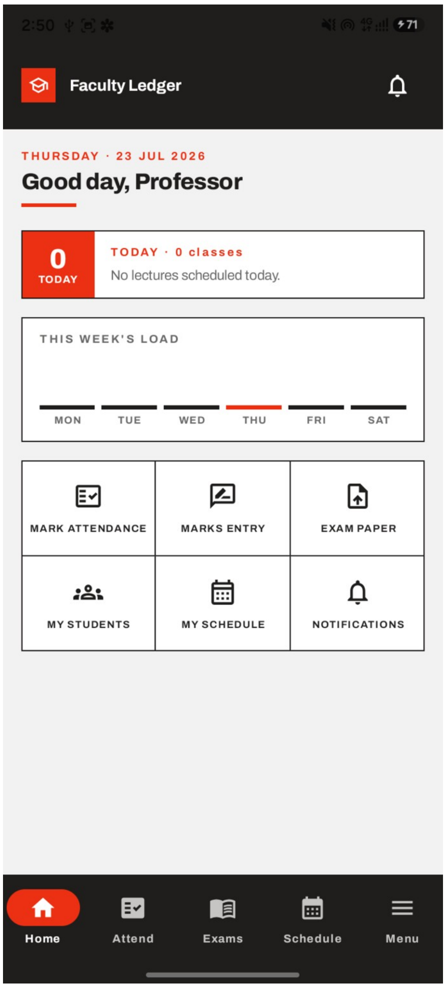

Figure 3-8 Teacher Mobile home screen

3.5.1.4 Teacher Desktop Application

The desktop teaching workspace exposes the same assigned work with room for larger rosters, mark-entry tables, schedules, and document actions.

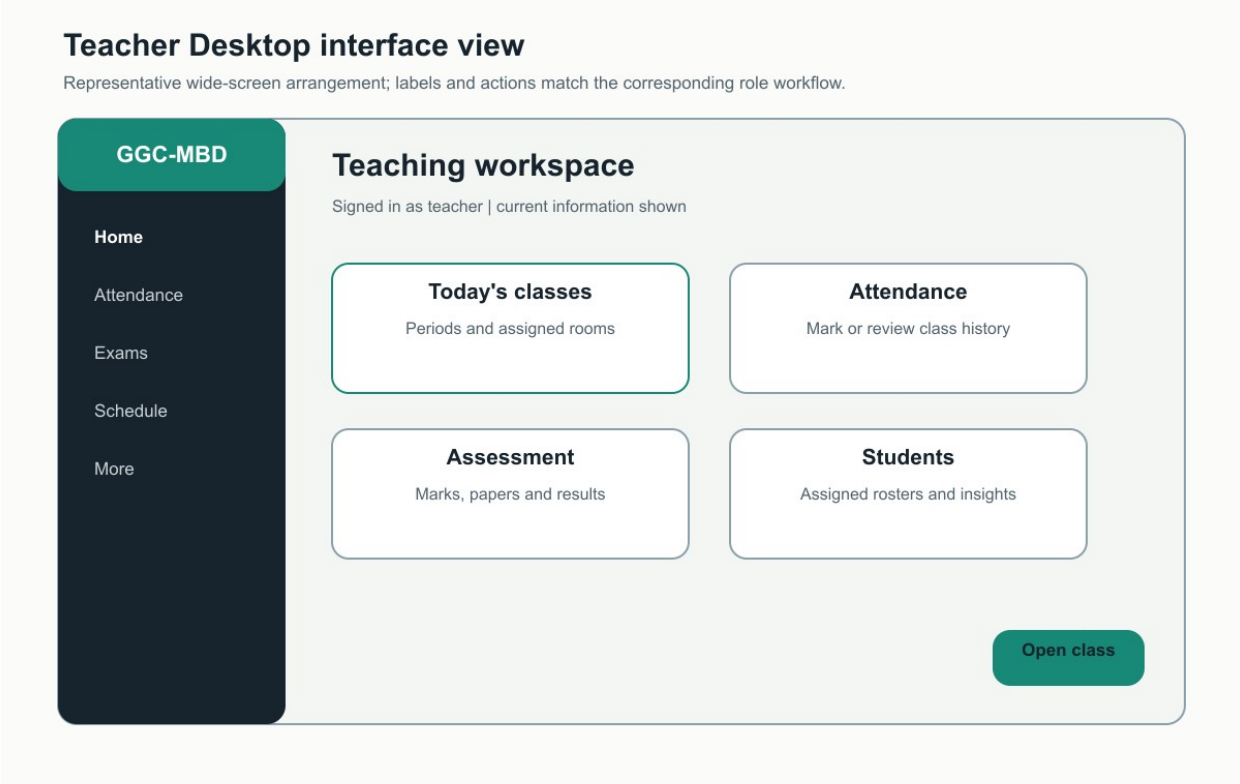

Figure 3-9 Teacher Desktop representative interface

3.5.1.5 Student Mobile Application

The student home screen summarises the current semester and provides direct access to personal academic information without exposing administrative controls.

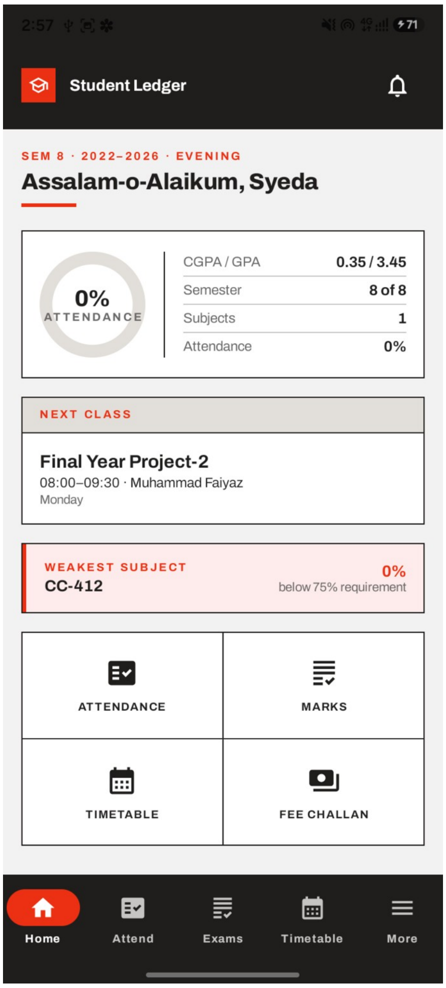

Figure 3-10 Student Mobile home screen

3.5.1.6 Student Desktop Application

The student desktop workspace presents the same personal information in a wider reading layout and adds appropriate open, save, and print actions.

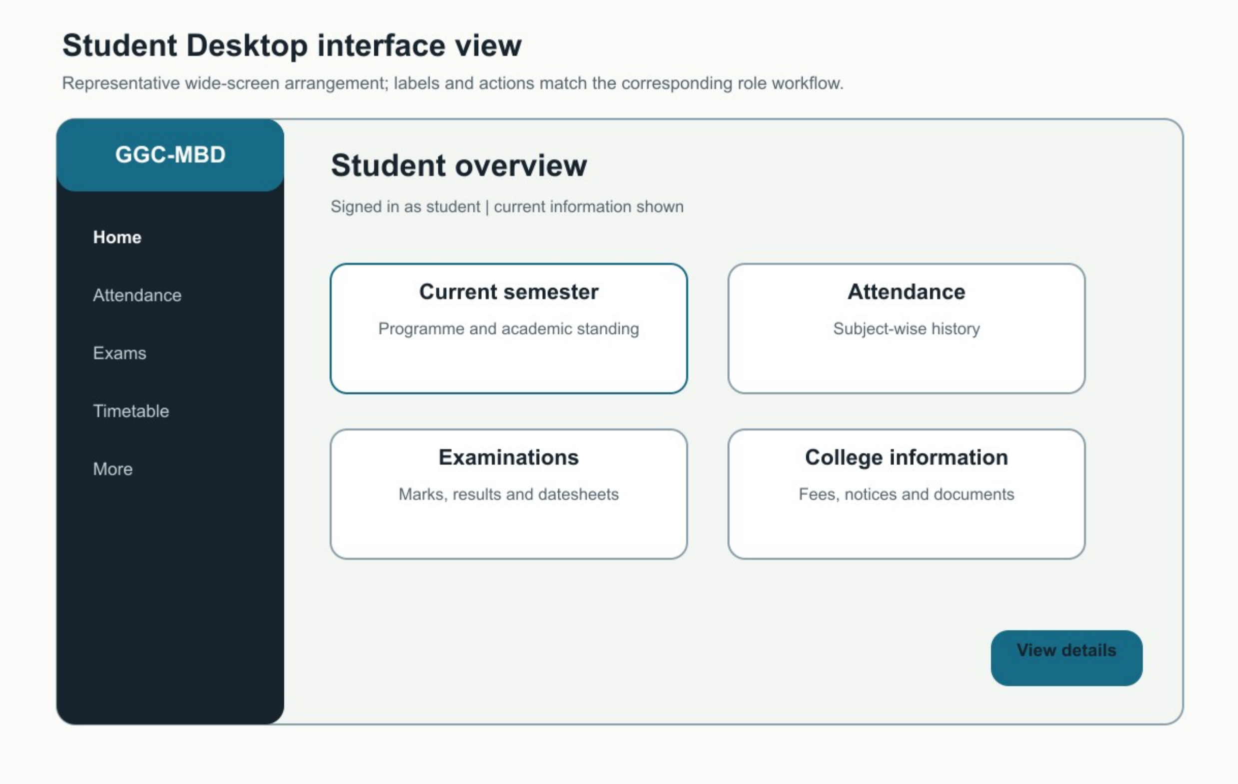

Figure 3-11 Student Desktop representative interface

3.5.2 Screen Objects and Actions

The following two use cases show how visible objects, user actions, and feedback work together. The same objects are available in the corresponding mobile and desktop application, although their position changes with the available space.

3.5.2.1 Use Case: Mark Class Attendance

Table 3-5 Screen Objects and Actions for Mark Class Attendance

Screen Object	User Action	Displayed Feedback
Class and subject selectors	The teacher chooses an assigned class meeting and subject.	Only authorised assignments are offered; missing context is explained before the roster opens.
Student roster cards or rows	The teacher marks each student as present, absent, or on leave and may record late arrival.	The selected state is shown immediately and the total marked count is updated.
Save Attendance action	The teacher submits the completed class record.	Incomplete or duplicate records are identified; a successful save shows confirmation and the recorded date.
Retry action	The teacher retries after an interrupted connection.	The unsaved selections remain visible so the class does not have to be marked again.

3.5.2.2 Use Case: Review a Mark Correction

Table 3-6 Screen Objects and Actions for Review a Mark Correction

Screen Object	User Action	Displayed Feedback
Pending request card or row	The administrator opens a request and reviews the student, subject, recorded mark, requested mark, and teacher's reason.	The original and requested values remain visible together to prevent an accidental decision.
Approve action	The administrator accepts a valid correction.	The mark is updated, the request becomes approved, and the item leaves the pending list.
Reject action	The administrator rejects the request and records a decision where required.	The mark remains unchanged, the request becomes rejected, and the decision is retained.
Empty or failure state	The administrator reviews the queue or retries loading it.	The screen distinguishes no pending requests from a connection or permission problem.

3.5.3 Common Feedback and Interaction Rules

Table 3-7 Common Interface Feedback

Situation	**Feedback Provided Across the Six Applications**
Loading	A clear progress state is shown without replacing the current screen with unexplained blank space.
Empty information	The screen explains that no records are available and, for authorised users, identifies the action that can create one.
Validation	The affected field and a plain-language correction are shown before submission.
Success	The completed action is confirmed and the visible information is refreshed or updated in place.
Connection failure	A short message explains that the request could not be completed and offers retry without displaying service addresses.
Permission restriction	The unavailable action is omitted or disabled, and direct access returns a clear access message.
Destructive action	A confirmation names the record and explains the effect before deletion, rejection, disabling, or unlinking.
Back and refresh	The top bar provides predictable back navigation and refresh access where current online information may have changed.

3.6 Behavioural Model

Behavioural models explain how the parts identified in the architecture cooperate over time. The interaction diagrams cover the main cross-role journeys and use application-neutral names so each flow applies equally to the mobile and desktop version of the role. The state diagrams then show how approval and publication records change when a decision is made.

3.6.1 Sign-In and Role Routing

A user begins in the selected role application. The account service verifies the details and returns the authorised role and account status. The application opens the matching workspace only when the account is active and the role is correct; otherwise it shows a clear sign-in or access message. The same sequence is used by all six applications.

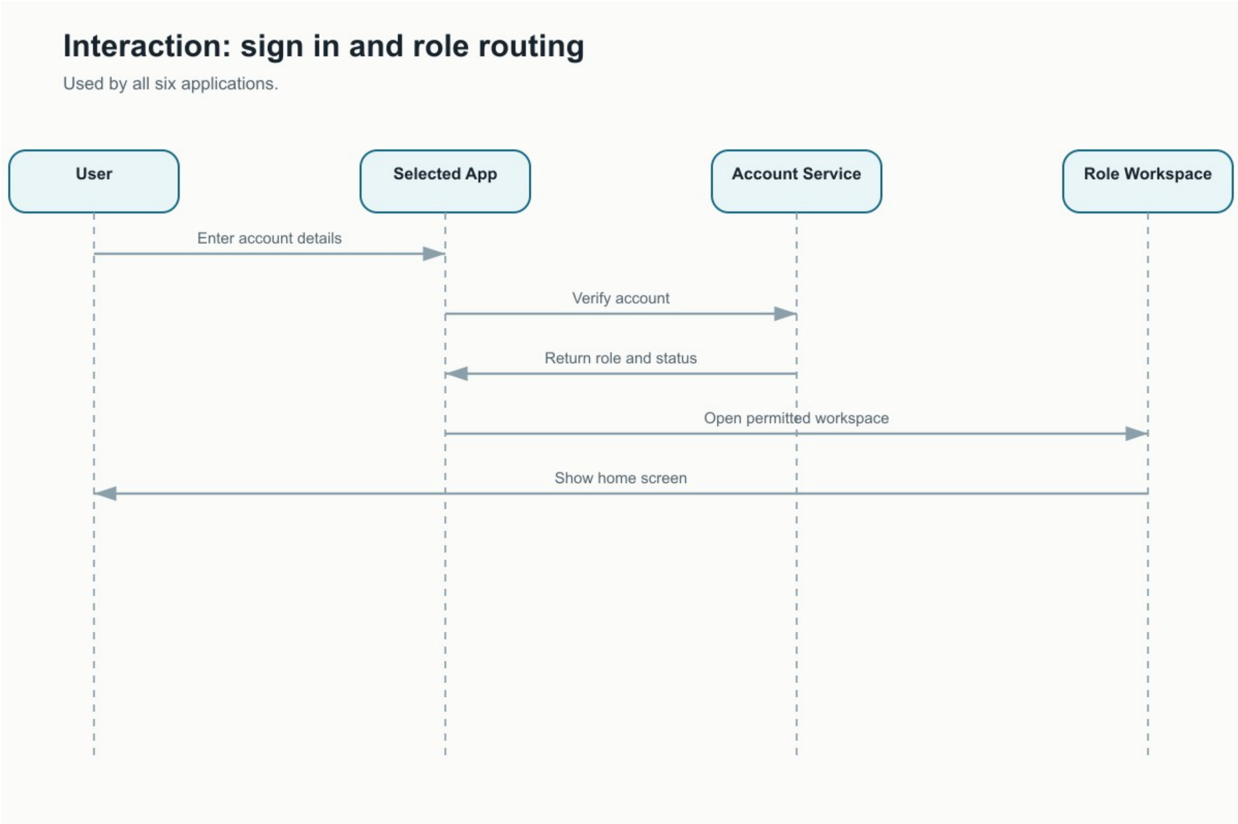

Figure 3-12 Sign-in and role-routing interaction diagram

3.6.2 Publish College Information

An administrator creates a notice, datesheet, event, or other college item in Admin Mobile or Admin Desktop. The application checks required information and audience selection before saving. Once publication is confirmed, only the intended roles or academic groups can see the item in their applications.

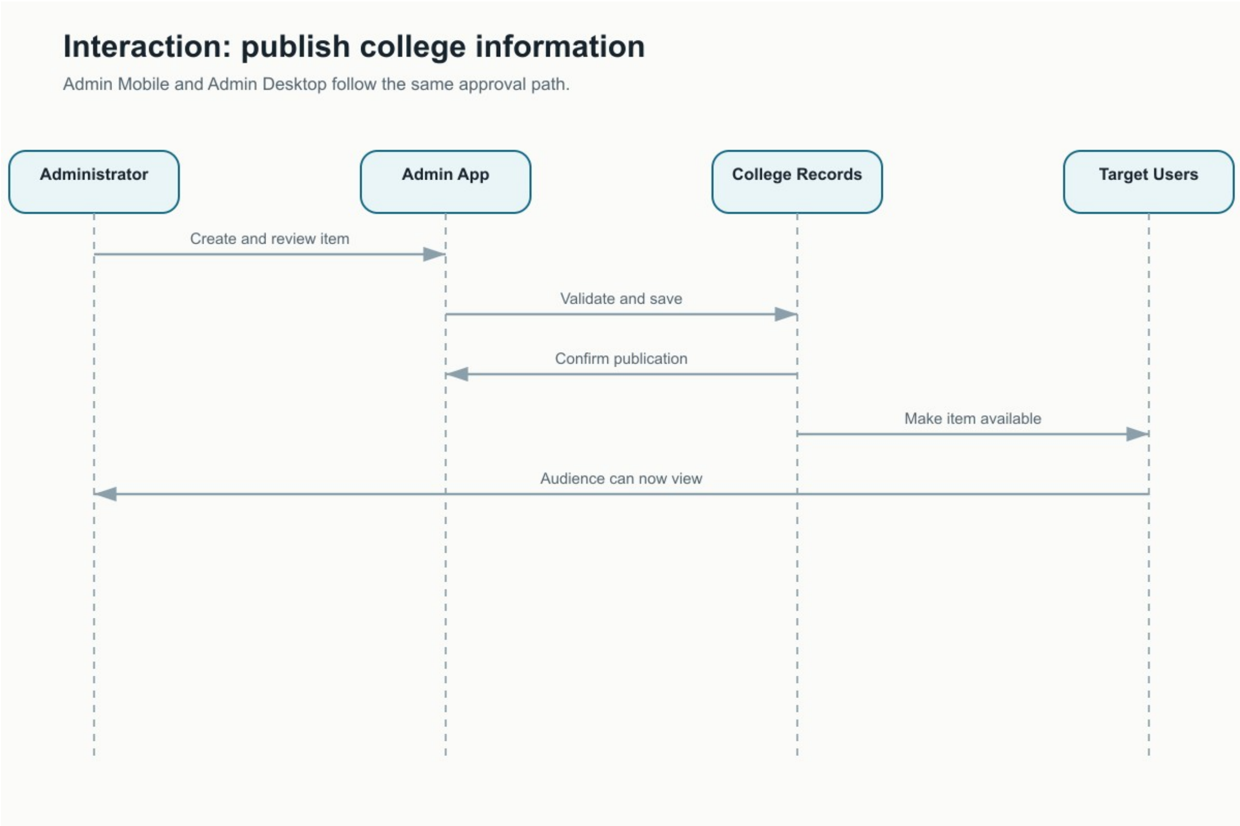

Figure 3-13 Administrative publication interaction diagram

3.6.3 Record Attendance and Assessment

A teacher chooses an assigned class and subject in Teacher Mobile or Teacher Desktop. The class roster is loaded from the authorised academic records. Attendance or first-time marks are checked and saved, and the teacher receives a clear result. If a recorded mark later needs correction, the teacher sends a review request rather than silently replacing it.

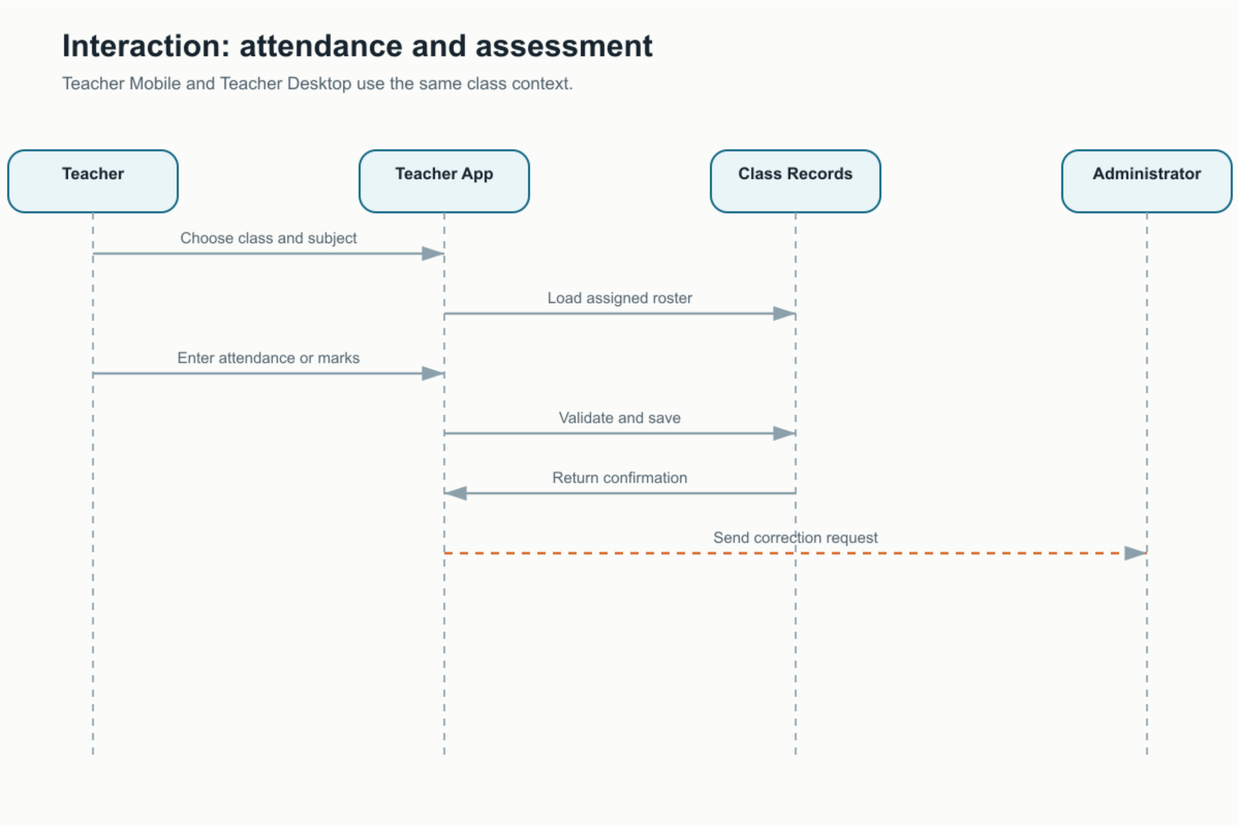

Figure 3-14 Teacher attendance and assessment interaction diagram

3.6.4 Register and Link a Student Account

A student registers through Student Mobile or Student Desktop and provides enough identity information to locate the college record. The link request is reviewed by an authorised administrator. Personal attendance, marks, results, fees, and timetable information remain unavailable until the account is linked to the matching student record.

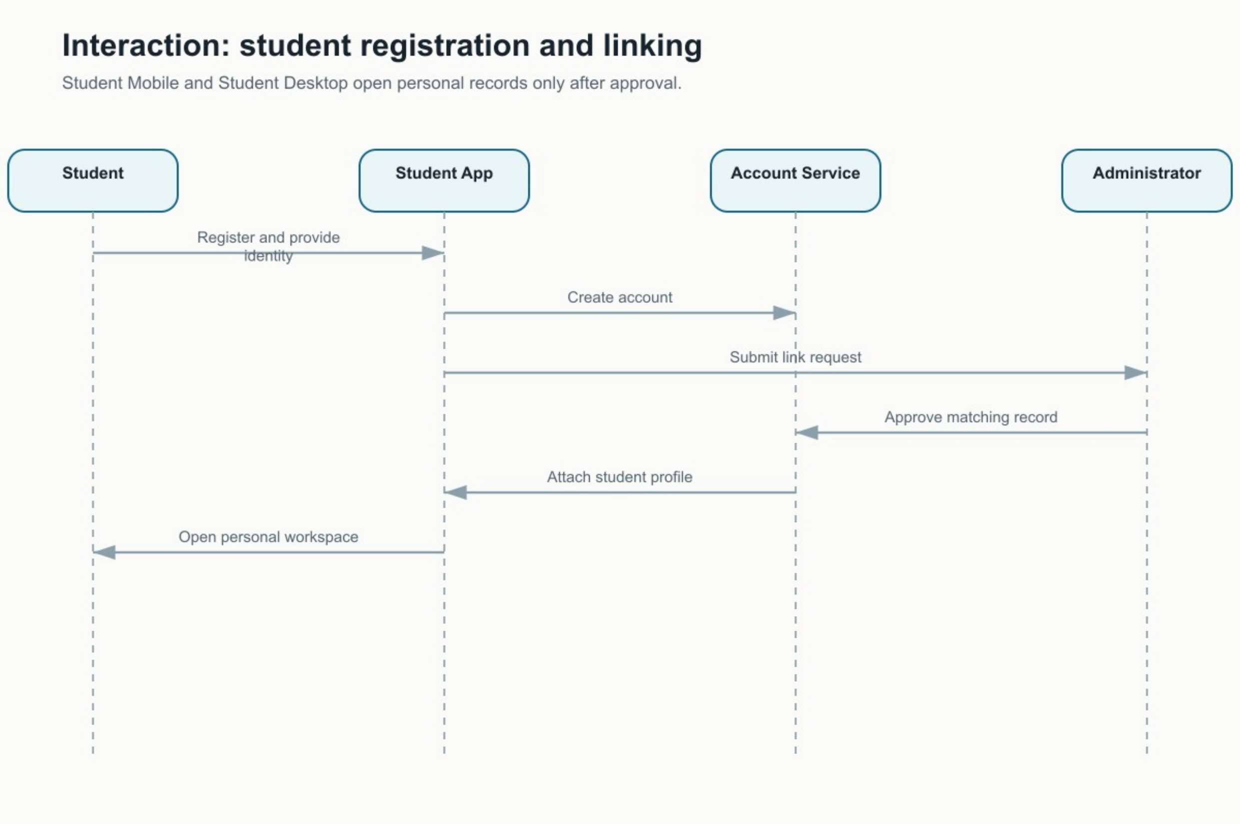

Figure 3-15 Student registration and linking interaction diagram

3.6.5 Publish and Open a Document

An authorised administrator or teacher selects a file, supplies its description and audience, and publishes it through the relevant mobile or desktop application. A permitted reader then requests the document from a Student, Teacher, or Admin application. Desktop applications may open, save, or print the returned file, while mobile applications use the equivalent device actions.

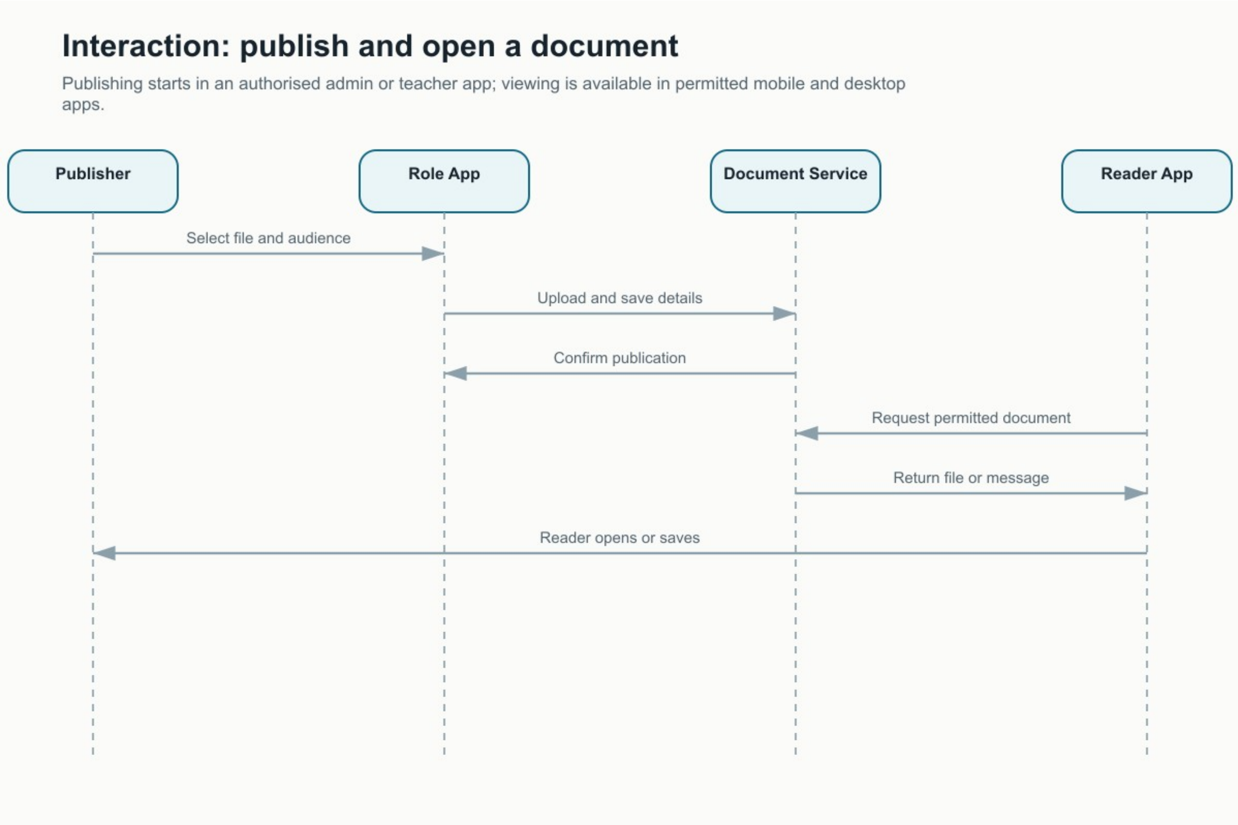

Figure 3-16 Document publication and opening interaction diagram

3.6.6 Student Link Request States

A student account begins unlinked. Submission creates a pending request, and an authorised reviewer either approves or rejects it. Approval opens the personal workspace; rejection keeps academic information closed but allows a corrected request. An authorised unlink action removes access without deleting the college student record.

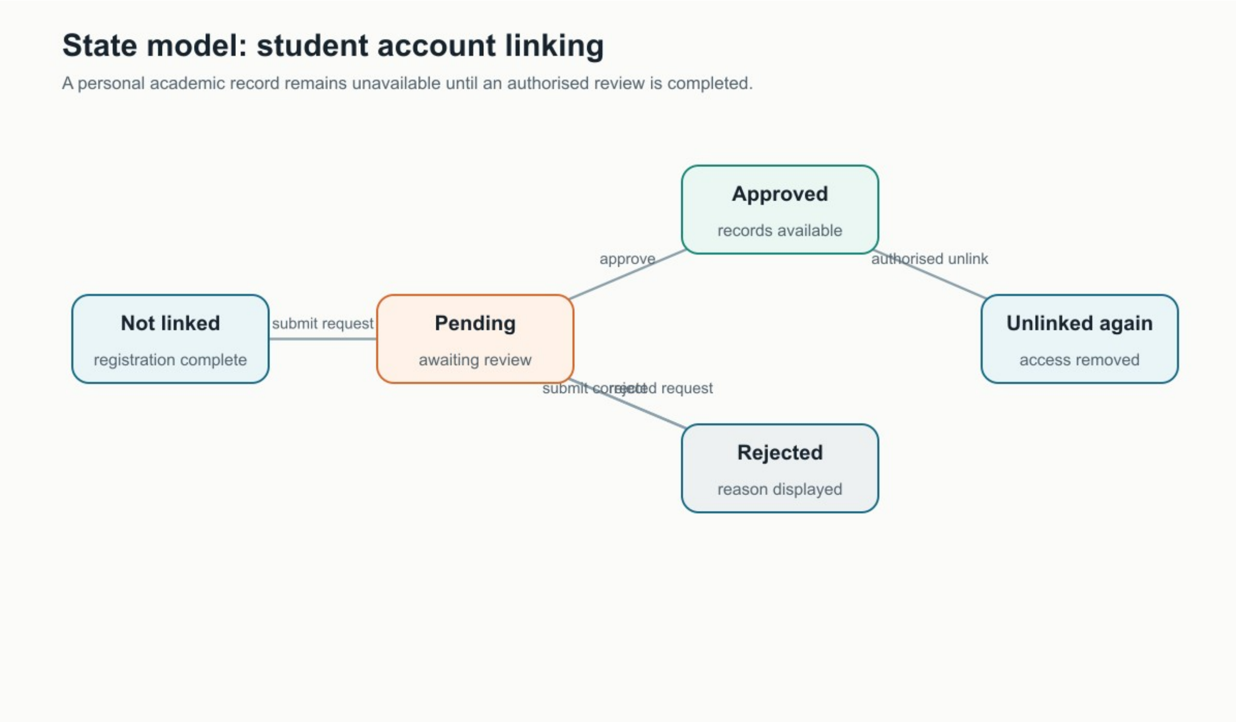

Figure 3-17 Student account-linking state-transition diagram

3.6.7 Mark Correction Request States

A saved mark is treated as recorded and locked. A teacher's correction creates a pending review. Approval updates the mark; rejection leaves it unchanged. In both cases the decision is closed and retained so the history can be understood later.

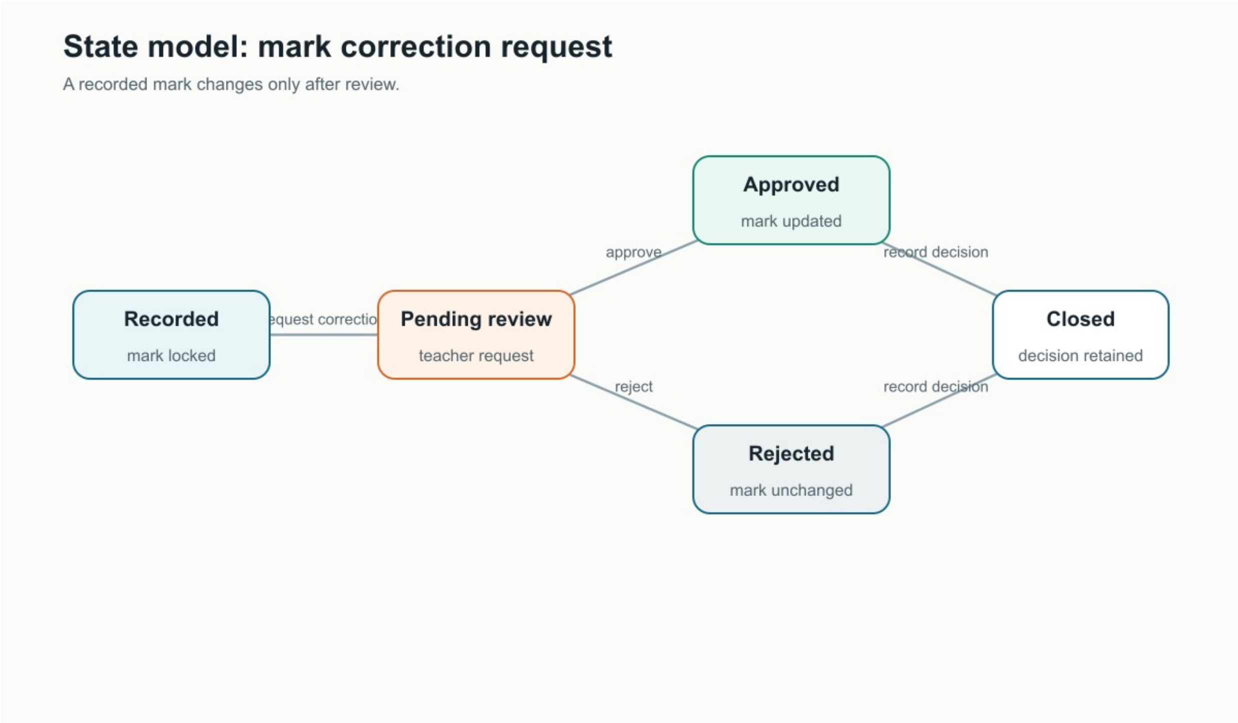

Figure 3-18 Mark-correction state-transition diagram

3.6.8 Publication States

Datesheets, documents, and targeted notices begin as drafts while their required information is prepared. A validated item can be published to its intended audience. When the information is no longer current, it is archived or removed from active views without changing unrelated college records.

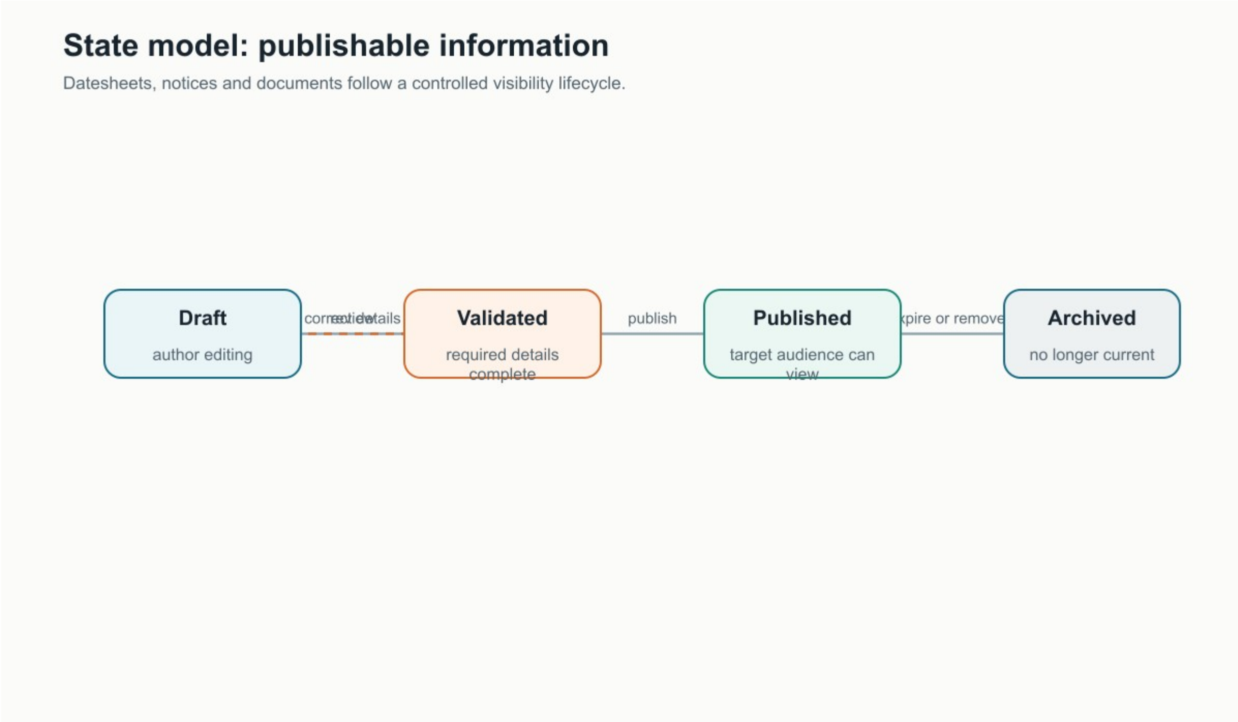

Figure 3-19 Publishable-information state-transition diagram

3.7 Design Decisions

The final design choices were based on the college's roles, the sensitivity of academic records, and the need to support both classroom phone use and longer office tasks. The table records the reason for each choice and the trade-off accepted with it.

Table 3-8 Major Design Decisions

Design Choice	Reason for the Choice	Accepted Trade-Off
Separate Admin, Teacher, and Student applications	Each role sees a smaller and clearer workspace, and sensitive administrative actions are not mixed into student or teacher navigation.	Six application packages must be maintained, tested, and released.
Mobile and desktop versions for every role	Phones support work during classes and movement around campus; Windows supports larger records, documents, printing, and office workflows.	Layouts require device-specific arrangement even though behaviour must remain consistent.
One shared source of college information	A record changed by an authorised user becomes available to every permitted application without manual transfer.	Most confirmed operations depend on an internet connection.
Layered client-server architecture	Presentation, role workflow, shared rules, and information storage can be understood and changed separately.	A small action passes through several responsibilities before it is confirmed.
Central role and permission enforcement	Confidential records remain protected even if a user attempts to bypass a hidden screen or action.	Access rules require careful testing whenever a new record type is introduced.
Relational and normalised academic data	Departments, sessions, subjects, students, and teaching records have clear relationships and avoid repeated copies of the same facts.	Reports sometimes combine several related records before presenting a summary.
Selective local continuity	Frequently opened teaching information can remain useful during a weak connection without trying to duplicate the entire college database on every device.	Some low-frequency screens still require a current connection.
Status-based approvals	Student linking, mark corrections, and publication follow visible states, making pending work and final decisions clear.	A correction may take longer because it waits for an authorised reviewer.
Administrative startup preparation	Departments, teachers, students, and supporting records are prepared when the Admin application opens, reducing repeated manual refreshes.	The first administrative startup may perform more background work.
Native desktop file and print actions	Windows users can select locations, open documents, and print through familiar operating-system controls.	The appearance of native windows is controlled by Windows rather than the application design.

3.8 Summary

This chapter transformed the requirements into a complete system design for Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, and Student Desktop. The domain, context, component, layered, and data models explained the system's structure. The interface views and screen-action tables showed how each role completes work and receives feedback. Interaction and state-transition diagrams then described the main sign-in, administration, teaching, student-linking, document, approval, and publication behaviours. The design decisions balance role separation, consistent mobile and desktop behaviour, controlled access, clear academic relationships, connection limits, and familiar device services. These models provide the design reference for the implementation discussed in Chapter 4.

Chapter 4

Implementation

4. Implementation

This chapter explains how the main features are carried out without reproducing source code. The algorithms are written as language-neutral pseudocode so their decisions, inputs, outputs, and processing cost can be reviewed independently of Android or Windows implementation details. The chapter then records the external services used by the six applications and the current state of the shared Git repository. The coding standards followed by the project are summarised in Appendix B.

4.1 Algorithm

The selected algorithms cover the work that is shared across all applications and the role-specific work most important to administrators, teachers, and students. Here, N represents students in a roster, P timetable periods considered for a conflict, S candidate students in a selected session, M published records, and F the size of a file.

Table 4-1 Algorithm Coverage Across the Six Applications

Algorithm	**Primary Applications**	**Main Purpose**
1. Start and route a signed-in user	All six applications	Restore a valid session and open only the workspace matching the account role.
2. Prepare administrative data	Admin Mobile, Admin Desktop	Load essential college reference information once for the signed-in administrator.
3. Validate a timetable period	Admin Mobile, Admin Desktop	Prevent teacher, room, and session clashes before saving.
4. Record class attendance	Teacher Mobile, Teacher Desktop	Validate and save one complete dated roster.
5. Record assessment marks	Teacher Mobile, Teacher Desktop	Validate first-time scores and preserve locked records.
6. Review a mark correction	Teacher Mobile/Desktop and Admin Mobile/Desktop	Route a correction through a visible request and authorised decision.
7. Link a student account	Student Mobile/Desktop and Admin Mobile/Desktop	Match a registered account to one approved college student record.
8. Build a student academic overview	Student Mobile, Student Desktop	Combine authorised personal attendance, schedule, marks, results, and fees.
9. Publish information for an audience	All six applications according to role	Publish and retrieve notices, datesheets, events, or documents for permitted users.
10. Complete a desktop file action	Admin Desktop, Teacher Desktop, Student Desktop	Open, save, export, print, or share an authorised file through Windows.

4.1.1 Start and Route a Signed-In User

Algorithm 1 Start and route a signed-in user

```
Input: saved account session and selected role application
Output: login screen, access message, or authorised role workspace
Time complexity: O(1) application decisions and indexed account lookups
```

```
1: START and show the splash screen while account checking is incomplete.
2: Restore the saved session, if one exists.
3: IF no valid session exists, open the login screen and END.
4: Read the last known role, then request the current role and account status.
5: IF the account is inactive, show an access message, clear the session, and END.
6: IF the role does not match the selected application, show the correct-role message and END.
7: IF the role is Administrator, run Algorithm 2 when essential data has not been prepared.
8: Open the authorised home workspace and END.
```

4.1.2 Prepare Administrative Data

Algorithm 2 Prepare administrative data

```
Input: administrator account and saved preparation status
Output: ready administrative workspace using current or previously prepared information
Time complexity: O(D + T + S + R), where the symbols represent loaded departments, teachers, students, and supporting records
```

```
1: START after the administrator role has been verified.
2: IF preparation is already complete for this account, open the workspace and END.
3: IF usable prepared information already exists, mark it ready, open the workspace, and END.
4: Show a preparation message and prevent duplicate preparation attempts.
5: Request departments, sessions, teachers, students, timetable, approvals, and supporting records.
6: Validate each returned group before replacing its previously available information.
7: IF essential preparation succeeds, remember completion for this administrator.
8: IF any request fails, retain safe existing information and make retry available.
9: Open the administrative workspace and END.
```

4.1.3 Validate a Timetable Period

Algorithm 3 Validate a timetable period

```
Input: session, day, start time, end time, teacher, room, and effective dates
Output: saved period or a clear conflict message
Time complexity: O(P) worst case for P relevant timetable periods
```

```
START when an administrator submits a timetable period.
IF the start time is not earlier than the end time, reject the period and END.
FOR EACH existing lecture on the same day and overlapping effective dates:
    IF its teacher matches and its time overlaps, reject with the teacher conflict and END.
    IF its room matches and its time overlaps, reject with the room conflict and END.
Check that the selected session does not already contain the same slot.
Save the period, include any approved shared sessions, return confirmation, and END.
```

4.1.4 Record Class Attendance

Algorithm 4 Record class attendance

```
Input: assigned class, subject, date, roster, attendance states, late flags, and notes
Output: one confirmed attendance record for every student in the selected class
Time complexity: O(N) time and O(N) temporary space for N students
```

```
START after the teacher selects an assigned class and subject.
Load the roster and any attendance already recorded for that date.
FOR EACH student, collect present, absent, or leave status and optional late or note information.
IF any required status is missing, identify the incomplete student and STOP without saving.
Create one dated attendance entry for every student.
Submit the complete class record as one operation.
IF confirmed, update the visible history and show success.
ELSE preserve the teacher's selections, show retry, and END.
```

4.1.5 Record Assessment Marks

Algorithm 5 Record assessment marks

```
Input: assigned class, subject, assessment type, roster, scores, and absence selections
Output: confirmed first-time marks or field-level correction messages
Time complexity: O(N) time and O(N) temporary space for N students
```

```
START after the teacher selects an assignment and assessment type.
Load the roster, previously recorded marks, and pending correction requests.
FOR EACH student without a locked mark:
    accept absence or a whole-number score within the assessment maximum.
    show a correction message for blank, non-numeric, negative, or excessive values.
IF any entered value is invalid, keep the form open and STOP.
Save only valid first-time marks and lock them after confirmation.
Direct any later change to Algorithm 6 and END.
```

4.1.6 Review a Mark Correction

Algorithm 6 Request and review a mark correction

```
Input: locked mark, requested mark, reason, teacher identity, and administrator decision
Output: approved updated mark or rejected request with the original mark retained
Time complexity: O(1) for one correction request
```

```
START when a teacher selects a locked mark.
Validate the requested value against the assessment maximum.
IF another request is already pending for this mark, show its status and END.
Create a pending request containing the original value, requested value, and reason.
Show the request in the authorised administrator's review list.
IF the administrator approves, update the mark and close the request as approved.
ELSE retain the original mark and close the request as rejected.
Record the reviewer and decision time, notify the affected view, and END.
```

4.1.7 Link a Student Account

Algorithm 7 Link a student account

```
Input: registered account, selected department and session, claimed roll number, and supporting identity
Output: approved personal workspace or rejected request with a reason
Time complexity: O(S) worst case for S students in the selected session
```

```
1: START after student registration is verified.
2: IF the account is already linked, open the personal workspace and END.
3: IF a request is pending, show its status and END.
4: Submit the claimed academic identity as a pending request.
5: An administrator reviews candidate students within the selected session.
6: Compare roll number and available identity information without exposing unrelated student details.
7: IF exactly one authorised record matches, link the account and approve the request.
8: ELSE reject with a clear reason and keep academic records closed.
9: Refresh the student's access state and END.
```

4.1.8 Build a Student Academic Overview

Algorithm 8 Build a student academic overview

```
Input: approved student link and current academic session
Output: personal home summary, attendance, schedule, marks, results, fees, and notices
Time complexity: O(C + A + M + R) for courses, attendance entries, marks, and result records used in the summary
```

```
1: START after the student account is linked.
2: Load the student's current session, semester, subjects, and timetable.
3: Load only that student's attendance, marks, semester results, fees, fines, and permitted publications.
4: Calculate attendance percentages and identify the next scheduled class.
5: Use the latest completed result for current academic standing.
6: IF one information group is unavailable, show the remaining confirmed groups and a retry message for the missing group.
7: Present the same information in the mobile or desktop student layout and END.
```

4.1.9 Publish Information for an Audience

Algorithm 9 Publish information for an audience

```
Input: authorised publisher, content type, title, details, audience, and optional file
Output: published item visible only to its intended audience
Time complexity: O(F + M): file transfer is proportional to F bytes and listing is proportional
to M returned records

 1: START when an authorised administrator or teacher prepares an item.
 2: Validate the title, required details, publication type, and target audience.
 3: IF a file is included, transfer it and confirm completion before creating the publication
record.
 4: Save the item with its role, department, or session audience.
 5: When an application requests current items, apply account role and academic scope before
returning results.
 6: Show only permitted items; explain empty information separately from a failed request.
 7: Allow the publisher to archive or remove an item within the assigned authority and END.
```

4.1.10 Complete a Desktop File Action

Algorithm 10 Complete a desktop file action

```
Input: authorised record, selected action, file details, and Windows destination or device
Output: opened, saved, exported, printed, shared, or safely cancelled result
Time complexity: O(F) time for a file of F bytes; constant additional memory when transferred
in parts

 1: START when a desktop user selects open, save, export, print, or share.
 2: Confirm that the signed-in role may access the selected record.
 3: Open the appropriate Windows selection, print, or application window.
 4: IF the user cancels, return to the application without changing college data and END.
 5: Transfer or generate the file at the approved destination.
 6: IF the operating-system action succeeds, show confirmation.
 7: ELSE show a plain-language file, permission, application, or printer message and END.
```

4.2 External APIs/SDKs

The six applications use the following external services and software development kits. Versions come from the project's shared dependency catalogue. The table names the public interface or action used, while internal implementation class names are omitted because they do not help a reader understand the system.

Table 4-2 Details of External APIs and SDKs

API / SDK and Version	Description	Purpose in CMS	Main Interface or Action	Applications
Supabase Authentication 3.1.4	Managed account and session service.	Verifies sign-in, restores sessions, supports recovery, and signs users out.	Sign in, restore session, recover account, sign out	All six
Supabase Database Client 3.1.4	Secure client for the managed relational college database.	Reads and updates departments, people, academic records, fees, publications, and reports.	Select, insert, update, delete, and approved server operations	All six
Supabase Storage 3.1.4	Managed file storage service.	Stores and retrieves college documents and examination-paper files separately from their descriptive records.	Upload, download, and delete file	All six according to role
Supabase Functions 3.1.4	Protected server-side operation service.	Carries out actions that require higher authority than a normal signed-in request.	Create account, change teacher status, promote session, archive session, revoke student link	Admin Mobile and Admin Desktop
Ktor Client 3.1.1	Network communication engine used by the managed-service client.	Sends secure requests from Android and Windows and returns success or failure results.	Android network engine and Windows network engine	All six
Kotlin Serialization 1.8.0	Structured information conversion library.	Converts exchanged college records between application objects and network messages.	Encode and decode structured messages	All six
Kotlin Coroutines 1.9.0	Asynchronous task library.	Keeps loading, saving, file transfer, and startup preparation away from the visible interaction thread.	Launch, wait, combine results, and cancel work	All six
Jetpack Compose BOM 2024.12.01 and Material 3	Android declarative interface toolkit.	Builds screens, navigation, dialogs, lists, cards, forms, and feedback for phones and tablets.	Compose screen and Material component APIs	Three mobile applications
Compose Multiplatform 1.7.3 and Material 3	Windows desktop interface toolkit.	Builds role workspaces, wide layouts, windows, dialogs, and shared visual components.	Desktop window and Compose component APIs	Three desktop applications
Room 2.6.1	Android local structured-storage toolkit.	Keeps authorised college records on the device for local-first reading, first-use preparation, and incremental refreshes.	Local read, write, observe, and transaction operations	Three mobile applications

API / SDK and Version	Description	Purpose in CMS	Main Interface or Action	Applications
SQLDelight 2.1.0	Desktop local structured-storage toolkit.	Keeps the corresponding authorised records in separate Windows application databases so desktop workspaces also reopen with their most recent successful data.	SQLite queries, migrations, transactions, and observable reads	Three desktop applications
Preferences DataStore 1.1.1	Small preference and state storage toolkit.	Remembers non-academic application state such as notification-view time and startup preparation status.	Read and update preferences	All six
Hilt 2.52 and Dagger 2.52	Application dependency assembly tools.	Provide each screen and service with the approved shared components for its platform.	Android and desktop component assembly	Hilt: mobile; Dagger: desktop
Apache PDFBox 3.0.3	PDF document creation library for Windows.	Creates exportable attendance and administrative record reports.	Create document, add pages and text, save PDF	Admin Desktop

Service addresses and access keys are supplied through protected application configuration and are not included in the report or repository. Failure messages shown to users describe the action that failed without revealing a service address or private request detail.

4.3 Code Repository

Git is used to track the complete project in one repository. The repository contains the three mobile applications, three desktop applications, shared modules, database changes, server operations, tests, build configuration, documentation sources, generated diagrams, and the final report. Keeping these parts together makes a change to shared behaviour visible to every application that depends on it.

Git Repository Link: https://github.com/hadiyasaleem/CMS

The figures below were checked on 15 August 2026. Commit, branch, and contributor statistics come from the current local history. Pull request, issue, review, and public-push figures come from the GitHub repository. The local history contains later project work than the last public push, so these two evidence sources are reported separately rather than forced into one misleading total.

4.3.1 Metrics of the Git Repository

Table 4-3 Git and GitHub Repository Metrics

Metric	**Verified Value**	**Interpretation**
Local commits	128	Commits reachable from the current local project history.
Maintained branch names	4	master, dev-hadia, dev-laraib, and dev-sharfa are present locally and/or on the configured remote.
Contributors	4 consolidated people	Repeated names using the same email identity are combined in Table 4-4.
Pull requests	20 total; 20 merged; 0 open	The public repository used pull requests for completed feature integration.
Standalone issues	0 open; 0 closed	The public GitHub issue tracker was not used as the team's main task record.
Submitted code reviews	1	One formal GitHub pull-request review is recorded; other checking may have occurred outside the formal review feature.
Latest local commit	26 July 2026	The current local history includes work after the public repository's last recorded push.
Latest public push	20 June 2026	The public GitHub snapshot is older than the local working history.
Default branch	master	The public repository identifies master as its default branch.

4.3.2 Contributor Statistics

Contributor aliases were consolidated by email address so different name spellings from different computers do not split one person's work into several rows. Added and removed line counts include generated and configuration files and should be read as activity evidence, not as a measure of code quality.

Table 4-4 Consolidated Contributor Statistics

Contributor	Commits	Lines Added	Lines Removed
Hadia / Hadiya Saleem	59	29,714	19,494
Sharfa Kiran	45	12,479	1,647
Mujahid Husnain	19	3,721	82
Syeda Laraib Qamar Kazmi	5	8	8
Total	128	45,922	21,231

The repository should continue using short, purpose-based commits and role or feature branches. Pull requests should record the affected applications, verification performed, and any shared behaviour that must remain consistent across mobile and desktop.

4.4 Summary

This chapter explained the implementation through ten high-level algorithms rather than source code. The pseudocode covered shared startup, administrative preparation and scheduling, teacher attendance and assessment, controlled mark correction, student linking and personal summaries, audience-based publication, and Windows file actions. The external-service table identified the versions and public operations used by all six applications. Finally, the repository section recorded local history, public collaboration activity, and consolidated contributor statistics while clearly separating the newer local work from the older public snapshot. Together these details show how the Chapter 3 design is carried out and provide a traceable basis for the testing discussed in Chapter 5.

Chapter 5

Testing and Evaluation

5. Introduction

Testing was used to check both verification and validation: whether the system follows its stated rules and whether its main role-based workflows produce the expected outcome. The chapter follows the Manual Template by presenting unit, functional, integration, and performance testing. Automated checks are used where results can be repeated reliably, while each test case records its setup, steps, input, expected result, actual result, remarks, and final status.

The evidence was refreshed on 15 August 2026. The shared academic test suite executed 127 checks in 50 groups, and the desktop suite executed 13 checks in 3 groups. All 140 checks passed with no failures, errors, or skipped tests. These checks support all six applications because the mobile and desktop versions use the same academic rules, while the desktop suite separately verifies the shared Windows startup and route coverage.

Table 5-1 Testing Practice and Evidence

Practice	How it was applied	Evidence
Independent tests	Each check prepares only the information it needs and does not depend on the outcome of an earlier check.	Tests can run in any order.
Meaningful results	Expected outcomes describe visible academic behaviour rather than internal implementation details.	Every completed case records a clear pass condition.
Controlled test data	Representative departments, sessions, teachers, students, attendance, marks, and publications are prepared for each scenario.	The same scenarios can be repeated.
No fixed live accounts	Automated checks use controlled examples instead of personal accounts or live college records.	No confidential credentials appear in the report.
Repeatable execution	Prior test outputs were cleared before the recorded run and the same project test runner was used for both suites.	140 passed, 0 failed, 0 skipped.

Table 5-2 confirms that no application is omitted. A shared check is counted for an application only where that application uses the tested academic rule; desktop-specific checks are additionally applied to all three Windows applications.

Table 5-2 Testing Coverage Across the Six Applications

Application	Representative coverage	Evidence in this chapter
Admin Mobile	Startup preparation, dashboard summaries, approvals, publications, and safe validation.	UT-1 to UT-3; FT-1; IT-1 to IT-3; PT-1
Admin Desktop	The same administrative rules plus desktop startup continuity, routes, and file actions.	UT-1 to UT-3; FT-1 and FT-4; IT-1 to IT-3; PT-1 and PT-2
Teacher Mobile	Schedule, roster, attendance, marks, correction requests, and role-scoped information.	UT-1 to UT-3; FT-2; IT-2 and IT-3; PT-1
Teacher Desktop	The same teacher workflows plus desktop route and continuity checks.	UT-1 to UT-3; FT-2 and FT-4; IT-2 and IT-3; PT-1 and PT-2
Student Mobile	Account linking, personal timetable, attendance, results, fees, fines, and publications.	UT-1 to UT-3; FT-3; IT-1 to IT-3; PT-1
Student Desktop	The same student workflows plus desktop route and continuity checks.	UT-1 to UT-3; FT-3 and FT-4; IT-1 to IT-3; PT-1 and PT-2

5.1 Unit Testing (UT)

Unit testing checks small, independent rules before they are used in a complete workflow. The selected cases focus on the rules most likely to affect every role: accepting valid information, rejecting unsafe input, presenting academic summaries correctly, and limiting information to the intended user.

Table 5-3 Testcase UT-1

Testcase ID	UT-1
Requirement ID	FR-1, FR-3, FR-4, FR-6, FR-8, FR-12, FR-20, FR-22, FR-24, FR-27, FR-29, FR-32
Title	Validate entered information and present safe error messages
Description	Checks the shared rules used by forms in all six applications.
Objective	Ensure invalid or incomplete information is rejected with a short, understandable message and without exposing a service address.
Driver/precondition	A controlled set of valid, missing, duplicated, out-of-range, and incorrectly formatted values is available.
Test steps	1. Submit each invalid value to its relevant rule. 2. Confirm the value is rejected. 3. Repeat with a valid boundary value. 4. Confirm unexpected service errors are converted into a safe user message.
Input	Blank names, invalid email and phone formats, duplicate identifiers, negative amounts, overlapping periods, out-of-range marks, and valid boundary values.
Expected Results	Every invalid value receives the correct field message; valid boundary values are accepted; no technical service address is shown.
Actual Result	All validation and safe-error checks completed successfully in the shared automated suite.
Remarks	This evidence applies to the common forms and messages used throughout all six applications.
Test Status	Pass

Table 5-4 Testcase UT-2

Testcase ID	UT-2
Requirement ID	FR-3, FR-4, FR-6, FR-7, FR-9, FR-10, FR-11, FR-12, FR-15, FR-16, FR-21, FR-22, FR-23, FR-31
Title	Prepare accurate academic summaries
Description	Checks calculations and summary cards used for departments, sessions, teachers, students, attendance, marks, results, fees, and insights.
Objective	Confirm totals, percentages, labels, warning indicators, and empty-state summaries remain accurate for each role.
Driver/precondition	Representative academic records are available, including normal values, empty lists, absent students, and incomplete marks.
Test steps	1. Prepare the controlled records. 2. Produce each role summary. 3. Compare totals and percentages with the expected hand-calculated values. 4. Repeat with empty and incomplete records.
Input	Departments, sessions, teacher assignments, student rosters, attendance statuses, marks, semester results, fee items, fines, and publication counts.
Expected Results	Every summary displays the expected values and handles empty or incomplete information without producing a misleading total.
Actual Result	All academic summary checks completed successfully in the shared automated suite.
Remarks	The same tested summaries support corresponding mobile and desktop screens.
Test Status	Pass

Table 5-5 Testcase UT-3

Testcase ID	UT-3
Requirement ID	FR-9, FR-11, FR-16, FR-19, FR-21, FR-23, FR-26, FR-30, FR-31
Title	Limit information to the correct role and audience
Description	Checks that administrator, teacher, and student summaries contain only the information intended for that role.
Objective	Prevent a user from receiving another role's controls or another student's personal academic information.
Driver/precondition	Separate administrator, teacher, linked-student, unlinked-student, and general-audience examples are prepared.
Test steps	1. Produce the result for each role. 2. Confirm allowed information is present. 3. Confirm restricted information is absent. 4. Repeat for general and role-specific publications.
Input	Role, teaching assignment, linked student identity, publication audience, and academic records belonging to multiple users.
Expected Results	Each role receives only its authorised actions and information; personal student data remains limited to the linked account.
Actual Result	All role, identity, and audience checks completed successfully in the shared automated suite.
Remarks	This case verifies the decision rules; complete live-service access control remains part of deployment verification.
Test Status	Pass

5.2 Functional Testing (FT)

Functional testing checks a complete feature from the user's point of view. The cases are organised by stakeholder so that the administrator, teacher, and student experience is represented on both mobile and desktop. A fourth case verifies that each Windows application provides the same required destinations and role gate as its mobile counterpart.

Table 5-6 Testcase FT-1

Testcase ID	FT-1
Requirement ID	FR-3, FR-4, FR-6, FR-7, FR-24, FR-25, FR-31
Title	Administrator opens a prepared college workspace
Description	Checks the first useful administrative view after a valid sign-in in Admin Mobile and Admin Desktop.
Objective	Confirm essential college information is prepared together so the administrator does not need to refresh every page separately.
Driver/precondition	A valid administrator account exists and representative departments, sessions, teachers, students, and requests are available.
Test steps	1. Sign in as an administrator. 2. Allow startup preparation to finish. 3. Open the dashboard, departments, people, sessions, and request areas. 4. Confirm the information remains available when moving between them.
Input	Representative administrative records, including at least one department, session, teacher, student, and pending request.
Expected Results	The workspace opens with accurate totals and prepared lists; moving between pages does not require a separate initial refresh.
Actual Result	The administrative preparation and dashboard checks passed in the shared automated suite.
Remarks	The same business outcome supports Admin Mobile and Admin Desktop; screen appearance is evaluated separately through parity review.
Test Status	Pass

Table 5-7 Testcase FT-2

Testcase ID	FT-2
Requirement ID	FR-9, FR-10, FR-11, FR-12, FR-13, FR-15, FR-17, FR-31
Title	Teacher completes attendance and assessment work
Description	Checks the main daily workflow shared by Teacher Mobile and Teacher Desktop.
Objective	Confirm an assigned teacher can view the correct class, record attendance and valid marks, review history, and request a correction when a saved mark must change.
Driver/precondition	A teacher has one assigned class with a timetable period, roster, subject, and assessment limit.
Test steps	1. Open the assigned class. 2. record attendance for the roster. 3. Review the saved attendance summary. 4. Enter valid marks. 5. Attempt an invalid mark. 6. Request correction of a previously saved mark.
Input	Present, absent, and leave statuses; valid and out-of-range marks; a corrected value and a reason.
Expected Results	Attendance and valid marks are accepted, invalid marks are rejected, history is accurate, and the correction remains pending until reviewed.
Actual Result	The roster, attendance, marks, history, and correction-request checks passed in the shared automated suite.
Remarks	This confirms the teacher workflow rules used by both teacher applications.
Test Status	Pass

Table 5-8 Testcase FT-3

Testcase ID	FT-3
Requirement ID	FR-9, FR-11, FR-16, FR-19, FR-21, FR-23, FR-26, FR-27, FR-28, FR-30, FR-32
Title	Student opens a linked personal academic workspace
Description	Checks the personal information shown by Student Mobile and Student Desktop after account linking.
Objective	Confirm the linked student sees only their own timetable, attendance, marks, results, fee information, fines, publications, and profile details.
Driver/precondition	A student account is linked to one roster record with representative academic and financial information.
Test steps	1. Sign in as the linked student. 2. Open the home summary. 3. Visit timetable, attendance, results, fees, fines, notifications, documents, and profile. 4. Confirm no other student's information appears.
Input	One linked student record plus unrelated records belonging to other students and sessions.
Expected Results	The correct personal information appears in every area, unrelated records remain hidden, and empty sections show a clear explanation rather than an error.
Actual Result	The student home, timetable, attendance, marks, results, fees, examinations, and profile checks passed in the shared automated suite.
Remarks	This verifies the shared personal-workspace behaviour used by both student applications.
Test Status	Pass

Table 5-9 Testcase FT-4

Testcase ID	FT-4
Requirement ID	FR-1, FR-33
Title	Desktop applications preserve role routes and Windows actions
Description	Checks Admin Desktop, Teacher Desktop, and Student Desktop against the required mobile destinations and desktop-only file actions.
Objective	Confirm each Windows application opens only the correct role workspace, includes every required destination, and connects file, open, print, and share actions without changing academic rules.
Driver/precondition	The desktop scenario register contains authentication, role gate, main routes, empty states, error states, and supported Windows actions.
Test steps	1. Review the registered scenarios for each desktop role. 2. Confirm every required role destination is represented. 3. Check sign-in, sign-out, back navigation, and role rejection. 4. Check that desktop actions call the intended Windows service.
Input	Administrator, teacher, and student route scenarios with signed-in, signed-out, allowed, and rejected states.
Expected Results	All required routes and gates are covered, and the correct Windows action is requested without exposing it inside the academic screen logic.
Actual Result	All 13 desktop continuity, route-coverage, and comparison checks passed.
Remarks	Native Windows title bars and operating-system dialogs remain outside visual equality, as defined in the project scope.
Test Status	Pass

5.3 Integration Testing (IT)

Integration testing checks that related features still produce one consistent result when they work together. The selected cases follow information across role boundaries: a student's link request moves to administrator review, a publication moves from an authorised publisher to its intended readers, and prepared information remains consistent between startup storage and the visible workspace.

Table 5-10 Testcase IT-1

Testcase ID	IT-1
Requirement ID	FR-6, FR-27, FR-28
Title	Student link request becomes an approved personal account
Description	Connects Student Mobile and Student Desktop request behaviour with Admin Mobile and Admin Desktop review behaviour.
Objective	Confirm one approved request links the account to the matching roster record and does not create two owners for the same student identity.
Driver/precondition	An unlinked student account and one matching roster record exist; no other account is approved for that record.
Test steps	1. Submit a link request as the student. 2. Open pending requests as an administrator. 3. Compare the claimed identity with the roster. 4. Approve the request. 5. Open the student workspace again.
Input	Student account, department, session, roll number, and matching roster identity.
Expected Results	The request becomes approved, the account is linked once, the pending item leaves the review queue, and the student sees the matching personal information.
Actual Result	Link identity, request-queue, approval, and linked student-home checks all passed in the shared automated suite.
Remarks	The case covers both student applications and both administrator applications through the same approval rules.
Test Status	Pass

Table 5-11 Testcase IT-2

Testcase ID	IT-2
Requirement ID	FR-18, FR-19, FR-24, FR-25, FR-26, FR-29, FR-30
Title	Published information reaches only its intended audience
Description	Connects publication by an administrator or authorised teacher with viewing in the relevant admin, teacher, and student applications.
Objective	Confirm draft or restricted information is not shown to the wrong role, while published information reaches every intended mobile and desktop reader.
Driver/precondition	Administrator, teacher, and linked-student examples exist with general and role-specific datesheets, events, documents, and notifications.
Test steps	1. Prepare a draft item. 2. Confirm it is absent for normal readers. 3. Publish it for a selected audience. 4. Open each role application. 5. Compare visible items with the selected audience.
Input	Draft and published datesheets, events, documents, and notifications for administrator, teacher, student, and general audiences.
Expected Results	Draft items remain hidden; each published item appears only for the selected audience; unrelated roles do not receive it.
Actual Result	Document, calendar, notification, examination, and role-audience checks passed in the shared automated suite.
Remarks	The evidence covers publication rules used by all six applications; live push delivery is not claimed because the applications refresh information when opened.
Test Status	Pass

Table 5-12 Testcase IT-3

Testcase ID	IT-3
Requirement ID	FR-3, FR-4, FR-6, FR-7, FR-9, FR-25, FR-26, FR-31
Title	Startup preparation and saved continuity produce one workspace
Description	Checks the connection between prepared academic information, saved continuity, and the visible role workspace on mobile and desktop.
Objective	Confirm previously prepared information can be used immediately and that missing essential information triggers preparation rather than an incomplete dashboard.
Driver/precondition	Complete, incomplete, and empty startup examples are available for administrative, teacher, and student roles.
Test steps	1. Start with complete saved information. 2. Confirm the workspace can use it. 3. Remove one essential group. 4. Start again. 5. Confirm preparation is requested and the resulting workspace contains a consistent set.
Input	Complete and incomplete combinations of departments, sessions, people, academic records, and publication information.
Expected Results	Complete information is reused; incomplete information is prepared before use; the visible workspace never mixes unrelated or partially prepared records.
Actual Result	Shared preparation checks and all desktop saved-continuity checks passed in the recorded test runs.
Remarks	Mobile and desktop use different local storage methods, but this case verifies the same user-facing continuity rule.
Test Status	Pass

Table 5-13 Testcase IT-4

Testcase ID	IT-4
Requirement ID	FR-3, FR-4, FR-24, FR-25, FR-26
Title	Local-first refresh retains authorised records across application restart
Description	Checks the common continuity rule used by Admin, Teacher, and Student applications on both Android and Windows.
Objective	Confirm a first refresh prepares locally readable records, a later refresh requests only changed records, and a failed refresh does not replace the last successful local view.
Driver/precondition	A role account has authority to read a prepared set of college records, including one changed record and one removed record.
Test steps	1. Open the role application and complete the first refresh. 2. Restart it without changing the account. 3. Confirm the prepared records are shown. 4. Change one authorised record and remove another on the service. 5. Refresh again. 6. Interrupt a later refresh before completion.
Input	Authorised role account, prepared records, one changed record, and one controlled deletion marker.
Expected Results	The first refresh prepares the local view; the later refresh updates only changed information; the removed record no longer appears in normal lists; and an interrupted refresh leaves the prior successful view and refresh point intact.
Actual Result	Shared checkpoint, Room migration, SQLite persistence, equal-timestamp boundary, local reset, and tombstone-handling checks passed in the recorded automated runs. Device-level interruption testing remains part of pilot acceptance.
Remarks	The test covers the same continuity rule across all six applications. It does not claim that local records override the authorised online database.
Test Status	Pass (automated evidence)

5.4 Performance Testing (PT)

Performance testing records how long important processing takes and whether it remains stable under the stated test conditions. The Gradle test runner was used because it provides repeatable timing for the verified academic and desktop checks. Prior test outputs were cleared, the two suites were run separately on the same project workstation, and their recorded execution times were compared with clear acceptance targets.

These measurements are deliberately limited. They establish a local processing and continuity baseline, but they do not represent Android device launch time, Windows interface drawing time, live college-network delay, or many simultaneous users. Those field measurements require production-sized records, connected devices, and the deployed service, and should be completed during acceptance testing rather than invented here.

Table 5-13 Performance Test Configuration

Item	Recorded configuration
Tool	Gradle test runner and its generated execution report
Execution date	15 August 2026
Environment	Project workstation with warmed project files; previous test outputs cleared
Shared workload	127 academic and role-behaviour checks across 50 groups
Desktop workload	13 continuity, route-coverage, and comparison checks across 3 groups
Acceptance targets	Shared workload below 5 seconds; desktop workload below 6 seconds
Excluded measurements	Physical-device startup, live-network delay, and concurrent-user load

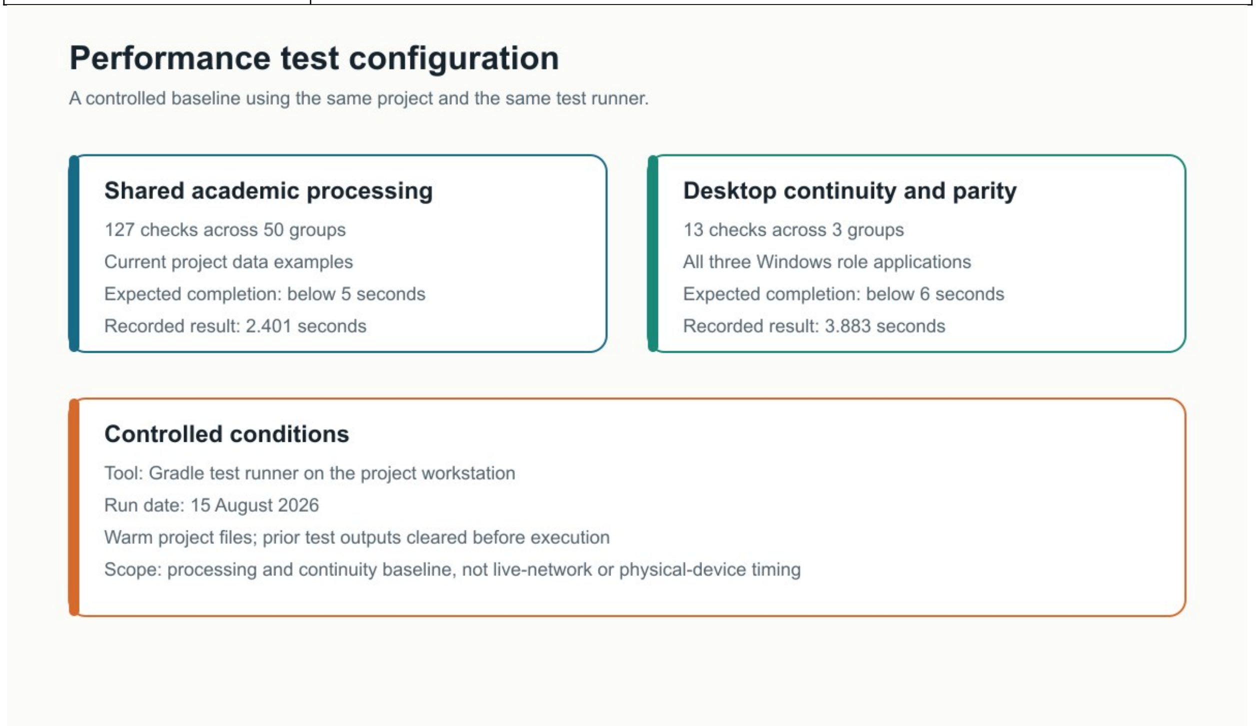

Figure 5-1 Performance test configuration

Table 5-14 Testcase PT-1

Testcase ID	PT-1
Requirement ID	Shared support for FR-1 to FR-32
Title	Shared academic processing baseline
Description	Measures the recorded execution time of the complete shared academic and role-behaviour suite used by all six applications.
Objective	Confirm the common decision and summary checks complete quickly and without unstable failures on the project workstation.
Driver/precondition	Compiled project files are available and prior shared test outputs have been cleared.
Test steps	1. Start the shared test suite. 2. Allow all 50 groups to complete. 3. Read the total checks, failures, skipped checks, and recorded execution time. 4. Compare the time with the 5-second target.
Input	127 controlled checks covering validation, summaries, role visibility, attendance, marks, linking, publications, reports, and startup preparation.
Expected Results	All checks pass with no skipped checks, and recorded test execution remains below 5 seconds.
Actual Result	127 of 127 checks passed; 0 failed, 0 errored, and 0 were skipped. Recorded test execution time was 2.401 seconds.
Remarks	The target was met with 2.599 seconds of margin. Gradle startup and compilation time are excluded because they are development-tool overhead, not application processing.
Test Status	Pass

Table 5-15 Testcase PT-2

Testcase ID	PT-2
Requirement ID	FR-1, FR-33 and desktop support for role requirements
Title	Desktop continuity and route-verification baseline
Description	Measures the recorded execution time of the shared desktop checks used by Admin Desktop, Teacher Desktop, and Student Desktop.
Objective	Confirm desktop continuity, required route coverage, and visual-comparison rules complete within the stated local baseline.
Driver/precondition	Compiled desktop project files are available and prior desktop test outputs have been cleared.
Test steps	1. Start the desktop test suite. 2. Allow all 3 groups to complete. 3. Read the total checks, failures, skipped checks, and execution time. 4. Compare the time with the 6-second target.
Input	13 controlled checks covering saved continuity, role-route coverage, authentication gates, and comparison rules.
Expected Results	All checks pass with no skipped checks, and recorded test execution remains below 6 seconds.
Actual Result	13 of 13 checks passed; 0 failed, 0 errored, and 0 were skipped. Recorded test execution time was 3.883 seconds.
Remarks	The target was met with 2.117 seconds of margin. This timing is a desktop processing baseline, not a complete measure of Windows screen rendering.
Test Status	Pass

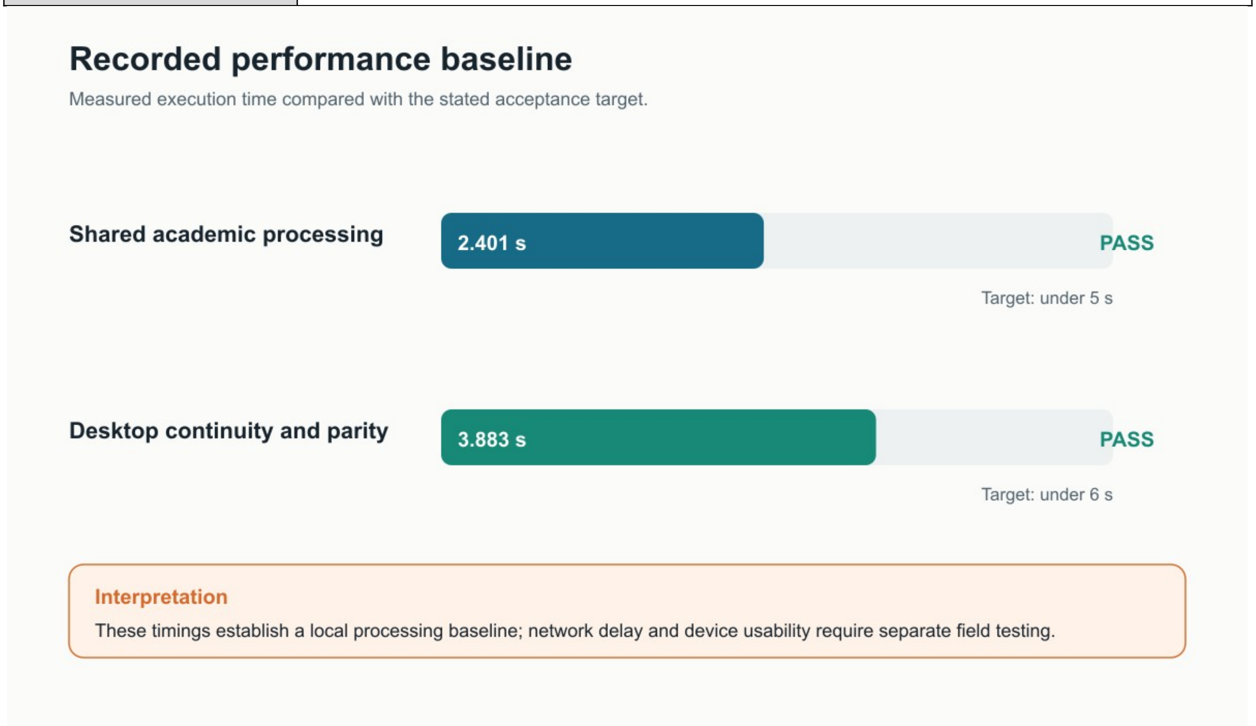

Figure 5-2 Recorded performance baseline and acceptance targets

5.5 Summary

This chapter evaluated the system through unit, functional, integration, and performance testing in the format required by the Manual Template. The recorded run completed 140 automated checks with no failures, errors, or skipped checks. Unit tests verified validation, academic summaries, and role visibility. Functional tests represented the complete administrator, teacher, and student experience on mobile and desktop. Integration tests followed student linking, controlled publication, and startup continuity across role boundaries. Performance tests recorded repeatable local baselines and documented their tool, configuration, targets, results, and limitations. Together, the evidence supports all six applications and the project's main objectives without claiming physical-device, live-network, or concurrent-user results that were not measured. Those remaining field checks belong to final deployment and user-acceptance testing.

Chapter 6

System Conversion

6. Introduction

System conversion explains how GGC-MBD can move from paper registers, office files, and separately communicated information to one coordinated College Management System. There is no earlier college database to transfer. Conversion therefore focuses on preparing reliable starting records, installing the six applications, introducing the new working procedures, training users, and confirming that daily work can continue safely.

The change covers Admin Mobile and Admin Desktop for college management, Teacher Mobile and Teacher Desktop for teaching activities, and Student Mobile and Student Desktop for access to personal academic information. The mobile and desktop versions support the same role responsibilities, while allowing each user to work from the device that is practical for them.

6.1 Conversion Method

A pilot conversion followed by phased expansion is the most suitable method. One department should first use the system with a small, checked set of real records. Its administrators, teachers, and students can confirm the complete journey before further departments are added. This limits disruption, gives staff time to learn the new process, and allows corrections to be made while the affected group is still small.

Table 6-1 Selection of the Conversion Method

Method	Assessment for GGC-MBD
Direct conversion	Not selected. Introducing all departments at once would make record correction, training, and user support difficult.
Parallel conversion	Not selected as the primary method. Paper records will remain available for checking during the pilot, but maintaining two complete operational systems would duplicate staff effort.
Pilot conversion	Selected for the first stage. One department provides a controlled setting for checking records, access, workflows, and training.
Phased conversion	Selected for expansion. Departments are added only after the pilot findings are resolved and an acceptance check is passed.

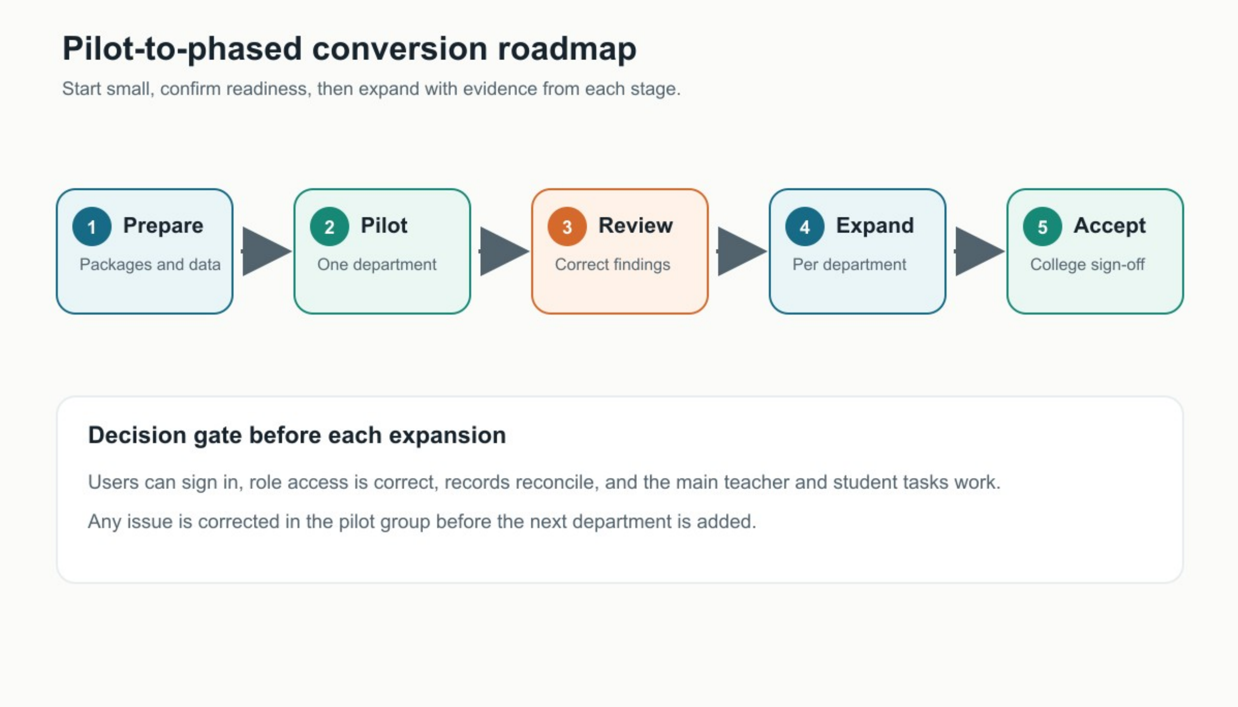

Figure 6-1 Pilot-to-phased conversion roadmap

6.2 Deployment

Deployment prepares the hosted college service and places the correct application on each approved device. Build readiness has been verified for all six applications: Android installation packages are available for the three mobile roles, and Windows installation packages are available for the three desktop roles. The mobile packages currently represent controlled test builds; signed release preparation and customer-site installation remain part of the pilot deployment. This distinction prevents build completion from being reported as completed operational deployment.

Table 6-2 Six-Application Deployment Coverage

Application	Primary users	Package and first-start check
Admin Mobile	Principal and authorised administrators	Android package; confirm sign-in, dashboard figures, people, academic structure, and approvals.
Admin Desktop	Office administrators	Windows installer; confirm the same admin records plus file selection, export, opening, and printing.
Teacher Mobile	Teaching staff	Android package; confirm assigned classes, attendance, marks, schedule, and notices.
Teacher Desktop	Teaching staff	Windows installer; confirm the same teacher responsibilities and desktop document actions.
Student Mobile	Enrolled students	Android package; confirm account linking and access to personal attendance, marks, timetable, fees, and notices.
Student Desktop	Enrolled students	Windows installer; confirm the same student information and desktop document actions.

All six applications use the same authorised college records. Deployment must therefore begin with the shared service and administrator access, continue with application installation, and end with role-based checks using pilot accounts.

Table 6-3 Deployment Procedure

Stage	Main activity	Completion evidence
1. Protect the current state	Create a recoverable copy of existing project data and record the approved release versions.	Backup is available and package versions are listed.
2. Prepare the service	Apply the approved data structure and security changes, then configure account communication and authorised college access.	Administrator sign-in and account recovery are checked.
3. Create initial access	Create the first authorised administrator and confirm that each role reaches only its own application area.	Admin, teacher, and student pilot accounts open the correct area.
4. Install applications	Install the three mobile packages on approved Android devices and the three desktop packages on approved Windows computers.	Each application opens and shows its correct name and role.
5. Load pilot records	Enter the college structure and load the checked records for the selected pilot department.	Counts and sample records agree with the approved source.
6. Prepare local continuity	Open each pilot application once while connected so it prepares its authorised local records. Then refresh one changed record and confirm a removed record leaves the normal list.	The initial local view, one incremental refresh, and one restart read are recorded for each role and platform.
7. Perform smoke checks	Complete one important admin, teacher, and student journey on both mobile and desktop.	The six application checks are signed by the pilot users.
8. Approve expansion	Record issues, correct them, repeat failed checks, and obtain approval before adding another department.	No unresolved critical issue remains for the next stage.

Operational settings and credentials are kept in controlled deployment records rather than in this report. This protects the college service while still giving the deployment team a complete, repeatable sequence to follow.

6.2.1 Data Conversion

The college is moving mainly from paper and office records rather than from an earlier database. Data conversion therefore means collecting approved source records, placing them in a consistent format, correcting obvious quality problems, entering or importing them, and comparing the result with the source before use.

Table 6-4 Data Conversion Controls

Record group	Preparation and loading	Acceptance control
Departments and sessions	Enter approved names, codes, study periods, and current semester in the Admin applications.	Compare the department and session totals with the approved college list.
Administrators and teachers	Create authorised accounts and record department, contact, status, and assigned responsibilities.	The college office confirms identity, role, department, and active status.
Students and rosters	Prepare names and class roll numbers in the provided spreadsheet format, then import the roster or enter a corrected record individually.	Reject missing names, invalid roll numbers, wrong-session records, and duplicates; reconcile the final roster count.
Curriculum and timetable	Enter approved courses, teaching assignments, rooms, periods, and weekly schedules.	Department staff compare a sample and the complete timetable with the signed academic plan.
Fees and publications	Enter the current fee structure and publish approved notices, events, datesheets, and documents.	The responsible office checks amounts, dates, audience, and published version.
Opening academic records	Where required, enter verified starting attendance, marks, and result information before the pilot begins.	Subject teachers and the department confirm totals and selected student records.

Student rosters can be prepared in CSV or Excel form for controlled import. Each row requires a student name and a class roll number that belongs to the target department and session. Invalid or repeated rows are reported before they are accepted. Other starting records are entered through the Admin Mobile or Admin Desktop application by authorised staff and checked by the relevant department.

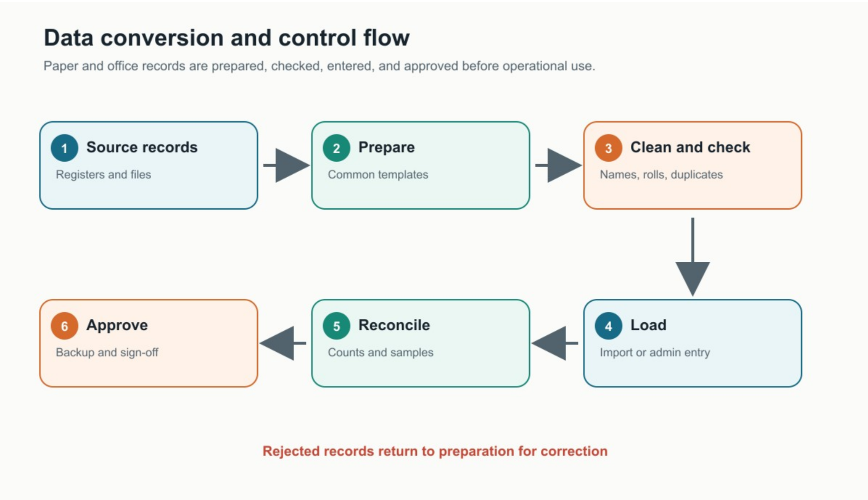

Figure 6-2 Data conversion and control flow

A backup is kept before loading. Afterwards, totals and selected records are checked against signed sources; accepted records are backed up and rejected records return for correction. A serious failure restores the last approved backup.

6.2.2 Training

Training is organised by role and by the work each person performs. Short demonstrations should use pilot records, followed by supervised practice and a simple completion check. Mobile and desktop users receive the same business instructions; only the method of selecting files, opening documents, or printing differs on Windows.

Table 6-5 Training Coverage for All Applications

Applications	Training audience	Required practice
Admin Mobile and Admin Desktop	Principal and authorised office staff	Create people and academic records, prepare rosters, manage schedules and publications, review requests, and confirm dashboard information.
Teacher Mobile and Teacher Desktop	Teaching staff	Open assigned classes, record attendance and marks, review schedules, submit required records, and read notices.
Student Mobile and Student Desktop	Students and support staff	Register and link an account, view personal attendance, marks, timetable, datesheet, fee information, notices, and documents.
Desktop-specific actions	Admin, teacher, and student desktop users	Choose or save a file, open a downloaded document, print an approved record, and confirm the selected destination.

Use Case 1: Record Class Attendance

This procedure is used in Teacher Mobile and Teacher Desktop. The same labels and decisions are presented in both versions; the wider desktop window only changes the available viewing space.

1. Sign in to the Teacher application and open Attendance.

2. Select the assigned class for the required date. If no class is selected, the application prompts the teacher to choose one.

3. Review the class name, date, roster total, and any existing attendance information.

4. Mark every student as Present, Absent, or Leave. Add late status or a note where required.

5. Check that every student has been marked, submit the register, and open Attendance History when an earlier record needs review.

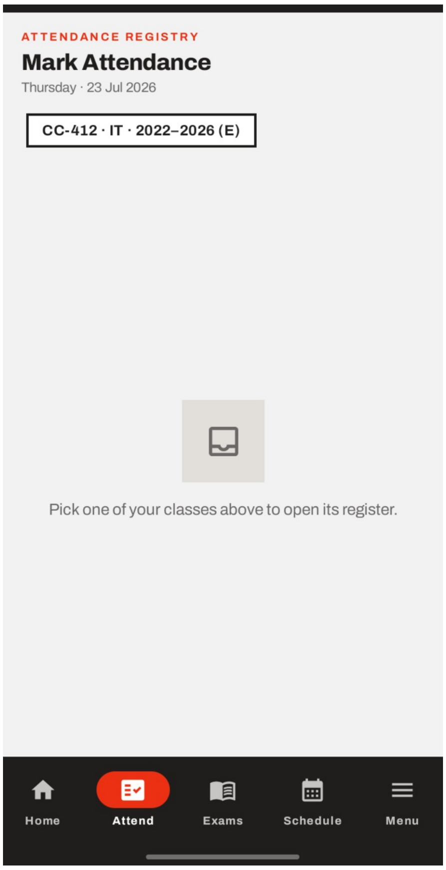

Figure 6-3 Selecting an assigned class

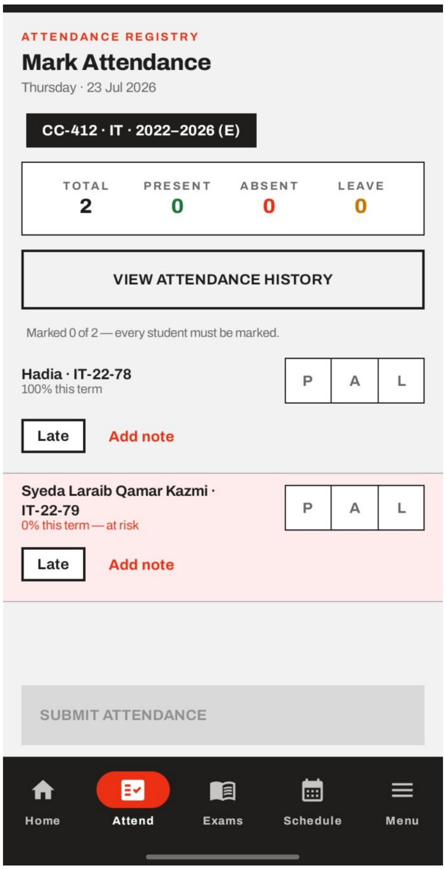

Figure 6-4 Completing the class register

Use Case 2: View Fee Challan

This procedure is used in Student Mobile and Student Desktop. Fee information is for guidance and payment is made through the college accounts office.

1. Sign in to the Student application using a linked student account.
2. Open More and select Fee Challan.
3. Review the fee period, listed charges, total amount, and due date when a structure has been published for the student's session.
4. If no fee structure is available, read the displayed message and contact the college office rather than assuming that no fee is due.
5. Use the displayed information when visiting the accounts office. The application does not collect an online payment.

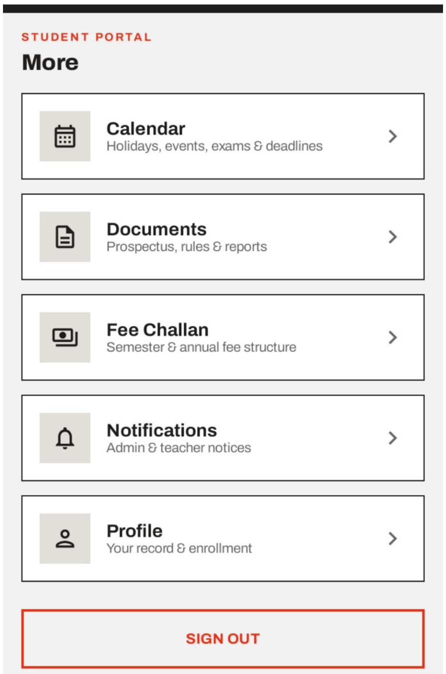

Figure 6-5 Opening Fee Challan from More

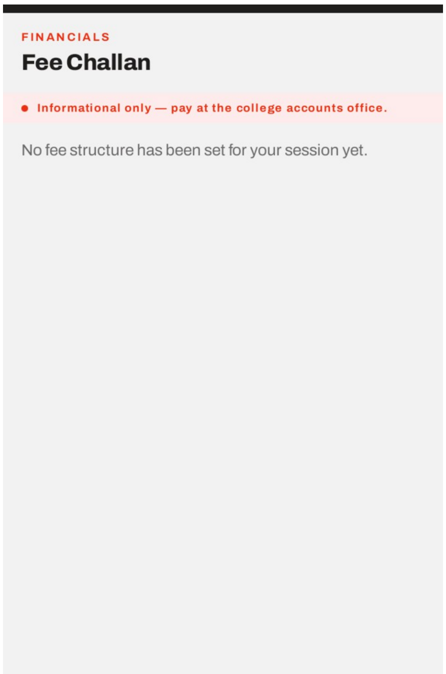

Figure 6-6 Fee Challan information or empty state

Training is complete when the learner can repeat the assigned task without guidance, understands any warning or empty-state message, and knows whom to contact when a record appears incorrect. Pilot feedback should be recorded and used to improve the next department's session.

6.3 Post Deployment Testing

Post-deployment testing is performed after the pilot applications and records have been placed in the college environment. Chapter 5 confirms the available automated baseline: 127 shared academic and role-behaviour checks and 13 desktop continuity checks passed. The checks below add the real accounts, devices, network, records, documents, and users that cannot be fully represented by development testing.

Customer-site execution and formal acceptance are not recorded as completed in the available project evidence. The following table is therefore the required pilot acceptance procedure, with results to be signed at the college rather than assumed in advance.

Table 6-6 Pilot Post-Deployment Acceptance Checks

Check	Applications covered	Required result
Installation and first start	All six applications	Each approved device opens the correct role application without a missing-resource or startup error.
Sign-in and role access	All six applications	Admin, teacher, and student pilot accounts reach only the information and actions intended for their role.
Administrative preparation	Admin Mobile and Admin Desktop	A department, session, people records, roster, curriculum, timetable, fee structure, and publication can be reviewed or maintained as authorised.
Teacher daily work	Teacher Mobile and Teacher Desktop	An assigned teacher can view a class, submit attendance, enter permitted marks, review schedule information, and cannot open another teacher's restricted work.
Student personal access	Student Mobile and Student Desktop	A linked student can view only their own attendance, marks, timetable, fee information, notices, and published documents.
Account linking and approval	Student and Admin applications	A student request reaches the Admin review area; approval links the correct record and rejection leaves other records unchanged.
Desktop document actions	All three desktop applications	The relevant user can choose a file, save or open an approved document, and print where the screen provides that action.
Connection interruption	All six applications	A clear message is shown, completed records are not falsely reported as saved, and normal work resumes after the connection returns.
Converted-record reconciliation	Admin applications with teacher and student samples	Department and roster totals match the approved source, and sampled users see the same accepted information.
Recovery and continuity	Shared service and all six applications	The authorised team can restore the approved backup in a controlled check and repeat the critical sign-in and record checks.

A failed check is recorded with the application, user role, device, affected record, and visible result. The rollout pauses for that area, the cause is corrected, and the same check is repeated. Expansion to another department requires all critical checks to pass and any minor issue to have an agreed owner and completion date.

6.4 Challenges

The main conversion challenges come from record quality, user readiness, and coordinating six separate applications. Development and real-device review have already shown why these controls are needed. Customer-site operational problems are not presented as completed experience because the available evidence does not confirm that the pilot has yet been formally deployed.

Table 6-7 Conversion Challenges and Responses

Challenge	Possible effect	Response and continuing control
Paper records may be incomplete or inconsistent	Names, roll numbers, class membership, or totals may be entered incorrectly.	Use one approved preparation format, report invalid and duplicate roster rows, compare totals, sample individual records, and obtain department sign-off.
Six applications must remain coordinated	A user may receive the wrong role package or one platform may behave differently from its matching application.	Maintain a release list for all three mobile and all three desktop packages, install by role, and repeat the same role journey on both platforms before expansion.
Reliable service and account setup is required	Users may be unable to sign in, recover access, or receive current information when configuration or connectivity is incomplete.	Use a controlled setup checklist, keep credentials outside the report, verify account communication, and test interruption and recovery during the pilot.
Staff and students are changing familiar routines	Users may continue keeping unofficial records or misunderstand warnings, approvals, and empty states.	Provide role-based demonstrations, supervised practice, named support contacts, and a short completion check before unsupervised use.
Some interface issues appear only on real devices	A screen may be difficult to use even when its underlying function is correct.	Review the important journeys on actual Android and Windows devices, collect screenshots and user feedback, correct the issue, and repeat acceptance before the next phase.
Deployment status can be overstated	Build packages may be mistaken for completed college adoption and acceptance.	Record package readiness, site installation, data approval, training, and acceptance as separate milestones, each supported by its own evidence.

These responses keep the conversion manageable and understandable for college staff. They also create a clear record of what is ready, what has been checked, and what must still be completed before the system is treated as operational across the college.

6.5 Summary

This chapter defined a pilot conversion followed by phased departmental expansion for the GGC-MBD College Management System. It covered deployment of Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, and Student Desktop; controlled preparation and reconciliation of starting records; role-based training with two illustrated user procedures; and the acceptance checks required after site installation. Package readiness and the Chapter 5 automated baseline are verified, while customer-site rollout and sign-off remain identified as the next controlled stage. The proposed controls address record quality, service readiness, user adoption, six-application coordination, real-device review, backup, and recovery without overstating the current deployment position.

Chapter 7

Conclusion

7. Introduction

This chapter evaluates the College Management System against the objectives agreed in Chapter 1. It also confirms that every stated requirement is connected to design, implementation, and testing evidence, explains the contribution made by the six applications, and identifies the work that should follow the project.

The evaluation separates a completed project feature from an activity that still requires college-site acceptance. This is important because the six applications have build and automated test evidence, while the pilot deployment, real-user acceptance, and production-scale measurements remain controlled next steps.

7.1 Evaluation

The ten objectives from Section 1.3 are evaluated below. A status of Met means the required capability is present and supported by the available project evidence. Partially met means meaningful evidence exists, but the full acceptance condition has not yet been completed.

Table 7-1 Evaluation of Project Objectives

No.	Objective from Chapter 1	Evaluation	Status
O1	Provide mobile and desktop applications for administrators, teachers, and students.	All six role applications are present. Android test packages and Windows installers have been produced.	Met
O2	Maintain one consistent set of college and academic records.	The applications use the same department, session, people, timetable, attendance, result, and fee records. Data reconciliation remains part of pilot acceptance.	Met
O3	Allow teachers to record attendance digitally and make it available to authorised users.	Teacher Mobile and Teacher Desktop support class attendance, while Admin and Student applications receive the relevant summaries after synchronisation.	Met
O4	Require administrator review before a saved mark is changed.	Teachers submit a correction request and administrators approve or reject it before the accepted record changes.	Met
O5	Let students view routine academic and fee information without an unnecessary office visit.	Student Mobile and Student Desktop present personal attendance, marks, results, timetable, datesheet, fee, fine, notice, and document information.	Met
O6	Publish notices, events, documents, and datesheets to intended users on both platforms.	The publication and audience rules passed the shared tests. Information is refreshed when the application opens; instant push alerts are not claimed.	Met
O7	Give administrators current summaries without manually combining registers.	Admin Mobile and Admin Desktop provide dashboard and insight summaries for departments, sessions, people, attendance, and performance.	Met
O8	Restrict information and actions according to role and identity.	Role, teaching-assignment, linked-student, and publication-audience checks passed. Live-site access remains part of pilot acceptance.	Met
O9	Keep each mobile and desktop role consistent in content, navigation, and major workflows.	Required destinations, role gates, and major workflows are represented on both platforms, but complete visual approval of every state on field devices remains outstanding.	Partially met
O10	Complete functional and user-interface testing for all six applications before submission.	All 140 recorded automated checks passed and representative mobile screens were reviewed. Full six-application device, network, and user-acceptance testing is still required.	Partially met

7.2 Traceability Matrix

The traceability matrix connects every functional requirement in Section 2.3 with its design area, representative source files, and the completed test cases reported in Chapter 5. The source-file column is included because it is required by the Manual Template; it identifies where the feature is represented without describing internal programming details.

A shared source file supports the academic rule used by matching mobile and desktop screens. The listed screen or navigation file identifies the visible application area. PT-1 applies to shared processing, while PT-2 applies to the three desktop applications.

Table 7-2 Complete Requirements Traceability Matrix

Req.	Requirement description	Design specification	Representative source file(s)	Test ID
FR-1	Sign in and route each user to the correct role area.	Authentication and role routing	AppRoot.kt; AdminNavHost.kt; TeacherNavHost.kt; StudentNavHost.kt	UT-3, FT-4, PT-1, PT-2
FR-2	Create and maintain authorised administrator accounts.	Administrator account management	AdministratorsController.kt; AdministratorsScreen.kt	UT-1, FT-1, PT-1
FR-3	Create, view, update, and remove departments.	Department management	DepartmentsActionController.kt; DepartmentsScreen.kt	UT-1, UT-2, FT-1, IT-3, PT-1
FR-4	Create and maintain academic sessions.	Session management	SessionDetailController.kt; SessionDetailScreen.kt	UT-1, UT-2, FT-1, IT-3, PT-1
FR-5	Maintain semester subjects and curriculum.	Curriculum management	SemesterSubjectsController.kt; SemesterSubjectsScreen.kt	UT-2, IT-3, PT-1
FR-6	Maintain students and session rosters.	Roster and student management	SessionStudentsController.kt; SessionStudentsScreen.kt	UT-1, UT-2, FT-1, IT-1, PT-1
FR-7	Maintain teachers, status, and assigned permissions.	Teacher and permission management	TeachersController.kt; TeachersScreen.kt	UT-2, FT-1, IT-3, PT-1
FR-8	Prepare and maintain college timetables.	Timetable planning	MasterTimetableController.kt; MasterTimetableScreen.kt	UT-1, UT-2, PT-1
FR-9	Show each teacher and student their personal schedule.	Role-based schedule views	TeacherScheduleController.kt; StudentTimetableController.kt	UT-3, FT-2, FT-3, IT-3, PT-1
FR-10	Record a complete class attendance register.	Attendance entry	MarkAttendanceController.kt; MarkAttendanceScreen.kt	UT-2, FT-2, PT-1
FR-11	Review attendance history and personal attendance summaries.	Attendance history and reporting	StudentAttendanceController.kt; AttendanceHistoryScreen.kt	UT-2, UT-3, FT-2, FT-3, PT-1
FR-12	Enter assessment marks within the permitted limits.	Assessment entry and validation	MarksEntryController.kt; MarksEntryScreen.kt	UT-1, UT-2, FT-2, PT-1
FR-13	Request a correction to a previously saved mark.	Mark-correction request	MarkEditRequestsController.kt; MarksEntryScreen.kt	UT-1, FT-2, PT-1
FR-14	Approve or reject a requested mark correction.	Mark-correction review	MarkEditRequestsController.kt; MarkEditRequestsScreen.kt	FT-2, PT-1
FR-15	Record semester results for an assigned class.	Semester-result entry	SemesterResultsController.kt; SemesterResultsScreen.kt	UT-2, FT-2, PT-1
FR-16	Show students their marks and semester results.	Personal marks and result views	StudentMarksController.kt; StudentResultsController.kt	UT-2, UT-3, FT-3, PT-1
FR-17	Submit an examination paper for the assigned subject.	Examination-paper submission	ExamPaperSubmissionController.kt; ExamPaperSubmissionScreen.kt	FT-2, IT-2, PT-1
FR-18	Prepare and publish examination datesheets.	Datesheet preparation and publication	DatesheetsController.kt; DatesheetsScreen.kt	IT-2, PT-1
FR-19	Show published datesheets to the relevant users.	Datesheet audience view	DatesheetsController.kt; DatesheetsScreen.kt	UT-3, FT-3, IT-2, PT-1
FR-20	Maintain the fee structure for an academic session.	Session fee management	SessionFeesController.kt; SessionFeesScreen.kt	UT-1, UT-2, IT-3, PT-1
FR-21	Show each student the fee information for their session.	Personal fee view	StudentFeeChallanController.kt; FeeChallanScreen.kt	UT-2, UT-3, FT-3, PT-1
FR-22	Record an authorised fine against a student.	Student fine management	StudentProfileController.kt; StudentProfileScreen.kt	UT-1, UT-2, PT-1
FR-23	Show fines to the affected student	Fine summary and role	StudentProfileController.kt;	UT-2, UT-3, FT-3,

Req.	Requirement description	Design specification	Representative source file(s)	Test ID
	and authorised staff.	access	StudentOwnProfileScreen.kt	PT-1
FR-24	Create and publish college calendar events.	Calendar and audience management	CalendarController.kt; CalendarScreen.kt	UT-1, IT-2, PT-1
FR-25	Publish approved college documents.	Document publication	DocumentsController.kt; DocumentsScreen.kt	FT-1, IT-2, IT-3, PT-1
FR-26	View, download, and open relevant documents.	Role-based document library	DocumentsController.kt; DocumentsScreen.kt	UT-3, FT-3, IT-2, IT-3, PT-1
FR-27	Submit a request to link an account with a student record.	Student account linking	LinkRequestsController.kt; LinkRequestScreen.kt	UT-1, FT-3, IT-1, PT-1
FR-28	Review and decide a student link request.	Link-request review	LinkRequestsController.kt; LinkRequestsScreen.kt	FT-1, FT-3, IT-1, PT-1
FR-29	Publish notifications to a selected audience.	Notification publication	NotificationsController.kt; NotificationsScreen.kt	UT-1, IT-2, PT-1
FR-30	Show notifications intended for the signed-in user.	Notification audience view	NotificationsController.kt; NotificationsScreen.kt	UT-3, FT-3, IT-2, PT-1
FR-31	Provide role-appropriate dashboards, insights, and reports.	Summaries and reporting	InsightsController.kt; InsightsScreen.kt	UT-2, UT-3, FT-1, FT-2, IT-3, PT-1
FR-32	View and maintain the signed-in user's profile.	Personal profile management	StudentProfileController.kt; ProfileScreen.kt; AdminProfileScreen.kt	UT-1, FT-3, PT-1
FR-33	Use Windows file selection, save, open, print, and share actions.	Desktop platform services	DesktopPlatformServices.kt; DocumentsScreen.kt	FT-4, PT-2

7.3 Conclusion

The project has produced one coordinated College Management System presented through six role applications. Admin Mobile and Admin Desktop support college structure, people, approvals, publications, and reports. Teacher Mobile and Teacher Desktop support attendance, assessment, schedules, student records, and examination work. Student Mobile and Student Desktop provide controlled access to personal academic, financial, timetable, and published information.

Its main contribution is not simply replacing a paper form with a screen. Information entered for one authorised purpose becomes useful to the other affected roles without requiring the college to prepare the same record repeatedly. A teacher's attendance entry, for example, contributes to the administrator's overview and the linked student's personal summary while remaining restricted from unrelated users.

Table 7-3 Correlation of Objectives and System Functionality

Objective	Functionality provided	Applications contributing
O1	Separate role experiences on Android and Windows	All six applications
O2	Shared academic structure and coordinated records	All six applications
O3	Attendance register, history, and personal summaries	Teacher, Admin, and Student applications
O4	Teacher correction request and administrator decision	Teacher Mobile/Desktop and Admin Mobile/Desktop
O5	Personal timetable, attendance, marks, results, fees, fines, and publications	Student Mobile and Student Desktop
O6	Audience-based notices, events, documents, and datesheets	All six applications
O7	Dashboard totals, attendance records, and academic insights	Admin Mobile and Admin Desktop
O8	Role, assignment, linked-student, and audience restrictions	All six applications
O9	Matching destinations and major role workflows across platforms	Each mobile application and its desktop counterpart
O10	Shared academic tests, desktop continuity tests, and representative screen review	All six applications

Eight objectives are met and two are partially met. The remaining gap is evidence rather than the absence of the main role functions: complete visual comparison and field acceptance must still cover every required state on representative Android and Windows devices. Chapter 6 therefore places customer-site pilot deployment and sign-off after package readiness, data preparation, and training.

Within this boundary, the project provides a practical foundation for GGC-MBD. It gives the college one place to manage routine academic work, reduces repeated manual preparation, improves visibility for authorised users, and keeps personal information tied to the correct role. The next stage should prove these benefits with real users and approved college records before phased expansion.

7.4 Future Work

Future work is prioritised by what the college needs for safe adoption, not by novelty. The first actions complete deployment evidence and strengthen continuity; optional service expansion should follow only after the pilot is stable.

Table 7-4 Prioritised Future Work

Priority	Enhancement	Reason and expected benefit	Completion evidence
1	Complete pilot deployment and user acceptance	Install signed mobile releases and Windows packages for one department, train users, reconcile approved records, and resolve findings before expansion.	Signed acceptance results for all six applications and the pilot data.
2	Extend offline work from reading to controlled queued writes	The current local-first design preserves authorised records and refreshes changed rows. A later phase can allow selected attendance and assessment entries to wait securely for submission when connectivity returns, with explicit conflict handling.	Interruption, recovery, duplicate-prevention, conflict-resolution, and user-message tests pass.
3	Add optional push alerts	Notify intended users when urgent notices, datesheets, events, or deadlines are published instead of relying only on opening or refreshing the application.	Delivery, audience, permission, and opt-out checks pass on Android and Windows.
4	Expand field and accessibility testing	Test every important state with representative administrators, teachers, and students, including keyboard use, readable contrast, text scaling, and assistive technology.	Approved usability and accessibility checklist with resolved critical findings.
5	Measure production-scale performance	Use realistic department, roster, publication, and concurrent-user volumes on the approved service plan before college-wide expansion.	Recorded response, reliability, storage, and recovery results meet agreed limits.
6	Keep an administrative change history	Record who approved sensitive changes such as role status, mark corrections, account links, and major publications, together with the time and reason.	Authorised audit reports reproduce each selected change without exposing restricted details.
7	Consider online fee payment	If college policy and banking arrangements permit, add secure payment confirmation while retaining the current information-only fee view.	Finance approval, provider security review, reconciliation, refund, and receipt tests pass.

The order can change if college policy or infrastructure requires it, but deployment acceptance should remain first. Adding more features before the pilot is stable would increase support and conversion risk without proving that the existing system is ready for routine use.

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Appendix-A

Fully Dressed Use Case Descriptions

Appendix-A Fully Dressed Use Case Descriptions

This appendix consolidates the final use cases for the College Management System. The descriptions are the same approved records used in Section 2.2.3, so their actors, flows, rules, requirement links, and application coverage remain consistent. Together they cover Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, and Student Desktop.

A.1 Fully Dressed Format

Table A-1 Fully Dressed Use Case Field Guide

Field	Purpose
Use Case ID	Provides a unique reference for the use case.
Use Case Name	States the user goal in a short action-based title.
Actors	Identifies the role or roles that take part.
Description	Summarises what the interaction achieves.
Trigger	States the event that starts the interaction.
Preconditions	Lists what must already be true before it begins.
Postconditions	States the expected result after completion or failure.
Normal Flow	Presents the main user and system sequence.
Alternative Flows	Covers valid variations, validation, empty information, and failure.
Business Rules	Records the college rules and access limits that govern the interaction.
Assumptions	Identifies information or conditions expected to be available.

A.2 Use Case Coverage Index

Table A-2 Use Case, Requirement, and Application Index

Use Case	Name	Requirements	Applications
UC-01	Sign In and Open the Correct Application Area	FR-1	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
UC-02	Manage Administrator Accounts	FR-2	Admin Mobile, Admin Desktop
UC-03	Manage Academic Structure	FR-3, FR-4, FR-5	Admin Mobile, Admin Desktop
UC-04	Manage Teachers and Students	FR-6, FR-7	Admin Mobile, Admin Desktop
UC-05	Prepare and Maintain Timetables	FR-8	Admin Mobile, Admin Desktop
UC-06	Review Approval Requests	FR-14, FR-28	Admin Mobile, Admin Desktop; Teacher Mobile and Teacher Desktop for link requests when permission is granted
UC-07	Publish College Information	FR-18, FR-20, FR-22, FR-24, FR-25, FR-29	Admin Mobile, Admin Desktop; permitted Teacher Mobile and Teacher Desktop for selected notices and datesheets
UC-08	Review Reports and Insights	FR-10, FR-11, FR-12, FR-15, FR-16, FR-31	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop
UC-09	Mark Class Attendance	FR-10	Teacher Mobile, Teacher Desktop
UC-10	Enter Assessment Marks	FR-12	Teacher Mobile, Teacher Desktop

Table A-2 Use Case, Requirement, and Application Index (continued)

Use Case	Name	Requirements	Applications
UC-11	Request a Mark Correction	FR-13	Teacher Mobile, Teacher Desktop
UC-12	Submit Examination Paper and Semester Results	FR-15, FR-17	Teacher Mobile, Teacher Desktop
UC-13	View Teaching Schedule and Assigned Students	FR-6, FR-9	Teacher Mobile, Teacher Desktop
UC-14	Register and Link a Student Account	FR-27, FR-28	Student Mobile, Student Desktop; reviewed in Admin Mobile, Admin Desktop, and permitted Teacher applications
UC-15	View Personal Attendance and Timetable	FR-9, FR-11	Student Mobile, Student Desktop
UC-16	View Marks, Results, and Datesheets	FR-16, FR-19	Student Mobile, Student Desktop
UC-17	View Fee Challan and Fines	FR-21, FR-23	Student Mobile, Student Desktop
UC-18	View Notices, Events, and Documents	FR-24, FR-26, FR-30	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
UC-19	Use Desktop File, Export, and Print Actions	FR-33	Admin Desktop, Teacher Desktop, Student Desktop
UC-20	View and Maintain Personal Profile	FR-32	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop

A.3 Detailed Use Case Descriptions

UC-01: Sign In and Open the Correct Application Area

Table A-3 Fully Dressed Description of UC-01

Use Case ID	UC-01
Use Case Name	Sign In and Open the Correct Application Area
Requirement Traceability	FR-1
Purpose	Allow a registered user to enter the system and reach the area assigned to their role.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
Actors	Administrator, Teacher, Student
Description	A registered user signs in and is directed to the correct role-based home area. A valid session may be reused when the application is opened again.
Trigger	The user opens an application or submits the sign-in form.
Preconditions	The user has a registered account. Administrator and teacher accounts have an active college status.
Postconditions	The correct home area is displayed and only authorised actions are available.
Normal Flow	1. The user opens the application. 2. The system checks for an existing valid session. 3. If needed, the user enters the registered email and password. 4. The system verifies the account and role. 5. The role-appropriate home area opens.
Alternative / Exception Flows	Invalid details produce a clear sign-in message. A disabled account is refused access. A role mismatch directs the user to the appropriate application. A connection failure shows a short retry message without displaying service addresses.
Business Rules	A user may access only the role and college records assigned to the account.
Assumptions	The account exists and the user has access to the registered sign-in details.
Includes / Extends	Included by every protected use case.

UC-02: Manage Administrator Accounts

Table A-4 Fully Dressed Description of UC-02

Use Case ID	UC-02
Use Case Name	Manage Administrator Accounts
Requirement Traceability	FR-2
Purpose	Allow an existing administrator to add and maintain other authorised administrators.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator
Description	An administrator views the administrator directory, creates an account, and maintains the account's college identity and status.
Trigger	The administrator opens People and selects Administrators.
Preconditions	The actor is signed in as an active administrator.
Postconditions	The administrator directory reflects the completed addition or update.
Normal Flow	1. The administrator opens the directory. 2. The system displays existing administrators. 3. The administrator selects Add Administrator. 4. Required identity and contact details are entered. 5. The system validates and creates the account. 6. The updated directory is shown.
Alternative / Exception Flows	Missing, invalid, or duplicate details are identified beside the relevant field. The current administrator cannot accidentally remove the only usable administrative access. A service failure leaves the directory unchanged and shows a safe message.
Business Rules	Only an administrator may create or maintain administrator accounts. Email and account identity must be unique.
Assumptions	The college has approved the person who is being granted administrative access.
Includes / Extends	Includes UC-01.

UC-03: Manage Academic Structure

Table A-5 Fully Dressed Description of UC-03

Use Case ID	UC-03
Use Case Name	Manage Academic Structure
Requirement Traceability	FR-3, FR-4, FR-5
Purpose	Maintain departments, academic sessions, semesters, and subjects as the foundation for all academic records.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator
Description	The administrator creates and reviews departments, sessions, semester progress, and the subjects taught in each semester.
Trigger	The administrator opens the Academics area.
Preconditions	The administrator is signed in. A department must exist before a session can be added, and a session must exist before its subjects can be entered.
Postconditions	The approved academic structure is available to timetable, attendance, marks, fee, and reporting activities.
Normal Flow	1. The administrator selects a department or creates one. 2. A session is created or opened. 3. Semester and session details are reviewed. 4. Subjects are added to the appropriate semester. 5. The system confirms the saved structure.
Alternative / Exception Flows	Duplicate identifiers, invalid years, missing names, or incomplete semester details are rejected with field-level guidance. Deletion is blocked or confirmed when dependent records exist.
Business Rules	Every session belongs to one department. Every subject belongs to a defined session and semester.
Assumptions	The administrator has the approved college programme and subject information.
Includes / Extends	Includes UC-01.

UC-04: Manage Teachers and Students

Table A-6 Fully Dressed Description of UC-04

Use Case ID	UC-04
Use Case Name	Manage Teachers and Students
Requirement Traceability	FR-6, FR-7
Purpose	Maintain the people connected with each academic session and the responsibilities granted to teachers.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator
Description	The administrator adds or imports students, reviews student profiles, creates teacher accounts, and manages teacher status and permissions.
Trigger	The administrator opens People, a session roster, or a student profile.
Preconditions	The administrator is signed in. The relevant department and session already exist.
Postconditions	Valid teacher and student records are available to authorised academic workflows.
Normal Flow	1. The administrator selects Teachers or a session's Students area. 2. Existing records are displayed. 3. The administrator adds, imports, or updates the required person. 4. The system validates identity and session information. 5. The directory or roster refreshes with the saved record.
Alternative / Exception Flows	Duplicate roll numbers or emails are rejected. Invalid import rows are reported without hiding successful rows. Disabling a teacher requires confirmation and does not delete historical records.
Business Rules	A roll number must be unique within its session. Teacher permissions are granted only by an administrator.
Assumptions	Approved staff and enrolment information is available to the administrator.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

UC-05: Prepare and Maintain Timetables

Table A-7 Fully Dressed Description of UC-05

Use Case ID	UC-05
Use Case Name	Prepare and Maintain Timetables
Requirement Traceability	FR-8
Purpose	Create an organised weekly schedule without assigning the same teacher or room to conflicting periods.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop
Actors	Administrator, authorised Teacher
Description	An authorised user creates or updates class periods for sessions, subjects, teachers, rooms, days, and times.
Trigger	The user opens Master Timetable or a session timetable.
Preconditions	Relevant sessions, subjects, and teachers exist. The user has timetable permission.
Postconditions	The approved timetable becomes available to affected administrators, teachers, and students.
Normal Flow	1. The user opens the timetable. 2. A day and period are selected. 3. Session, subject, teacher, room, and time are entered. 4. The system checks for conflicts. 5. The period is saved and the timetable refreshes.
Alternative / Exception Flows	A teacher, room, or session conflict prevents saving and identifies the conflict. Incomplete times or missing assignments are rejected. Cancelling leaves the timetable unchanged.
Business Rules	A teacher, room, or session cannot be assigned to overlapping periods.
Assumptions	The college has agreed its teaching periods and room information.
Includes / Extends	Includes UC-01.

UC-06: Review Approval Requests

Table A-8 Fully Dressed Description of UC-06

Use Case ID	UC-06
Use Case Name	Review Approval Requests
Requirement Traceability	FR-14, FR-28
Purpose	Give authorised staff a controlled way to approve student account links and corrections to locked marks.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop; Teacher Mobile and Teacher Desktop for link requests when permission is granted
Actors	Administrator, authorised Teacher
Description	The reviewer examines a pending request, compares its supporting information with college records, and approves or rejects it with a recorded outcome.
Trigger	The reviewer opens a pending request from the People or Requests area.
Preconditions	A pending request exists and the reviewer has the required permission.
Postconditions	The request is marked approved or rejected. Approved changes are applied to the relevant student link or mark record.
Normal Flow	1. The reviewer opens the request queue. 2. A pending request is selected. 3. Submitted and existing information is compared. 4. The reviewer chooses Approve or Reject. 5. The system asks for confirmation and records the decision.
Alternative / Exception Flows	The reviewer may return without deciding. An already-decided request cannot be decided again. A rejection reason is required where the screen requests one. Failure leaves the request pending.
Business Rules	The person who submitted a mark-correction request cannot approve it. Each pending request may receive only one final decision.
Assumptions	The reviewer can verify the request against college information.
Includes / Extends	Includes UC-01; may extend UC-11 or UC-14.

UC-07: Publish College Information

Table A-9 Fully Dressed Description of UC-07

Use Case ID	UC-07
Use Case Name	Publish College Information
Requirement Traceability	FR-18, FR-20, FR-22, FR-24, FR-25, FR-29
Purpose	Create and publish datesheets, fees, fines, calendar events, documents, and notifications for the intended users.
Priority	High
Supported Applications	Admin Mobile, Admin Desktop; permitted Teacher Mobile and Teacher Desktop for selected notices and datesheets
Actors	Administrator, authorised Teacher
Description	An authorised user prepares college information, selects its audience or academic scope, reviews it, and publishes it for viewing in the relevant applications.
Trigger	The user selects an Add, Create, Upload, Issue, or Publish action.
Preconditions	The user is signed in with the required permission. Any referenced session or student exists.
Postconditions	The new information is stored and becomes visible to its intended users according to publication status and audience.
Normal Flow	1. The user opens the relevant records area. 2. The content and dates are entered or a document is selected. 3. Audience and academic scope are chosen. 4. The system validates the information. 5. The user confirms publication. 6. The published item appears in the list.
Alternative / Exception Flows	Missing dates, invalid amounts, unsupported files, or incomplete audience information prevent publication. The user may save or cancel where the feature allows. Upload failure keeps the item unpublished.
Business Rules	Only authorised users may publish. Students see only information addressed to them or their academic group. Fines must identify a student and amount.
Assumptions	The publisher has approved and accurate content.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

UC-08: Review Reports and Insights

Table A-10 Fully Dressed Description of UC-08

Use Case ID	UC-08
Use Case Name	Review Reports and Insights
Requirement Traceability	FR-10, FR-11, FR-12, FR-15, FR-16, FR-31
Purpose	Help authorised staff review attendance, academic progress, and college summaries without combining separate registers.
Priority	Medium
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop
Actors	Administrator, Teacher
Description	The user opens a report or insight, applies available academic filters, and reviews totals, trends, or students needing attention within the user's authority.
Trigger	The user opens Attendance Records, Insights, or an academic summary.
Preconditions	The user is signed in and relevant academic records exist.
Postconditions	The requested report is displayed. Viewing does not change academic records.
Normal Flow	1. The user opens a reporting area. 2. Available scope and filters are shown. 3. The user selects a department, session, semester, subject, or period where applicable. 4. The system prepares the authorised summary. 5. The user reviews or exports the result.
Alternative / Exception Flows	No matching records produces an informative empty state. Invalid filters are ignored or corrected. A loading failure offers retry without exposing service details.
Business Rules	Administrators may review college-wide information. Teachers may review only their assigned academic records.
Assumptions	Reports reflect the latest successfully stored attendance and assessment records.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

UC-09: Mark Class Attendance

Table A-11 Fully Dressed Description of UC-09

Use Case ID	UC-09
Use Case Name	Mark Class Attendance
Requirement Traceability	FR-10
Purpose	Record one class meeting's attendance accurately for every enrolled student.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects an assigned class and date, marks each student, reviews totals, and submits the register.
Trigger	The teacher selects Mark Attendance.
Preconditions	The teacher is signed in, assigned to the class, and the class roster is available.
Postconditions	One attendance record is stored for each student and becomes available to authorised users.
Normal Flow	1. The teacher selects an assigned class. 2. The date and roster are displayed. 3. Each student is marked present, absent, late, or on leave. 4. The teacher reviews the totals. 5. The register is submitted and confirmation is shown.
Alternative / Exception Flows	An incomplete register is identified before submission. A duplicate class register is opened for review rather than silently duplicated. A connection failure keeps the screen state and offers retry.
Business Rules	A teacher may mark only assigned classes. One final status is recorded per student for a class meeting.
Assumptions	The roster and teacher assignment are correct.
Includes / Extends	Includes UC-01.

UC-10: Enter Assessment Marks

Table A-12 Fully Dressed Description of UC-10

Use Case ID	UC-10
Use Case Name	Enter Assessment Marks
Requirement Traceability	FR-12
Purpose	Record valid marks for students in an assigned subject and examination type.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects a class and assessment, enters each student's marks, resolves validation messages, and submits the completed award list.
Trigger	The teacher selects Marks Entry.
Preconditions	The teacher is assigned to the subject. The class roster and allowed maximum marks exist.
Postconditions	Valid marks are stored and become read-only to the teacher unless an approved correction is granted.
Normal Flow	1. The teacher selects session, subject, semester, and assessment type. 2. The roster is displayed. 3. Marks are entered for each student. 4. The system checks values and completeness. 5. The teacher submits the list. 6. Confirmation and locked status are shown.
Alternative / Exception Flows	Negative marks, values above the maximum, and missing required entries are rejected. An already-submitted list opens as locked. A failed submission leaves the entries available for correction and retry.
Business Rules	Marks must be within the permitted range. A submitted mark cannot be changed directly.
Assumptions	The teacher has completed the assessment and has the approved award information.
Includes / Extends	Includes UC-01; may be extended by UC-11.

UC-11: Request a Mark Correction

Table A-13 Fully Dressed Description of UC-11

Use Case ID	UC-11
Use Case Name	Request a Mark Correction
Requirement Traceability	FR-13
Purpose	Allow a teacher to request a justified correction without directly changing an approved mark.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects a locked mark, enters the proposed value and reason, and submits a request for administrative review.
Trigger	The teacher selects the correction action beside a locked mark.
Preconditions	The teacher entered or is responsible for the mark, and the mark is locked.
Postconditions	A pending request is available to an administrator while the original mark remains unchanged.
Normal Flow	1. The teacher opens the locked award list. 2. A student mark is selected. 3. The proposed mark and reason are entered. 4. The system validates the range and checks for another pending request. 5. The request is submitted and its pending status is shown.
Alternative / Exception Flows	An invalid proposed mark or missing reason prevents submission. A second pending request for the same mark is refused. Cancelling leaves the mark unchanged.
Business Rules	Only an authorised reviewer may apply the requested change. The original value remains active until approval.
Assumptions	The teacher can explain why the correction is required.
Includes / Extends	Extends UC-10 and may lead to UC-06.

UC-12: Submit Examination Paper and Semester Results

Table A-14 Fully Dressed Description of UC-12

Use Case ID	UC-12
Use Case Name	Submit Examination Paper and Semester Results
Requirement Traceability	FR-15, FR-17
Purpose	Allow a teacher to submit required examination material and record final semester outcomes for assigned students.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher selects the relevant academic context, uploads an examination paper or records semester results, and submits the completed information.
Trigger	The teacher opens Submit Exam Paper or Semester Results.
Preconditions	The teacher is assigned to the relevant session or subject. Required students and semester information exist.
Postconditions	The paper or semester result is stored and available to authorised users according to its status.
Normal Flow	1. The teacher selects the academic session and semester. 2. For a paper, the subject and file are selected; for results, each student's result information is entered. 3. The system validates completeness. 4. The teacher reviews and submits. 5. Confirmation is displayed.
Alternative / Exception Flows	Unsupported or missing files, incomplete student results, and invalid academic values prevent submission. Cancelling leaves existing records unchanged. A transfer failure offers retry.
Business Rules	Teachers may submit only for assigned academic work. Result values must follow the college's allowed grading information.
Assumptions	The paper and result information have been approved for submission.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

UC-13: View Teaching Schedule and Assigned Students

Table A-15 Fully Dressed Description of UC-13

Use Case ID	UC-13
Use Case Name	View Teaching Schedule and Assigned Students
Requirement Traceability	FR-6, FR-9
Purpose	Give a teacher a reliable view of assigned classes, periods, and students.
Priority	High
Supported Applications	Teacher Mobile, Teacher Desktop
Actors	Teacher
Description	The teacher reviews the weekly schedule, upcoming class, assigned sessions, and student rosters permitted by the timetable.
Trigger	The teacher opens Home, Schedule, or My Students.
Preconditions	The teacher is signed in and has at least one assignment.
Postconditions	The requested schedule or roster is displayed without changing it.
Normal Flow	1. The teacher opens the relevant area. 2. The system identifies the teacher's assignments. 3. Classes or students are grouped by schedule and session. 4. The teacher selects an item for details. 5. The requested information is displayed.
Alternative / Exception Flows	A teacher with no assignment receives a clear empty state. Missing or unavailable information offers refresh. The teacher cannot open an unrelated roster by changing a selection.
Business Rules	Teachers may view only students and schedules connected with their assigned teaching work.
Assumptions	The administration has prepared the current timetable and roster.
Includes / Extends	Includes UC-01.

UC-14: Register and Link a Student Account

Table A-16 Fully Dressed Description of UC-14

Use Case ID	UC-14
Use Case Name	Register and Link a Student Account
Requirement Traceability	FR-27, FR-28
Purpose	Connect a student's sign-in account with the correct college roll-number record after review.
Priority	High
Supported Applications	Student Mobile, Student Desktop; reviewed in Admin Mobile, Admin Desktop, and permitted Teacher applications
Actors	Student, Administrator, authorised Teacher
Description	A student registers, enters identifying college information, submits a link request, and waits for an authorised decision before personal records are opened.
Trigger	A new or unlinked student selects Register or Link College Record.
Preconditions	The student has a valid email and an existing college enrolment record.
Postconditions	A pending request is created. After approval, the account is linked to one student record; after rejection, the reason or status is shown.
Normal Flow	1. The student registers or signs in. 2. The system identifies the account as unlinked. 3. The student enters roll number and required identity details. 4. The system validates the form and submits a request. 5. An authorised reviewer verifies it through UC-06. 6. The student sees the final status and gains access after approval.
Alternative / Exception Flows	Invalid identity details, an unknown roll number, or an already-linked record prevents submission. A duplicate pending request is not created. Rejected requests may be corrected and resubmitted where allowed.
Business Rules	One account may link to one student record, and one student record may have one approved account.
Assumptions	The college roster contains accurate identity information for comparison.
Includes / Extends	Includes UC-01 and UC-06.

UC-15: View Personal Attendance and Timetable

Table A-17 Fully Dressed Description of UC-15

Use Case ID	UC-15
Use Case Name	View Personal Attendance and Timetable
Requirement Traceability	FR-9, FR-11
Purpose	Give a linked student a current view of class schedule and personal attendance history.
Priority	High
Supported Applications	Student Mobile, Student Desktop
Actors	Student
Description	The student reviews the weekly timetable, next class, attendance totals, subject percentages, and available attendance history for the linked record.
Trigger	The student opens Attendance or Timetable.
Preconditions	The student is signed in and linked to a college record.
Postconditions	The authorised personal schedule or attendance information is displayed without modification.
Normal Flow	1. The student opens the required area. 2. The system identifies the linked session and roll number. 3. Timetable or attendance information is loaded. 4. The student selects a day, subject, or summary where available. 5. Details are displayed.
Alternative / Exception Flows	No published timetable or attendance produces an informative empty state. A connection failure offers retry. The student cannot select another student's record.
Business Rules	Students may view only their own attendance and the timetable of their linked session.
Assumptions	Teachers and administrators have entered the relevant source records.
Includes / Extends	Includes UC-01.

UC-16: View Marks, Results, and Datesheets

Table A-18 Fully Dressed Description of UC-16

Use Case ID	UC-16
Use Case Name	View Marks, Results, and Datesheets
Requirement Traceability	FR-16, FR-19
Purpose	Allow a linked student to review personal assessment information and published examination schedules.
Priority	High
Supported Applications	Student Mobile, Student Desktop
Actors	Student
Description	The student views subject marks, semester results, progression information, and datesheets published for the linked academic session.
Trigger	The student opens Exams, Marks, Results, or Datesheets.
Preconditions	The student is signed in and linked. Relevant records have been entered or published.
Postconditions	The authorised academic information is displayed without allowing student changes.
Normal Flow	1. The student opens the Exams area. 2. Marks, Results, or Datesheets is selected. 3. The system loads information for the linked record and session. 4. Available semester or examination details are shown. 5. The student reviews the selected item.
Alternative / Exception Flows	Unpublished or unavailable information produces a clear message. Incomplete historical data is shown only where valid. A loading failure offers retry.
Business Rules	A student may view only personal marks and results, and only datesheets published for the relevant audience.
Assumptions	Teachers and administrators have completed the relevant academic records.
Includes / Extends	Includes UC-01.

UC-17: View Fee Challan and Fines

Table A-19 Fully Dressed Description of UC-17

Use Case ID	UC-17
Use Case Name	View Fee Challan and Fines
Requirement Traceability	FR-21, FR-23
Purpose	Provide a student with current personal fee information without a routine information visit to the accounts office.
Priority	High
Supported Applications	Student Mobile, Student Desktop
Actors	Student
Description	The student opens the fee area to review the session fee structure, personal challan information, due dates, payment note, and recorded fines.
Trigger	The student selects Fee Challan or Fines.
Preconditions	The student is signed in and linked. A fee structure or fine record exists where applicable.
Postconditions	The student's current fee and fine information is displayed without recording payment.
Normal Flow	1. The student opens the fee area. 2. The system identifies the linked session and roll number. 3. Fee heads, totals, due information, and fines are prepared. 4. The student reviews the details. 5. On desktop, the available record may be opened or printed.
Alternative / Exception Flows	If no fee structure or fines exist, the system shows an informative empty state. Missing required information is not replaced with an invented amount. Loading failure offers retry.
Business Rules	Students see only their own fee and fine records. Displaying a challan does not mark it paid.
Assumptions	The accounts information has been entered by authorised college staff.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

UC-18: View Notices, Events, and Documents

Table A-20 Fully Dressed Description of UC-18

Use Case ID	UC-18
Use Case Name	View Notices, Events, and Documents
Requirement Traceability	FR-24, FR-26, FR-30
Purpose	Give users one place to read college announcements, calendar events, and published documents intended for them.
Priority	Medium
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
Actors	Administrator, Teacher, Student
Description	The user opens notifications, calendar, datesheets, or documents and views items addressed to the user's role or academic group.
Trigger	The user selects a notice, event, or document from the relevant area or notification entry point.
Preconditions	The user is signed in. The item has been published for the user's audience.
Postconditions	The item is displayed and may be marked read where supported. Published content remains unchanged.
Normal Flow	1. The user opens the relevant information area. 2. The system lists authorised published items. 3. The user selects an item. 4. Details or the document are displayed. 5. The user returns to the list or performs an allowed file action.
Alternative / Exception Flows	An expired, removed, or unavailable item produces a safe message. Unsupported opening is handled through download or an explanatory message. Empty lists explain that nothing has been published.
Business Rules	Only items addressed to the user's role, session, or general college audience may be displayed.
Assumptions	Publishers have selected the correct audience and content.
Includes / Extends	Includes UC-01 and may include UC-19 on desktop.

UC-19: Use Desktop File, Export, and Print Actions

Table A-21 Fully Dressed Description of UC-19

Use Case ID	UC-19
Use Case Name	Use Desktop File, Export, and Print Actions
Requirement Traceability	FR-33
Purpose	Allow Windows users to complete role-appropriate document and reporting actions through familiar desktop services.
Priority	Medium
Supported Applications	Admin Desktop, Teacher Desktop, Student Desktop
Actors	Administrator, Teacher, Student
Description	A desktop user selects a file, chooses a save location, opens a document, prints available information, or invokes a supported sharing action from an existing workflow.
Trigger	The user selects Upload, Open, Save, Export, Print, or Share where that action is available.
Preconditions	The user is signed in, authorised for the underlying record, and using a Windows desktop application.
Postconditions	The selected operating-system action completes or returns a clear cancellation or failure result to the application.
Normal Flow	1. The user opens an authorised record or form. 2. A desktop action is selected. 3. The application opens the appropriate Windows selection, save, open, or print service. 4. The user completes the operating-system step. 5. The application confirms completion or returns to the original screen.
Alternative / Exception Flows	The user may cancel without changing data. Missing file access, unsupported file type, or unavailable printer produces a clear message. The application does not expose internal service information.
Business Rules	Desktop actions do not bypass the user's role or record permissions. Native Windows windows may differ visually from the application.
Assumptions	Windows has an appropriate file association or printer where the selected action requires one.
Includes / Extends	Extends UC-04, UC-07, UC-08, UC-12, UC-17, or UC-18.

UC-20: View and Maintain Personal Profile

Table A-22 Fully Dressed Description of UC-20

Use Case ID	UC-20
Use Case Name	View and Maintain Personal Profile
Requirement Traceability	FR-32
Purpose	Allow each signed-in user to review account identity, update permitted profile details, and sign out safely.
Priority	Medium
Supported Applications	Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop
Actors	Administrator, Teacher, Student
Description	The user opens the profile area, reviews account and college identity, changes allowed contact or profile information, and may sign out.
Trigger	The user selects Profile or Sign Out.
Preconditions	The user is signed in.
Postconditions	Valid permitted changes are stored, or the session is ended and protected information is no longer accessible after sign-out.
Normal Flow	1. The user opens Profile. 2. Account and college identity information is displayed. 3. The user edits an allowed field if needed. 4. The system validates and saves the change. 5. The user may return or sign out. 6. Sign-out returns to the authentication screen.
Alternative / Exception Flows	Invalid details are identified without replacing valid information. Protected identity and role fields remain read-only. If saving fails, the previous profile remains. Cancelling sign-out keeps the session active.
Business Rules	Users may change only fields permitted for their role. Role, status, and protected college identity require an authorised administrative process.
Assumptions	The account and related college profile are available.
Includes / Extends	Includes UC-01.

Appendix B

Coding Standards

Appendix-B General Coding Standards and Guidelines

The project applies one set of readability, organisation, access, reuse, and error-handling rules across Admin Mobile, Admin Desktop, Teacher Mobile, Teacher Desktop, Student Mobile, Student Desktop, and the shared project areas. The purpose is not to prescribe a particular tool to an examiner; it is to keep the six applications understandable, consistent, and safe to maintain.

Table B-1 Coding Standards Applied Across the Project

Standard	How It Is Applied
Meaningful names	Names describe the college record or action they represent. Short or unclear abbreviations are avoided unless they are already familiar college terms.
Consistent naming	The same naming pattern is used for data, actions, screens, and reusable parts throughout the project.
Read-only information	Information that should not change during an operation is treated as read-only so that accidental changes are less likely.
Controlled access	Information is read or changed through a defined responsibility rather than being exposed widely across the system.
Clear action names	Each operation is named for the result it produces, such as signing in, publishing a notice, or marking attendance.
Useful comments	Comments explain a non-obvious rule, limitation, or decision. They do not repeat a clear statement in different words.
Consistent layout	Indentation, line spacing, and formatting follow one project style so that work from different modules remains easy to read.
File and folder organisation	Files are named after their main responsibility and are grouped by role, feature, or shared purpose.
Logical class structure	Related information and actions are kept together, while unrelated responsibilities are separated.
Modular organisation	Authentication, academic structure, people, attendance, assessment, records, and reports are organised as distinct areas that cooperate through clear boundaries.

Table B-1 Coding Standards Applied Across the Project (continued)

Standard	**How It Is Applied**
Minimum necessary scope	A value or operation is available only where it is needed. Project-wide access is avoided when a smaller scope is sufficient.
Appropriate permissions	Administrative, teaching, student, and shared responsibilities remain separate. A visible button is never treated as the only access check.
High cohesion	Each unit concentrates on one related purpose, such as validating a form or preparing an attendance summary.
Low coupling	A screen depends on the information and actions it needs rather than on the internal details of storage or communication.
Avoided repetition	Rules used by more than one application, such as validation and safe error messages, are defined once where practical.
Short focused operations	Long activities are divided into understandable steps so that each part can be reviewed and tested independently.
Single responsibility	A unit should have one main reason to change. Display, validation, data access, and operating-system actions are kept distinct.
Extension without disruption	New departments, sessions, subjects, roles, or records should be added through existing patterns without rewriting unrelated work.
Exception handling	Expected failures are handled at the nearest responsible boundary. Users receive a short corrective message instead of a service address or internal failure detail.
Diagnostic records	Development and support messages record the affected operation and safe context without storing passwords, private tokens, or unnecessary personal information.

These standards are checked during implementation review and testing. A justified exception may be used when a platform requires different handling, but the user-facing rule and expected result remain consistent across mobile and desktop.

Appendix C

Application Prototype

Appendix-C Application Prototype

The prototype is a six-application system organised around three user roles and two device types. The mobile figures are cropped captures from working Android builds; the crop removes device and former title-bar content that is not needed to explain the workflow. The desktop figures are representative wide-screen views prepared from the implemented navigation and screen structure. They are labelled separately so the evidence is not overstated.

Table C-1 Prototype Coverage Across All Six Applications

Application	Evidence	Representative Scope
Admin Mobile	Working mobile build capture	College summary, academic structure, people, approvals, records, and reports.
Admin Desktop	Representative desktop interface view	The same administrative areas arranged for wider office work and desktop actions.
Teacher Mobile	Working mobile build capture	Assigned classes, attendance, assessment, students, schedule, and notices.
Teacher Desktop	Representative desktop interface view	The same teaching work with wider rosters, schedules, and document actions.
Student Mobile	Working mobile build capture	Personal attendance, timetable, marks, results, fees, notices, and profile.
Student Desktop	Representative desktop interface view	The same personal information with wider reading and suitable file or print actions.

C.1 Administrator Applications

Both administrator applications provide the same college-wide responsibilities. The mobile arrangement favours quick cards and short routes, while the desktop arrangement uses the available width for office tasks.

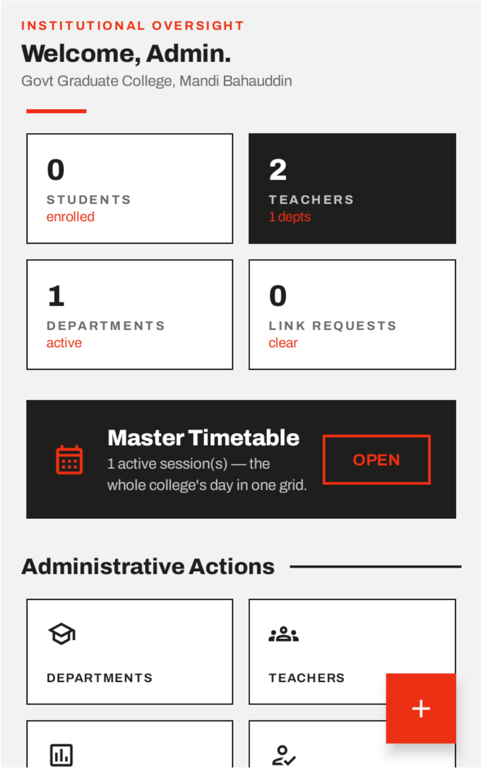

Figure C-1 Admin Mobile working dashboard area

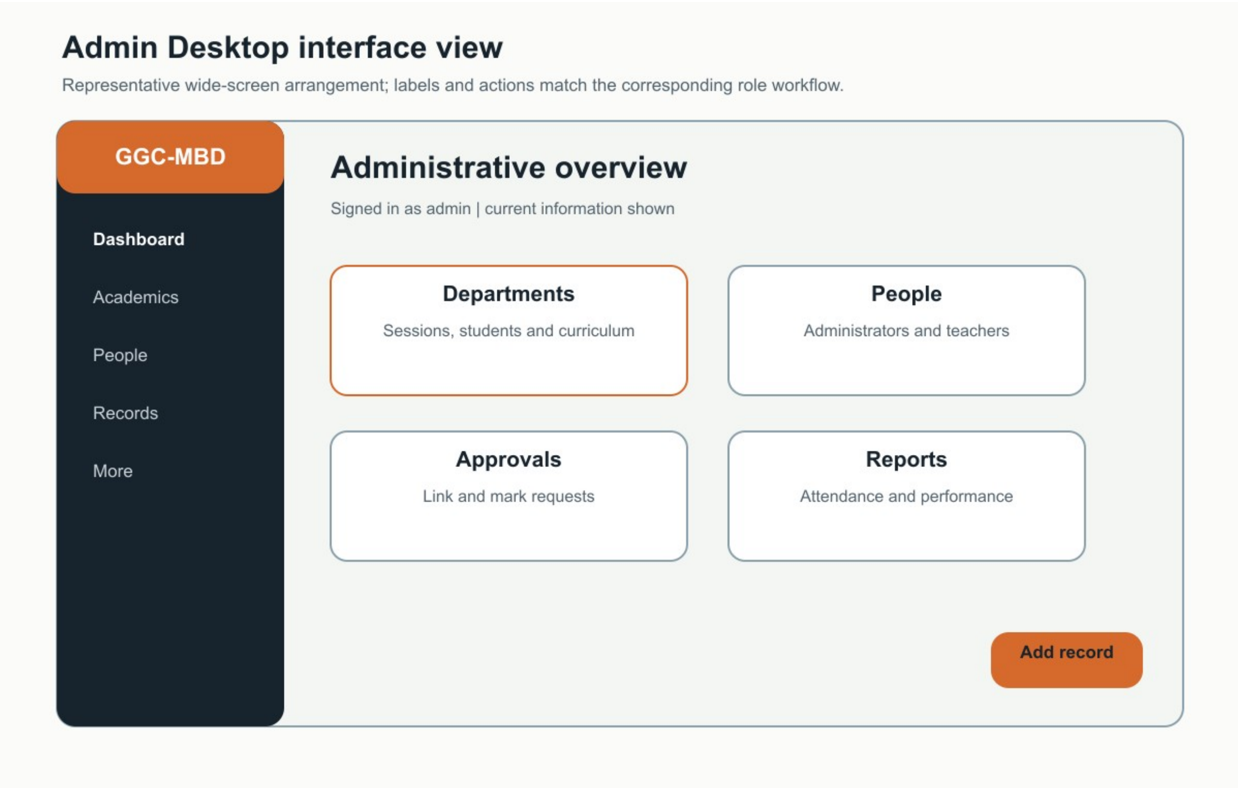

Figure C-2 Admin Desktop representative interface

C.2 Teacher Applications

The teacher applications centre on assigned classes. Attendance, assessment, schedules, students, and permitted college information keep the same meaning on mobile and desktop.

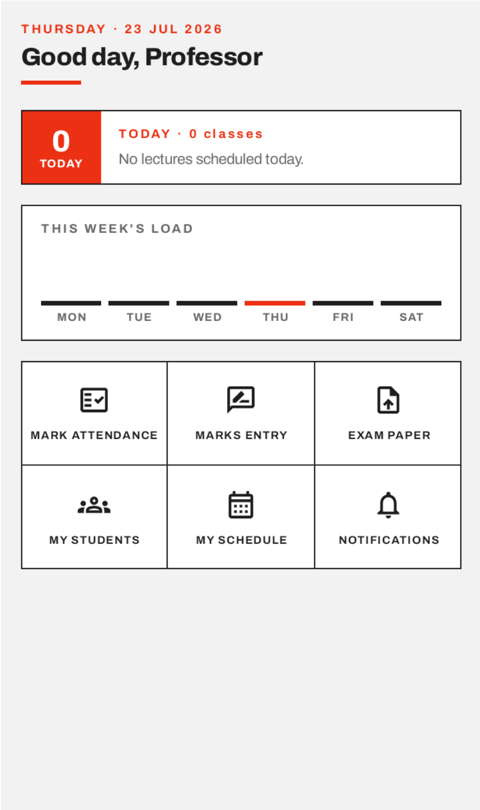

Figure C-3 Teacher Mobile working home area

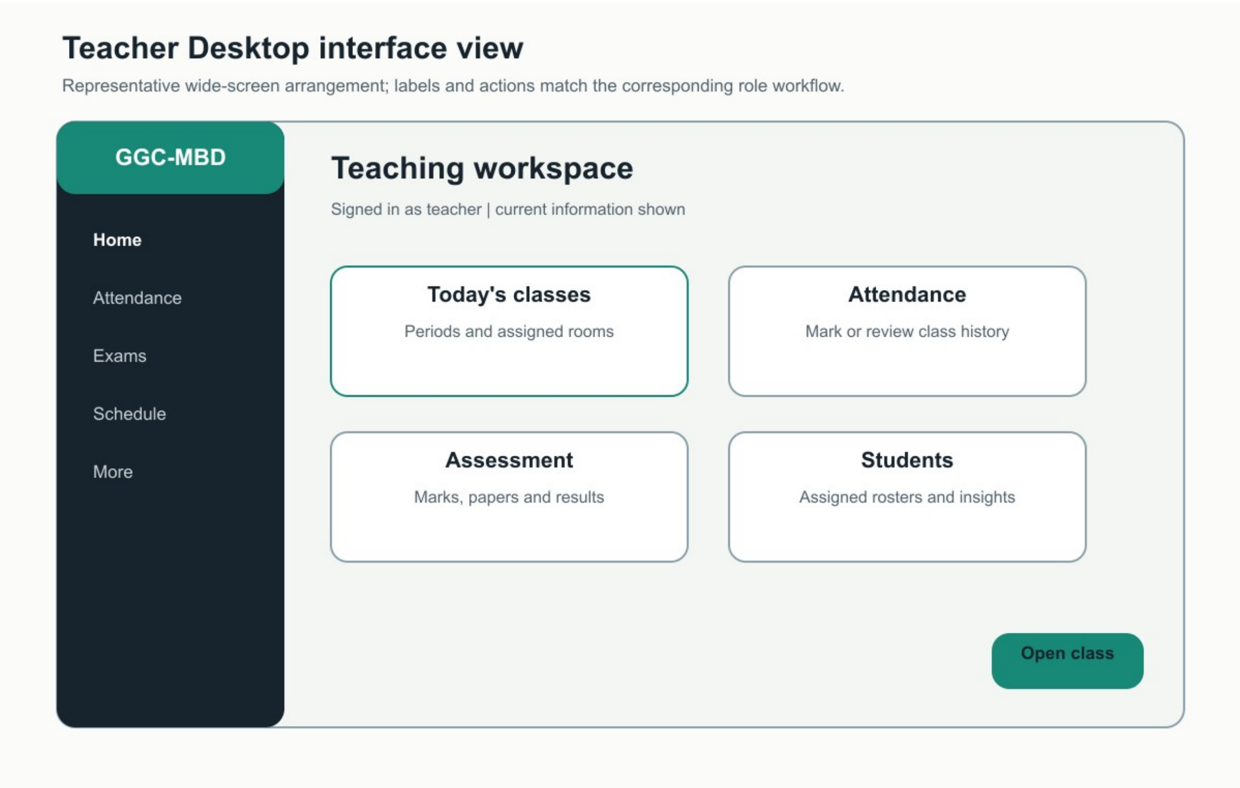

Figure C-4 Teacher Desktop representative interface

C.3 Student Applications

The student applications show only the signed-in student's linked academic information. The prototype includes personal attendance, timetable, examinations, fee information, college notices, documents, and profile access.

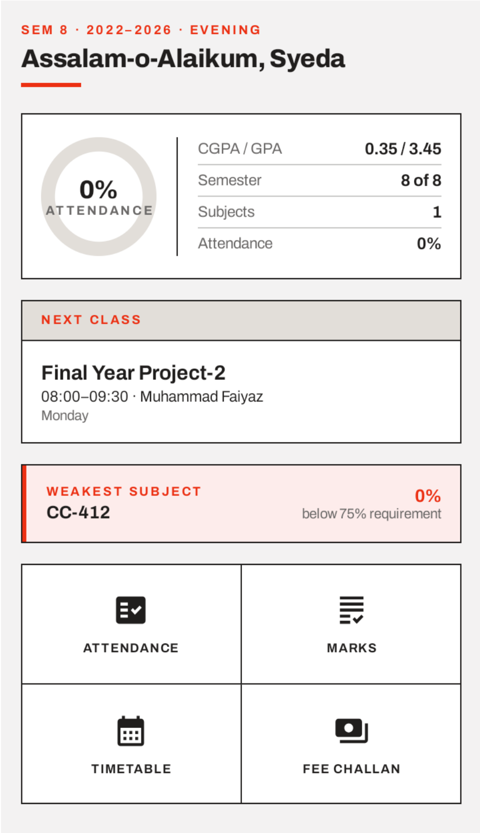

Figure C-5 Student Mobile working home area

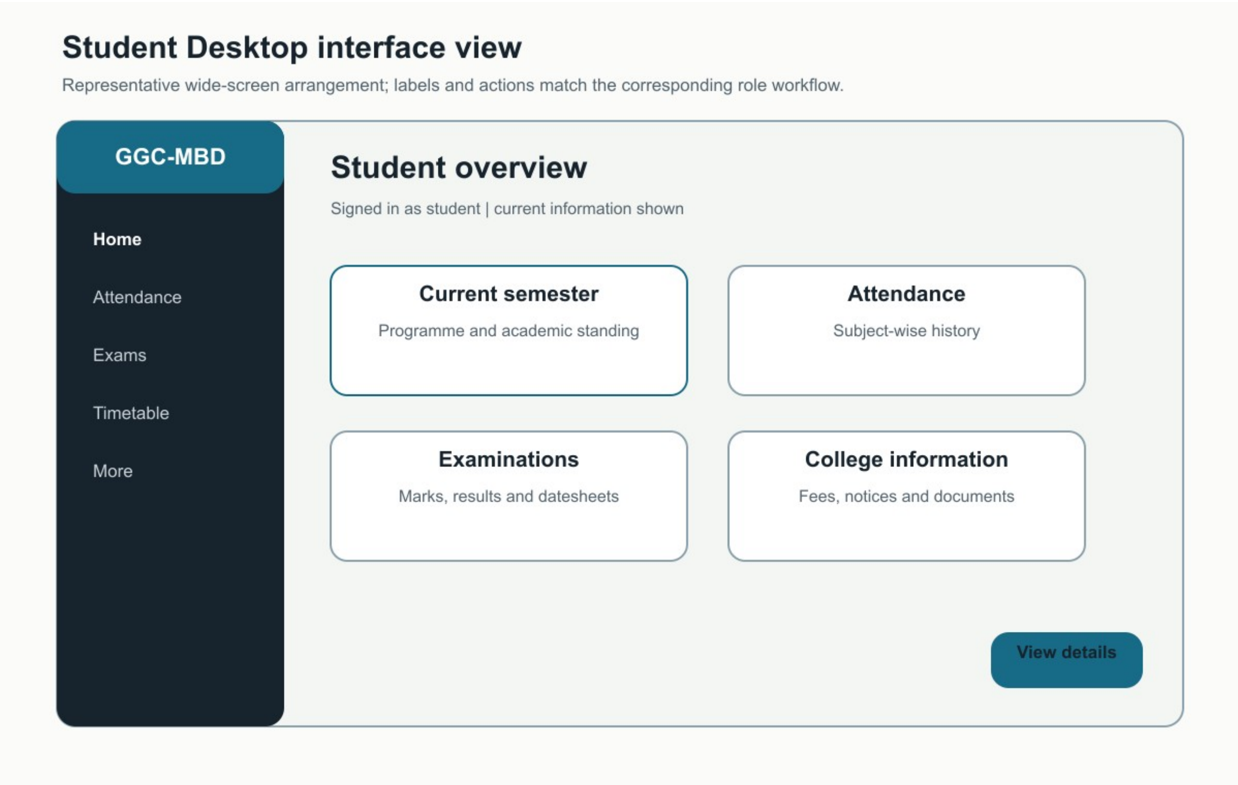

Figure C-6 Student Desktop representative interface